

A RESEARCH MANUSCRIPT FOR THE SURREAL PROJECT

Infinite-Dimensional Hahn Spectral Theory

Part I: infinitesimal thickening and range defects
in Hahn–Hilbert spaces

Part II: exact synthesis and algebra-relative spectra
for row- and column-finite Hahn matrices

Part III: Drazin halos and ramified spectral mapping
in arbitrary Hahn operator algebras

Part IV: Fredholm windows for compact self-adjoint
operators under noncommuting Hahn perturbations

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Status. Four candidate original theorem packages, each with explicit proofs and its own documented literature comparison. None claims to solve a named published open problem. Priority is not certified for any of them. The general proofs have not been independently refereed or checked by a proof assistant. The four accompanying programs check finite algebraic identities, not the infinite-dimensional theorems. Classical Hilbert, Drazin, Fredholm, Hahn–Neumann, Higman and surreal normal-form inputs are identified rather than presented as new.

Abstract

This report assembles four infinite-dimensional spectral theories over one Hahn scalar field $K = \mathbb{C}((t^\Gamma))$. They are printed together because they continue the same non-claim of the collection's finite-dimensional spectral report, and they are printed as separate parts because the word *spectrum* does not mean the same thing in them: the vector spaces and operator algebras are different. Parts I–III allow arbitrary nonzero set-sized ordered value groups; Part IV assumes Γ divisible.

Part I fixes an ordinary complex Hilbert space H and a nonzero set-sized ordered abelian group Γ , and studies $\mathcal{H}_\Gamma = H((t^\Gamma))$ with its coefficientwise-convolution inner product. An automatic-structure theorem identifies every everywhere-defined adjointable operator, with no assumed continuity or support condition, with a Hahn series of bounded operators on H . For an ordinary bounded normal T , the set of $\lambda \in K$ for which $T - \lambda I$ fails to be bijective on $\mathcal{H}_\Gamma$ is exactly $\sigma_{\mathbb{C}}(T) \cup (\sigma_{\mathbb{C}}(T)' + \mathfrak{m}_K)$, where the prime denotes ordinary accumulation points and $\mathfrak{m}_K$ the infinitesimals. Accumulation points thicken into whole infinitesimal monads; isolated points do not thicken; no new eigenvalues appear. An injective ordinary leading operator retains its entire algebraic range defect under every positive-order Hahn perturbation, so a coercive positive self-adjoint operator can fail to be onto with a cokernel of dimension at least the continuum.

Part II uses the same generality for Γ and studies the algebra $\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$ of Hahn series of row- and column-finite matrices acting on $\mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma))$. For $A = D + B$ with $D = \text{diag}(d_i)$ of pairwise distinct ordinary real entries and $B = B^*$ of globally well-ordered positive support, there is a unique near-identity normalized Hahn-unitary diagonalization, exact vectorwise spectral synthesis, a full commutant and spectral-projection classification, and a coherent resolvent at every noneigenvalue. The spectrum relative to $\mathcal{A}_I$ is exactly the labeled eigenvalue set, even when the real residues accumulate. Weighted walks control every coefficient, and a sharp order $2r + 2$ finite-section threshold holds for one-scale locally finite graphs.

Part III takes any nonzero unital complex algebra $\mathfrak{A}$ and proves that the spectrum of a constant $T \in \mathfrak{A}$ relative to $\mathfrak{A}((t^\Gamma))$ is $\sigma_{\mathfrak{A}}(T) \cup (\sigma_{\text{D}, \mathfrak{A}}(T) + \mathfrak{m}_K)$, with $\sigma_{\text{D}, \mathfrak{A}}$ the finite-index Drazin spectrum, without normality, norm or divisibility. For $\mathfrak{A} = \mathcal{B}(H)$ this is Part I's adjointable-algebra spectrum, and it recovers that half of Part I's normal formula; for $\mathfrak{A} = \text{RCF}_I(\mathbb{C})$ it is Part II's spectrum of constant matrices. A ramification criterion decides exactly when polynomial spectral mapping holds over a nondivisible Γ .

Part IV returns to $H((t^\Gamma))$ with Γ divisible. For an injective compact self-adjoint T and a self-adjoint Hahn perturbation E of positive order that need not commute with T , the spectrum of $T + E$ is the whole monad $\mathfrak{m}_K$ together with finite Hermitian eigenvalue clusters over the nonzero eigenvalues of T . These local pieces do not assemble: a trace-class example has no Hahn-unitary diagonalizer with bounded coefficients and no coefficientwise eigenvalue trace sum or determinant product, although a coefficientwise Fredholm determinant with an exact Fredholm alternative exists.

Section 3 states the interface: what the spectra are, why the accumulation statements of the parts are not in conflict, and which hypothesis is doing the work in each.

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Shared front matter

1 Why this is one report, and why it is four parts

1.1 The non-claim that the parts continue

The collection already contains a substantial finite-dimensional spectral report: Hermitian diagonalization, singular values, conditioning, determinantal scales, spectral descent, and ramification of positive matrix roots [2]. Every dimension there is an ordinary finite integer. That report closes by listing, in its own words, what it does *not* contain:

“There is no theorem here about infinite-dimensional Hilbert spaces over $\mathbf{No}[i]$, bounded-operator spectra, compact or trace-class operators, or spectral measures.”

The first two manuscripts assembled here (Parts I and II) supply the first two of those items and the compact half of the third. The two later ones add to the second and third items: Part III computes the algebra-relative spectrum of every constant bounded operator, normal or not, and Part IV treats compact self-adjoint operators under noncommuting infinitesimal perturbations and supplies a coefficientwise trace and Fredholm determinant. A general trace-class theory and a classical Hilbert-space spectral-measure theory are still not developed; Part II supplies an algebraic projection calculus with vectorwise Hahn additivity.

- *Infinite-dimensional Hilbert spaces over the surcomplex numbers.* Sections 5 and 6 construct one explicit such space and determine its adjointable operator algebra exactly.
- *Bounded-operator spectra.* Theorem 8.2 computes the spectrum of the scalar extension of every ordinary bounded normal operator. Theorem 32.1 computes, for every constant element of any unital complex algebra, its spectrum relative to the Hahn extension of that algebra; for bounded operators this is the adjointable-algebra spectrum of Part I, and the Volterra operator shows that isolated spectral points of nonnormal operators can acquire whole halos (section 35.2).
- *Compact operators.* Corollary 9.2 does the same for ordinary compact normal operators, and states explicitly that “compact” there describes the original operator on the ordinary Hilbert space and asserts nothing about compactness or compactoidness in the Hahn valuation topology. Theorem 42.1 adds an arbitrary noncommuting self-adjoint positive-order Hahn perturbation of an injective compact self-adjoint operator, with the same meaning of “compact”.
- *Trace-class operators and spectral measures.* Parts I–III contain no trace-class theory. Part IV defines a coefficientwise trace on $\mathcal{S}_1(H)((t^\Gamma))$ and a Fredholm determinant on its nonnegative-valuation component, with an exact Fredholm alternative (theorem 44.4), and proves that ordinary coefficientwise eigenvalue sums and products need not represent them (theorems 45.1 and 45.2). This is not a general trace-class or nuclear-operator theory. Theorem 19.2 gives a complete Boolean algebra of spectral projections with vectorwise strong Hahn additivity. This does not assert topological countable additivity: remark 19.4 distinguishes the two and supplies a counterexample even for a rank-one value group.

The finite-dimensional report is already a long treatise on determinantal finite-matrix theory. Absorbing these manuscripts into it would bury them. This report is its sibling, not its continuation, and section 3 records what the reports may and may not be read as saying jointly. Parts III and IV arrived after Parts I and II and are built on them: Part III generalizes the algebra half of Part I’s normal spectral formula, and Part IV treats the case that Part I

sets aside as requiring a different theorem (section 15.2). Appendix E records how the four manuscripts were assembled.

1.2 The central hazard: several senses of “spectrum”

The parts use the word *spectrum* for different things, on different vector spaces and in different operator algebras. Quoted side by side without their categories, the conclusions of Parts I and II about accumulation read as a contradiction. They are not one. Parts I–III allow arbitrary nonzero set-sized ordered abelian groups; the coefficient spaces and operator classes, rather than divisibility, account for the different spectral conclusions. The original Part I manuscript assumed divisibility. The proof review removes that unused restriction, as explained in section 5.1. Part IV keeps its source’s divisibility hypothesis.

There are exactly two kinds of spectrum in this report, and the parts use three instances of them. A *bijection spectrum* is the set of λ for which $X - \lambda I$ fails to be bijective on a named vector space: this is $\sigma_\Gamma^{\text{alg}}$ of Part I, also used in Part IV. An *algebra-relative spectrum* is the set of λ for which $X - \lambda 1$ has no two-sided inverse in a named unital algebra: this is $\sigma_\Gamma^{\text{adj}}$ (the algebra $\mathcal{A}_\Gamma(H)$) in Parts I and IV, $\sigma_{\mathcal{A}_I}$ (the algebra $\mathcal{A}_I$) in Part II, and $\Sigma_\Gamma^{\mathfrak{A}}$ (the algebra $\mathfrak{A}_\Gamma = \mathfrak{A}((t^\Gamma))$, for an arbitrary unital $\mathfrak{A}$) in Part III. The third notion, $\Sigma_\Gamma^{\mathfrak{A}}$, is defined for constant elements $T \in \mathfrak{A}$ and contains the constant cases of the other two algebra-relative spectra: $\Sigma_\Gamma^{\mathcal{B}(H)} = \sigma_\Gamma^{\text{adj}}$ and $\Sigma_\Gamma^{\text{RCF}_I(\mathbb{C})} = \sigma_{\mathcal{A}_I}$ (proposition 3.1). No statement of this report identifies a bijection spectrum with an algebra-relative one except where a theorem proves it for the operators in question (theorems 8.2 and 42.1, and the Volterra example section 35.2).

Section 3 works the apparent conflict between Parts I and II out on a single operator, says exactly why both conclusions hold, and shows that Part III’s formula produces both of them.

The inner-product conventions also differ: Parts I and IV are linear in the first variable, whereas Part II is conjugate-linear in the first. To translate a scalar pairing, conjugate its value (equivalently, exchange its arguments); do not copy a variational functional without translating its linearity. Positivity and the adjoint identity are unchanged by this conversion. Part III uses no inner product.

1.3 Part I at a glance

Throughout Part I, $H \neq 0$ is a complex Hilbert space, Γ is nonzero, ordered, abelian and set-sized, and

$$K = \mathbb{C}((t^\Gamma)), \quad \mathcal{H}_\Gamma = H((t^\Gamma)).$$

The valuation is the least support exponent. A prime on an ordinary compact set denotes its accumulation set, not a derivative and not an essential spectrum.

Theorem 1.1 (Part I theorem package). *The following assertions hold in this model.*

- (i) *Every everywhere-defined adjointable K -linear operator on $\mathcal{H}_\Gamma$ has a unique expansion in $\mathcal{B}(H)((t^\Gamma))$. Its coefficient operators are ordinary bounded operators and have one common well-ordered support.*
- (ii) *For an ordinary bounded normal T , its coefficientwise extension has*

$$\sigma_\Gamma(T) = \sigma_{\mathbb{C}}(T) \cup (\sigma_{\mathbb{C}}(T)' + \mathfrak{m}_K), \quad \sigma_{p,\Gamma}(T) = \sigma_{p,\mathbb{C}}(T). \quad (1.1)$$

Here σ_Γ is the failure-of-bijection spectrum on $\mathcal{H}_\Gamma$; it also equals the spectrum in the adjointable algebra.

	Part I	Part II	Part III	Part IV
Vector space	$\mathcal{H}_\Gamma = H((t^\Gamma))$, H an ordinary complex Hilbert space; every ordinary square-summable vector is a constant of $\mathcal{H}_\Gamma$	$\mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma))$; each Hahn coefficient has <i>finite</i> coordinate support, but a vector may have infinitely many nonzero Hahn coordinates	none is needed; for $\mathfrak{A} = \mathcal{B}(H)$ the algebra acts on $\mathcal{H}_\Gamma$, for $\mathfrak{A} = \text{RCF}_I(\mathbb{C})$ on $\mathcal{H}_I$	$\mathcal{H}_\Gamma$ of Part I, with H infinite-dimensional and separable
Operator class	$\mathcal{A}_\Gamma(H) = \mathcal{B}(H)((t^\Gamma))$, Hahn series of <i>ordinary bounded</i> operators on H	$\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$, Hahn series of row- and column-finite matrices, with no ordinary boundedness at all	<i>constant</i> elements T of $\mathfrak{A}_\Gamma = \mathfrak{A}((t^\Gamma))$, $\mathfrak{A}$ any nonzero unital complex algebra	$A = T + E \in \mathcal{A}_\Gamma(H)$, T injective compact self-adjoint, $E = E^*$ of positive order, not commuting with T
“Spectrum” means	$\sigma_\Gamma^{\text{alg}}(A) = \{\lambda \in K : A - \lambda I \text{ is not bijective on } \mathcal{H}_\Gamma\}$; for $A \in \mathcal{A}_\Gamma(H)$ also $\sigma_\Gamma^{\text{adj}}(A)$, failure of invertibility in $\mathcal{A}_\Gamma(H)$	$\sigma_{\mathcal{A}_I}(X) = \{z \in K : X - zI \text{ has no two-sided inverse inside } \mathcal{A}_I\}$; a spectrum relative to a named unital K -algebra	$\Sigma_\Gamma^{\mathfrak{A}}(T)$: no two-sided inverse of $T - \lambda I$ inside $\mathfrak{A}_\Gamma$; equal to $\sigma_\Gamma^{\text{adj}}$ for $\mathfrak{A} = \mathcal{B}(H)$ and to $\sigma_{\mathcal{A}_I}$ for $\mathfrak{A} = \text{RCF}_I(\mathbb{C})$	both Part I spectra, proved equal for this class; also the algebraic Fredholm spectrum $\sigma_{\text{ess,alg}}$
Hypothesis on Γ	nonzero, set-sized, ordered abelian, not assumed divisible ; only the metric-completeness subsection 12.1 assumes $\Gamma \subseteq \mathbb{R}$	nonzero, set-sized, ordered abelian, not assumed divisible ; the source manuscript records the absence of that hypothesis as a feature, and only near-one binomial roots are used	nonzero, set-sized, ordered abelian, not assumed divisible ; the polynomial image depends on divisibility (theorem 34.7)	nonzero, set-sized, ordered abelian and divisible
What accumulation does	accumulation points of $\sigma_\mathbb{C}(T)$ acquire whole complex infinitesimal monads (theorem 8.2); isolated points do not thicken	accumulation of the ordinary residues d_i does not enlarge $\sigma_{\mathcal{A}_I}(A)$ beyond $\{\lambda_i\}$ (theorem 20.2 and example 20.3)	every point of the Drazin spectrum acquires a whole halo; in a Banach algebra these are the accumulation points and the isolated non-poles (theorem 32.1, (33.1))	the accumulation point 0 keeps the whole monad $\mathfrak{m}_K$, with no eigenvalues in it; each nonzero eigenvalue becomes a finite Hermitian cluster (theorem 42.1)

Table 1: The senses of “spectrum” used in this report. Every spectral statement below names its vector space or algebra and which of these definitions it uses.

- (iii) If $S : V \rightarrow V$ is injective and E is an endomorphism-valued Hahn series of strictly positive order, then

$$\text{coker}(S + E) \cong W((t^\Gamma)), \quad V = \text{Ran } S \oplus W,$$

noncanonically. No commutation hypothesis is required.

- (iv) For $De_n = n^{-1}e_n$ and $s = t^\eta$, $\eta > 0$, the operator $C = D^2 + s^2I$ on $\ell^2((t^\Gamma))$ is bounded, positive, self-adjoint and coercive, but not surjective. Its cokernel has K -dimension at least $2^{\aleph_0}$. At every rank its range is closed and proper in the valuation topology, with zero orthogonal complement.
- (v) The extension of an ordinary bounded T has a least F -valued norm bound if and only if T attains its ordinary Hilbert operator norm. When the least bound exists, it is the ordinary norm itself.

Each assertion is proved in Part I. The result about the least norm bound is not a claim that these operators are unbounded: all operators in (i) have field-valued bounds. The failure is the absence of a *least* such bound. Likewise, a spectral point in (ii) need not be an eigenvalue.

1.4 Part II at a glance

Throughout Part II, Γ is a set-sized ordered abelian group with no divisibility assumption, I is a nonempty index set, and

$$F = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)), \quad \mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma)), \quad \mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma)).$$

Let $D = \text{diag}(d_i)$ with the $d_i \in \mathbb{R}$ pairwise distinct, let $B = B^* \in \mathcal{A}_I$ have positive support, and set $A = D + B$.

Theorem 18.2 constructs a unique pair (U, Λ) with

$$AU = U\Lambda, \quad U^*U = UU^* = \mathbf{1}, \quad U = \mathbf{1} + \text{positive-support terms},$$

where $\Lambda = \text{diag}(\lambda_i)$, $\lambda_i - d_i$ has positive support, and every U_{ii} is positive real Hahn. All exponents of the corrections belong to the additive monoid generated by $\text{supp } B$; the construction uses no extension of the value group. The distinct-real-label hypothesis entails $|I| \leq |\mathbb{R}|$; no claim of a larger residue index set is implicit.

Theorems 19.1, 19.2 and 20.2 then give exact eigenvector expansions, classify every commuting operator and spectral projection, and prove $\sigma_{\mathcal{A}_I}(A) = \{\lambda_i : i \in I\}$, the spectrum relative to the specified unital K -algebra and not a Banach-algebra spectrum. Theorems 21.2 to 21.4 and 21.6 determine how much of an infinite operator a single coefficient can see: an eigenvalue coefficient is supported by closed weighted walks, a change outside a finite section can first affect it only through a walk that leaves the section and returns, and the corresponding valuation threshold is exact in a family of path examples.

1.5 Part III at a glance

Throughout Part III, $\mathfrak{A}$ is any nonzero unital complex algebra and Γ any nonzero set-sized ordered abelian group; no norm, normality or divisibility is assumed. For a constant $T \in \mathfrak{A}$, theorem 32.1 proves

$$\Sigma_\Gamma^\mathfrak{A}(T) = \sigma_\mathfrak{A}(T) \cup (\sigma_{D,\mathfrak{A}}(T) + \mathfrak{m}_K),$$

where $\sigma_{D,\mathfrak{A}}(T)$ is the set of $a \in \mathbb{C}$ at which $T - a\mathbf{1}$ is not finite-index Drazin invertible. The mechanism is an exact inverse descent (theorem 31.3): a strong automorphism of the Hahn field moves any nonzero infinitesimal to a monomial, a cyclic-support projection reduces a

putative inverse to a finite-tail Laurent inverse, and such an inverse exists exactly at Drazin points, with exact valuation $-\nu v(\varepsilon)$ at index ν (theorem 32.2). In a Banach algebra the Drazin spectrum consists of the accumulation points and the isolated non-poles (33.1), which recovers the adjointable-algebra half of Part I's normal formula (corollary 33.1 and remark 33.2); the Volterra operator has spectrum $\{0\}$, no accumulation point, and $\Sigma_\Gamma^{\mathcal{B}(H)}(\mathcal{V}) = \mathfrak{m}_K$. A polynomial germ of ramification index e maps a halo, after subtracting its ordinary image, onto zero and exactly the nonzero infinitesimals with valuation in $e\Gamma$, which gives a necessary and sufficient criterion for spectral mapping over every value group (theorems 34.5 and 34.7).

1.6 Part IV at a glance

Throughout Part IV, H is infinite-dimensional and separable, Γ is *divisible*, and the space and algebra are $\mathcal{H}_\Gamma$ and $\mathcal{A}_\Gamma(H)$ of Part I. For $A = T + E$ with T ordinary injective compact self-adjoint and $E = E^* \in \mathcal{A}_\Gamma(H)$ of positive order, not necessarily commuting with T , theorem 42.1 proves

$$\sigma_\Gamma^{\text{adj}}(A) = \sigma_\Gamma^{\text{alg}}(A) = \mathfrak{m}_K \cup \bigcup_{c \in \sigma_{\mathbb{C}}(T) \setminus \{0\}} \{\lambda_{c,1}, \dots, \lambda_{c,m_c}\},$$

with the second set exactly the point spectrum, and $\sigma_{\text{ess,alg}}(A) = \mathfrak{m}_K$. The finite clusters come from coefficientwise Riesz projections and a direct rotation. They do not assemble into a global spectral theorem in $\mathcal{A}_\Gamma(H)$: for $A_\star = \text{diag}(2^{-n}) + t^\eta Y$ with a trace-class weighted path Y , no Hahn unitary in $\mathcal{A}_\Gamma(H)$ diagonalizes $A_\star$ (theorem 43.1), and the coefficientwise eigenvalue sum and product diverge although the coefficientwise trace and Fredholm determinant exist (theorems 44.4, 45.1 and 45.2). No nonzero coherent Hahn analytic function detects the whole monad at the accumulation point (theorem 46.4).

1.7 What is and is not claimed as new

No part claims a classical input as new, and the four parts have separate novelty boundaries and separate literature comparisons.

For Part I: the ordinary spectral theorem, Baire category, the closed graph theorem, Hilbert norm estimates and Hahn support lemmas are classical inputs [8, 6]. The non-Archimedean failure of orthogonal decomposition is known in other models, including work of Aguayo and Nova [10] and Aguayo, Nova and Shamseddine [11]; we do not claim to have discovered that failure in general. The proposed contribution is the particular combination of automatic adjoint structure, the exact accumulation-monad spectrum, the arbitrary positive-support range-defect formula, and the explicit coercive example with a continuum-dimensional defect in $H((t^\Gamma))$. In the repository sections and primary literature inspected, these statements were not found in this form. This is a qualified report of the search, not a proof of priority; a theorem may have predecessors under terminology not located by that search.

For Part II: simple-eigenvalue perturbation theory and its normalization conventions are classical, and Greenbaum, Li and Overton give an accessible account of both eigenvalue and eigenvector perturbation [15]; fixed-point and near-diagonal formulations are also established, including the work of Kenmoe, Smerlak and Zadorin [16]. The recursive Rayleigh–Schrödinger mechanism and the usual first corrections are not claimed as new. The support argument uses Neumann's ordered-series machinery [6] and Higman's word-order theorem [7]. Matrix products and walks have a long-established connection; the path-sum framework of Giscard, Thwaite and Jaksch is an explicit reference [17], and the walk theorem below is not a claim to have discovered that connection. The coefficient algebra is likewise not new: Ara, Goodearl and O'Meara study real and complex row- and column-finite matrices and emphasize that positivity in that algebra does not force the existence of eigenvalues or square roots [18], which is why an infinite-dimensional result must state both its vector space and its operator algebra. The

elementary no-eigenvector example in section 24 is a boundary test, not a new phenomenon. The proposed contribution is the combined arbitrary-rank Hahn theorem: a globally coherent normalized diagonalizer for infinitely many simple residue levels, exact synthesis on a specified Hahn module, coherent resolvents at every noneigenvalue, and marked-walk stability with sharp finite-section thresholds. Neither the finite report inspected in [3] nor the cited perturbation and row-finite sources state that package. This is a targeted comparison, not an exhaustive theorem of bibliographic absence; some components may have predecessors in formal operator or valued-algebra literature not located in this search.

For Part III: Drazin inverses and their finite-index identities, the resolvent-pole characterization, Banach-algebra Drazin spectra, the empty-Drazin-spectrum characterization of algebraic elements, ordinary polynomial spectral mapping and the nilpotent ceiling formula are classical [19, 20, 13]; generalized Drazin invertibility [21] is a different condition. Strong Hahn-automorphism constructions belong to the surrounding literature [23]. The constant-normal formula is Part I's. The proposed contribution is the arbitrary-infinitesimal inverse descent in the original value group, the resulting halo formula for every unital complex algebra, the exact inverse valuation, and the ramification criterion identifying every missing polynomial-image valuation. The algebraicity criterion is a Hahn interpretation of a classical fact.

For Part IV: compact spectral theory, isolated-cluster perturbation [26], the direct rotation of two projections [27], ordinary trace-class determinant identities [28], Hahn-field algebraic closedness [25] and non-Archimedean Fredholm theory in Serre's different category [29] are credited, and the range-defect factorization is Part I's. The proposed contribution is the exact arbitrary-rank spectrum and algebraic Fredholm spectrum for the stated noncommuting compact-residue class, the trace-class example with no bounded-coefficient Hahn-unitary diagonalizer and no coefficientwise eigenvalue trace sum or determinant product, the support-controlled Hahn Fredholm alternative with its finite-coupling domain and accumulation obstruction, and the separation of point-spectrum invariance from monad growth under enlargement of the value group. Both comparisons were targeted, at the pin recorded in appendices C and D, and certify no priority.

1.8 Relation to the nearby non-Archimedean literature

Aguayo–Nova–Shamseddine work with free Banach spaces of countable type over the complex Levi–Civita field, represented by weighted c_0 -type spaces [11]. Ishizuka proves spectral results for self-adjoint compactoid operators over valued fields with formally real residue field [12]. These are nearby but different settings. The constant copy of H in the Part I space retains every ordinary square-summable vector, while its valuation topology is discrete; the extension of an ordinary compact operator is not thereby compactoid in the new topology.

The June 2026 preprint of Miabey treats diagonal operators and finite-rank perturbations in a non-Archimedean Hilbert-space setting [14]. It is used only to delimit related contemporary work; none of its theorems is an input to the proofs here. Formal resolvent expansions also have a classical relationship with pole and Drazin-spectrum questions; see Boasso [13]. The normal-operator formula of Part I is proved directly, so no Drazin-theoretic equivalence is needed as an unproved bridge. Part III now proves that equivalence for constant elements of any unital algebra (theorem 31.3), and so gives a second proof of the algebra-relative half of Part I's formula (remark 33.2); the Part I proof does not depend on it.

For Part II, the relevant nearby object is the algebra of ordinary row- and column-finite matrices itself [18]. The reference concerns ordinary such matrices and is not a source for the Hahn diagonalization asserted here.

Strand	Established input or predecessor	Contribution here
Finite surcomplex matrices	Repository spectral chapter	Infinite coefficient Hilbert spaces (Part I); arbitrary index sets (Part II)
Hahn formal calculus	Neumann support lemma	Automatic global support for adjoints (Part I)
Ordinary normal spectra	Hilbert spectral theorem	Exact infinitesimal thickening formula (Part I)
Formal perturbation	Neumann inversion near the identity	Persistence of arbitrary injective range defects (Part I)
Non-Archimedean orthogonality	Known nonorthomodular examples	A closed coercive range with continuum defect (Part I)
Operator norm bounds	Ordinary norm and attainment	Exact least-bound criterion over the Hahn field (Part I)
Perturbation recursions	Rayleigh–Schrödinger corrections	Global support certificate for all orders at once (Part II)
Row- and column-finite matrices	Ara–Goodearl–O’Meara	Hahn diagonalization under simple ordinary residues (Part II)
Matrix walks	Path-sum framework	Marked-walk stability and sharp section thresholds (Part II)
Drazin inverses	Finite-index identities, resolvent poles	Inverse descent at every Hahn infinitesimal; halo formula in any unital algebra (Part III)
Polynomial spectral mapping	Ordinary Drazin spectral mapping	Exact ramified image and value-group criterion (Part III)
Compact perturbation	Riesz projections, direct rotation	Exact spectrum under noncommuting Hahn perturbations (Part IV)
Fredholm determinants	Ordinary trace-ideal identities	Hahn Fredholm alternative; failure of spectral sums and products (Part IV)

Table 2: The novelty boundary: classical tools are not renamed as new theorems.

1.9 Set-sized mathematics and terminology

All index sets, exponent groups, supports and families in the proofs are sets, and we work in ordinary ZFC. The passages to $\mathbf{No}[i]$ in sections 2.4 and 25 are transports of set-sized Hahn objects, not formations of a set containing all surreal numbers. The full surreal class offers additional scales but is not a set-sized coefficient field in ZFC; the proofs below avoid class-indexed summations and class-sized choices by fixing the workspace at the outset.

“Positive support” means that every exponent is strictly greater than zero; it does not mean that the complex coefficients are positive. “Hermitian” means $A^* = A$. In Part II, “unitary” means two-sided algebraic unitarity in $\mathcal{A}_I$, and a “Hahn-orthonormal basis” means a family with the strong-summation synthesis property proved in theorem 19.1, not a Hamel basis and not an ordinary Hilbert-space basis.

2 Shared Hahn preliminaries

This section fixes the Hahn conventions used by all four parts and states the support lemma once. Each part specifies its own coefficient space, operator algebra and, where present, inner product; these are not shared merely because the scalar field is the same.

2.1 Support conventions and strong summability

A Hahn series with coefficients in a complex vector space V is a family

$$x = \sum_{\gamma \in \Gamma} x_\gamma t^\gamma, \quad x_\gamma \in V,$$

whose support $\{\gamma : x_\gamma \neq 0\}$ is well ordered in the given order of Γ . Write $V((t^\Gamma))$ for the resulting set. The zero series has empty support. For $x \neq 0$ put $v(x) = \min \text{supp } x$, and put $v(0) = +\infty$.

A family $(x_a)_{a \in J}$ in $V((t^\Gamma))$ is *Hahn summable*, also called *strongly Hahn summable*, when its union of supports is well ordered and, for every γ , only finitely many a have $(x_a)_\gamma \neq 0$. Its sum, written $\sum_a^H x_a$, is defined coefficientwise by those finite sums. All unqualified infinite formal sums in this report mean this notion. It is not convergence of ordinary partial sums in the full surreal fine topology, and proposition 23.1 exhibits a strongly summable family whose partial sums do not converge even in the intrinsic valuation topology of K .

2.2 The support lemma, stated once

The first two source manuscripts open with the Hahn–Neumann support lemma, one citing it and supplying the two-set finiteness argument, the other reproving Higman’s word-order input by a minimal bad sequence. The lemma is classical, and it is already printed with a full proof elsewhere in this collection: the finite-dimensional spectral report proves the word-order statement as its `spec:lem:words` and the positive-monoid statement as its `spec:lem:neumann`, in its section “A self-contained support engine” [2]. We therefore state it here once, in the form all four parts use, and do not reprint that proof.

Lemma 2.1 (Classical Hahn–Neumann support lemma). *Let Γ be an ordered abelian group.*

- (i) *If $A, B \subseteq \Gamma$ are well ordered, then $A + B$ is well ordered and each γ has only finitely many representations $\gamma = a + b$ with $a \in A$ and $b \in B$. A finite union of well-ordered subsets is well ordered, and the corresponding statement holds for any finite number of summands.*
- (ii) *If $S \subseteq \Gamma_{>0}$ is well ordered, then the additive monoid*

$$S^* = \{s_1 + \cdots + s_n : n \geq 0, s_j \in S\},$$

including the empty sum 0, is well ordered, and for each $\gamma \in \Gamma$ there are only finitely many ordered words $(s_1, \dots, s_n)$ over S with sum γ , the length n being allowed to vary.

This is the standard support lemma [6], whose word form rests on Higman’s theorem that finite words over a well-quasi-ordered alphabet are well-quasi-ordered [7]. Part (i) is elementary and both source manuscripts give the argument: infinitely many distinct a in a fixed-sum family yield an increasing subsequence in A and hence a strictly decreasing subsequence $\gamma - a$ in B , a contradiction; and every sequence in a well-ordered set has an infinite nondecreasing subsequence, obtained by successively choosing an occurrence of the minimum of the remaining tail. Part (ii) is the usual Neumann lemma in its positive-support form. Both are stated classical inputs here, not new combinatorial claims.

Part (ii) from Higman’s theorem, as the Part IV source deduces it. Order finite words over the well-ordered alphabet S by subsequence embedding with coordinatewise increase; Higman’s theorem makes this a well-quasi-order [7]. In any infinite sequence of words there are indices $i < j$ with the i th word embedded in the j th, and positivity of the letters makes the total of the first at most that of the second; so no infinite strictly decreasing sequence of totals exists. For a fixed total, a strict embedding either inserts a positive letter or strictly increases a letter, and so strictly increases the total; hence infinitely many distinct words cannot share one total. This deduction does not assume that $n \min S$ is cofinal in Γ .

Remark 2.2 (Why the full word form is needed). All four parts use the finite-word assertion of lemma 2.1(ii), not only the assertion that each fixed power is meaningful. The latter alone would not justify a geometric series. The estimate $v(X^n) \geq n\varepsilon$ is also not a substitute: in a value group of higher rank the sequence $n\varepsilon$ may be bounded above, as section 23 exhibits concretely.

Lemma 2.1(i) licenses scalar multiplication on $V((t^\Gamma))$ and products of operator-valued Hahn series, even when the coefficient algebra is noncommutative; a finite number of supports is combined by iteration.

2.3 Support-certified inversion and formal substitution

Lemma 2.3 (Support-certified Neumann inverse). *Let $\mathcal{R}$ be a unital complex algebra and let $E \in \mathcal{R}((t^\Gamma))$ have $v(E) > 0$. Then*

$$(I + E)^{-1} = \sum_{n \geq 0} (-E)^n \quad (2.1)$$

is a well-defined two-sided inverse in $\mathcal{R}((t^\Gamma))$, with support in $(\text{supp } E)^$.*

Proof. Expand each coefficient of E^n as a sum over ordered words in $\text{supp } E$. Lemma 2.1 makes the union well ordered and all coefficients finite, including the sum over n ; the products of the resulting series are the formal convolution coefficients, and the geometric identity is a formal one-variable identity. Noncommutativity causes no problem, because every factor is a power of the same E . $\square$

Remark 2.4 (A second proof, from the first source). The Part I manuscript proves lemma 2.3 by a different route, which is recorded here because it uses no formal-identity bookkeeping. After lemma 2.1 has made every coefficient of $\sum_{n \geq 0} (-E)^n$ a finite sum, the finite identity $(I + E) \sum_{n=0}^N (-E)^n = I + (-1)^N E^{N+1}$ telescopes coefficientwise across all word lengths, since at a fixed exponent only finitely many word lengths occur; right multiplication telescopes as well. This argument uses associativity but not commutativity of $\mathcal{R}$.

Corollary 2.5 (Evaluation of homogeneous constructions). *Let $\mathcal{R}$ be a unital complex algebra, let $S \subseteq \Gamma_{>0}$ be well ordered, and let $X_n \in \mathcal{R}((t^\Gamma))$ satisfy $\text{supp } X_n \subseteq nS$ for every $n \geq 1$. Then $(X_n)_{n \geq 1}$ is strongly summable. Finite sums, products, coefficientwise linear maps preserving supports, and regroupings of such homogeneous constructions commute with their evaluation $\sum_{n \geq 1}^H X_n$. The same holds with $\mathcal{R}$ replaced by a module over it, or by K itself.*

Proof. The union of supports lies in S^* . At a fixed exponent, lemma 2.1(ii) bounds the possible lengths and the possible ordered exponent words. All coefficient operations within one word are finite, so the claimed interchanges are finite rearrangements coefficient by coefficient. $\square$

Corollary 2.6 (Formal power series at a positive-support argument). *If $X \in \mathcal{R}((t^\Gamma))$ has positive support, then any one-variable formal power series with complex coefficients can be evaluated at X , and formal identities between such series remain valid whenever their substitutions have zero constant term. In particular $(I + X)^{-1}$ is given by (2.1) and $(I + X)^{-1/2}$ is defined by its binomial series.*

Proof. Take $S = \text{supp } X$ in corollary 2.5. $\square$

2.4 The surcomplex realization

Suppose an ordered additive embedding $j : \Gamma \hookrightarrow \mathbf{No}$ is fixed; when Γ is itself a set-sized ordered subgroup of the surreal additive group, take j to be the inclusion. Conway normal forms give the maps

$$\sum_{\gamma} r_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} r_{\gamma} \omega^{-j(\gamma)}, \quad r_{\gamma} \in \mathbb{R}, \quad (2.2)$$

and their coefficientwise complexification. These are an ordered-field embedding $F \hookrightarrow \mathbf{No}$ and a field embedding $K \hookrightarrow \mathbf{No}[i]$, preserving conjugation and the prescribed strong sums. The classical normal-form and multiplication facts are standard surreal theory [5].

The sign in (2.2) matters. A well-ordered set of Hahn exponents becomes a reverse-well-ordered set of Conway exponents, exactly the normal-form condition, and positive Hahn exponents become infinitesimal monomials. All supports are sets, so the normal forms define actual surreal or surcomplex numbers, and the vector spaces below never become proper classes. In particular, $\Gamma = \mathbb{Q}$ and $t = \omega^{-1}$ suffice for all explicit counterexamples of Part I.

Warning 2.7 (Distinct completions and topologies). The Part I construction $H((t^{\Gamma}))$ is neither the algebraic tensor product $H \otimes_{\mathbb{C}} K$ nor a silently chosen sequence space $\ell^2(K)$. The Part II space $\mathbb{C}^{(I)}((t^{\Gamma}))$ is likewise not ordinary ℓ^2 , and its vectors have finite-support *coefficients* without having finitely many nonzero Hahn coordinates. Nor does the intrinsic valuation topology of either space equal the topology induced by embedding its scalar field into the full surreal fine topology, in which the set-sized scalar workspace is discrete [1]. All of these distinctions matter to the results.

The closure of $H \otimes_{\mathbb{C}} K$ in $H((t^{\Gamma}))$ is determined exactly by `duals:thm:completion` and `duals:thm:cofinality-completion` of `docs/surcomplex/three-duals-of-hahn-vector-spaces/`.

3 The interface between the parts

3.1 What is shared and what is not

Parts I and II share the scalar field K , the support conventions of section 2.1, the support lemma lemma 2.1, the inversion tools of section 2.3, and the surcomplex embedding (2.2). Nothing else is shared. In particular no theorem of Part I is used in the proof of any theorem of Part II, and no theorem of Part II is used in Part I. Parts III and IV use the same shared tools. Part IV works in Part I's space and algebra and uses Part I's range-defect theorem theorem 7.1, for which its source gives a second proof (remark 40.1). The main theorems of Part III use no theorem of Parts I, II or IV; its comparison with them is section 3.3. No theorem of Part III or IV is used in Part I or Part II. Each part has its own inner product, its own operator algebra, its own spectral definition, and its own coefficient-space requirements, and each proves its own positivity statement from scratch: proposition 5.1 constructs the positive square root $\|x\| = \sqrt{\langle x, x \rangle}$ from the even leading exponent of $\langle x, x \rangle$ and a near-one binomial root; this works for every ordered value group. Likewise, proposition 17.3 proves positivity without assuming that every positive scalar has a square root. Part II uses no norm, and corollary 18.3 explicitly allows nondivisible groups.

Neither construction needs divisibility, and neither does Part III, whose straightening automorphism adjoins no fractional exponent (lemma 31.1). Part IV assumes it, to diagonalize finite Hermitian clusters inside the field (lemma 41.3). In Part I all field-valued roots used in norms and moduli have even leading exponent, and all other scalar roots are near-one binomial roots; ordinary positive operator square roots are taken in $\mathcal{B}(H)$, before Hahn extension. General

positive Hahn scalars and squared norms have different root requirements: t has no square root in $\mathbb{R}((t^{\mathbb{Z}}))$, whereas $\|te\| = t$ for any ordinary unit vector e .

3.2 The accumulation statements are about different objects

Read without their categories, the two parts appear to disagree about accumulation. Part I proves that accumulation points of the ordinary spectrum thicken into infinitesimal monads. Part II proves that the algebra-relative spectrum is exactly the labeled eigenvalue set even when the real residues accumulate. One operator makes the difference visible.

Let $I = \mathbb{N}_{\geq 1}$ and consider the diagonal data $d_n = 1/n$, whose residues accumulate at 0.

- As an *ordinary bounded operator* on $H = \ell^2(\mathbb{N}_{\geq 1})$, the map $De_n = n^{-1}e_n$ is compact, normal and injective, and its ordinary spectrum is $\{1/n : n \geq 1\} \cup \{0\}$ with 0 as its only accumulation point. Corollary 9.2 then gives, for the failure-of-bijectivity spectrum on $\mathcal{H}_\Gamma = \ell^2((t^\Gamma))$,

$$\sigma_\Gamma(D) = \mathfrak{m}_K \cup \{1/n : n \geq 1\}.$$

The whole infinitesimal monad of 0 is in the spectrum. The witness is concrete: a putative solution of $Dy = f$ with $f = (n^{-1})_n$ would need leading coefficient $(1)_n$, which is not in ℓ^2 .

- As an *element of $\mathcal{A}_I$* with $I = \mathbb{N}_{\geq 1}$ and $B = 0$, the same matrix $\text{diag}(1/n)$ has $\text{diag}(n) \in \text{RCF}_I(\mathbb{C}) \subseteq \mathcal{A}_I$ as a two-sided inverse, because a diagonal matrix of unbounded complex entries is a perfectly good row- and column-finite matrix. Hence $0 \notin \sigma_{\mathcal{A}_I}(D)$, and example 20.3 records this.

Both statements are true. They are not statements about the same operator: one is about a bounded operator on a Hilbert space whose vectors are square-summable, the other about a matrix acting on finite-support coefficient vectors, and invertibility is being tested in two different algebras. The unbounded diagonal inverse is excluded in Part I precisely because theorem 6.1 forces the coefficients of an adjointable operator to be ordinary *bounded* operators; it is admitted in Part II precisely because row- and column-finiteness replaces boundedness there. Section 14 makes the same point from the finite-section side, and proposition 24.1 makes the converse point: the Part II diagonalizer has a first coefficient with unbounded entries, so that theorem does not secretly solve the ordinary bounded-operator problem. Theorem 43.1 of Part IV makes the same point for an arbitrary Hahn unitary in $\mathcal{A}_\Gamma(H)$ and a trace-class perturbation.

3.3 One formula for the constant algebra-relative spectra

Part III's spectrum is defined for any unital algebra, and two of this report's algebra-relative spectra are instances of it. The following statement is an observation made in assembling this report: it follows from the definitions and theorem 32.1 and is not a theorem of any of the four source manuscripts.

Proposition 3.1 (The algebra-relative spectra of constant elements). (i) *For $\mathfrak{A} = \mathcal{B}(H)$ one has $\mathfrak{A}_\Gamma = \mathcal{A}_\Gamma(H)$, so for every ordinary bounded T*

$$\sigma_\Gamma^{\text{adj}}(T) = \Sigma_\Gamma^{\mathcal{B}(H)}(T) = \sigma_{\mathbb{C}}(T) \cup (\sigma_{\mathcal{D}, \mathcal{B}(H)}(T) + \mathfrak{m}_K) \supseteq \sigma_\Gamma^{\text{alg}}(T).$$

For normal T the inclusion is an equality by theorem 8.2.

- (ii) *For $\mathfrak{A} = \text{RCF}_I(\mathbb{C})$ one has $\mathfrak{A}_\Gamma = \mathcal{A}_I$, so for every constant row- and column-finite matrix T , not necessarily diagonal or Hermitian, $\sigma_{\mathcal{A}_I}(T) = \Sigma_\Gamma^{\text{RCF}_I(\mathbb{C})}(T)$ is given by (32.1).*

Proof. Both algebra identities are the definitions of $\mathcal{A}_\Gamma(H)$ and $\mathcal{A}_I$, and the spectra are then defined by the same condition. The inclusion in (i) holds because a two-sided inverse in $\mathcal{A}_\Gamma(H)$ acts as an inverse map on $\mathcal{H}_\Gamma$. $\square$

Applied to the operator of section 3.2, $d_n = 1/n$, the one formula (32.1) produces both of that section's conclusions, and shows that the difference lies entirely in which algebra is named.

- In $\mathfrak{A} = \mathcal{B}(H)$, $\sigma_{\mathbb{C}}(D) = \{1/n\} \cup \{0\}$ and 0 is an accumulation point, hence not a finite pole, so $\sigma_{\mathcal{D}, \mathcal{B}(H)}(D) = \{0\}$ by (33.1). Then $\Sigma_{\Gamma}^{\mathcal{B}(H)}(D) = \{1/n : n \geq 1\} \cup \mathfrak{m}_K$, which is corollary 9.2.
- In $\mathfrak{A} = \text{RCF}_I(\mathbb{C})$, $D - a$ is invertible for every $a \notin \{1/n\}$, including $a = 0$, because every diagonal matrix is row- and column-finite; so $\sigma_{\text{RCF}_I(\mathbb{C})}(D) = \{1/n\}$. Each $D - 1/m$ is Drazin invertible of index one, with $P_{\text{nil}} = E_{\{m\}}$ and the diagonal Drazin inverse with entries $(1/n - 1/m)^{-1}$ for $n \neq m$ and 0 at m . So the Drazin spectrum is empty and $\Sigma_{\Gamma}^{\text{RCF}_I(\mathbb{C})}(D) = \{1/n\}$, which is example 20.3.

The same computation applies to any constant diagonal $D = \text{diag}(d_i)$ in $\text{RCF}_I(\mathbb{C})$, distinct labels or not, and gives $\{d_i : i \in I\}$; for distinct real labels this is the case $B = 0$ of theorem 20.2. Part III does not treat the perturbed, nonconstant operators $D + B$ of Part II, nor the nonconstant operators of Part IV. Neither does it speak about bijectivity on a vector space: the statement $\sigma_{\Gamma}^{\text{alg}} = \sigma_{\Gamma}^{\text{adj}}$ is proved in this report for constant normal operators (theorem 8.2), for the Volterra operator (section 35.2), and for the operators of Part IV (theorem 42.1), and nowhere else.

3.4 Reading one part alone

Each part is self-contained given section 2, except that Part IV uses Part I's space, algebra and range-defect theorem, and Part III's comparison remarks refer to Parts I and II. A reader interested in the Hilbert-space obstructions can read Part I and stop; a reader interested in exact infinite diagonalization can read Part II and stop; a reader interested in nonnormal constant operators in an arbitrary algebra can read Part III; a reader interested in noncommuting compact perturbations and determinants can read Part I's sections 5 and 7 and then Part IV. Section 49 describes the four verification suites together, and appendices A to D record the separate repository and literature audits, which were carried out against three different pinned revisions.

Part I. Hahn–Hilbert spaces: infinitesimal spectral thickening and closed-range defects

Standing hypotheses for Part I. $H \neq 0$ is an ordinary complex Hilbert space and Γ is a nonzero set-sized ordered abelian group. Divisibility is not assumed. Only section 12.1, which uses a real-valued metric, additionally assumes $\Gamma \subseteq \mathbb{R}$; the subsequent closed-range and duality statements use the valuation topology at any rank. “Spectrum” in this part means failure of bijectivity on $\mathcal{H}_\Gamma = H((t^\Gamma))$, or failure of invertibility in $\mathcal{A}_\Gamma(H)$, and theorem 8.2 proves that for constant normal operators the two agree.

4 Why infinite dimension is the next substantial question

This part asks what happens when an ordinary infinite-dimensional Hilbert space is retained as the coefficient space, rather than replacing a finite matrix by a larger finite matrix. Over $\mathbb{C}$, positivity, completeness and adjoints support a familiar chain of implications: self-adjoint spectra are real, coercive operators are invertible, and closed subspaces admit orthogonal projections. The construction below retains a positive-definite inner product, all ordinary Hilbert vectors as constants, and a large adjointable operator algebra; in rank one it is complete. Nevertheless several of those implications fail, for an exactly identifiable reason: *a leading coefficient must itself be an ordinary Hilbert vector.*

The central results are not a list of isolated anomalies. An automatic operator-structure theorem makes the admissible algebra intrinsic. An injective-leading-term theorem explains the persistence of range defects. Together with ordinary normal spectral theory, these give an exact spectral classification and quantitative counterexamples to coercive inversion.

5 The Hahn–Hilbert space

5.1 The field and the roots actually needed

Define

$$F = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)) = F[i].$$

Order F by its first nonzero coefficient and conjugate K coefficientwise. Let $\mathcal{O}_K = \{z : v(z) \geq 0\}$ and $\mathfrak{m}_K = \{z : v(z) > 0\}$, including zero in both sets. Coefficient extraction at zero gives $\text{st} : \mathcal{O}_K \rightarrow \mathbb{C}$. Every finite scalar has a unique expression $z = a + \varepsilon$ with $a \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_K$.

The source manuscript assumed divisibility to obtain square roots of all positive elements of F . That assumption is stronger than the proofs need. For a positive scalar $r = ct^\delta(1 + u)$ with $c > 0$ and $v(u) > 0$, a square root exists in F if and only if $\delta \in 2\Gamma$. Necessity follows by taking valuations; if $\delta = 2\beta$, then

$$\sqrt{r} = \sqrt{c}t^\beta \sum_{n \geq 0} \binom{1/2}{n} u^n.$$

The support lemma and the binomial identity verify this positive root. Squared vector lengths and scalar moduli always have even leading exponent, as the next proposition proves. The roots of ordinary positive operators used later are formed in the ordinary bounded-operator algebra. Thus no Part I proof requires roots of arbitrary positive Hahn scalars, and the standing divisibility restriction can be removed.

5.2 The Hahn–Hilbert inner product

We take ordinary and extended inner products to be linear in the first variable and conjugate-linear in the second. On $\mathcal{H}_\Gamma = H((t^\Gamma))$ define

$$\langle x, y \rangle = \sum_{\alpha, \beta} \langle x_\alpha, y_\beta \rangle_H t^{\alpha+\beta}. \quad (5.1)$$

Each ordinary Hilbert inner product in the coefficient is evaluated *before* the Hahn convolution. For example, a constant vector $(1/n)_{n \geq 1} \in \ell^2$ has its ordinary finite real squared norm, even though the separate family of its constant-coordinate squared magnitudes is not Hahn summable. We are not using that inadmissible family as a Hahn sum.

Proposition 5.1 (Positive-definite scalar extension). *For any ordered abelian group Γ , formula (5.1) defines a positive-definite Hermitian form. Its norm $\|x\| = \sqrt{\langle x, x \rangle}$ exists in F , satisfies the Cauchy–Schwarz and triangle inequalities, and*

$$v(\langle x, x \rangle) = 2v(x), \quad v(\|x\|) = v(x) \quad (x \neq 0). \quad (5.2)$$

If $x = t^\delta(u + \text{positive-order terms})$ with $u \neq 0$, then $\|x\| = t^\delta(\|u\|_H + \text{positive-order terms})$.

Proof. Lemma 2.1(i) gives well-definedness and sesquilinearity. The coefficient at 2δ , where $\delta = v(x)$, is exactly $\|x_\delta\|_H^2 > 0$; all other contributions have larger exponent. This proves positivity. For $x \neq 0$ write

$$\langle x, x \rangle = \|x_\delta\|_H^2 t^{2\delta}(1 + u), \quad u = 0 \text{ or } v(u) > 0.$$

The binomial root of corollary 2.6 gives

$$\|x\| = \|x_\delta\|_H t^\delta \sum_{n \geq 0} \binom{1/2}{n} u^n.$$

This is a positive square root in F for every Γ ; set $\|0\| = 0$. Its leading term proves both valuation assertions. The same argument for $H = \mathbb{C}$ defines the scalar modulus $|z| = \sqrt{z\bar{z}}$ without divisibility. Completing a square for $y \neq 0$ (the case $y = 0$ is immediate), using $\langle x - ay, x - ay \rangle \geq 0$ with $a = \langle x, y \rangle / \langle y, y \rangle$, gives $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$. The usual expansion of $\|x + y\|^2$, followed by this inequality, gives the triangle inequality. These are finite algebraic arguments over the ordered field F . $\square$

6 Every adjoint has a global Hahn expansion

6.1 The apparently smaller coefficient algebra

Let $\mathcal{A}_\Gamma(H) = \mathcal{B}(H)((t^\Gamma))$. Thus an element $A = \sum_\gamma A_\gamma t^\gamma$ has bounded ordinary operator coefficients and one well-ordered support. There is no uniform ordinary bound on the norms of all its coefficients. It acts by

$$(Ax)_\delta = \sum_{\gamma+\alpha=\delta} A_\gamma x_\alpha. \quad (6.1)$$

Lemma 2.1 proves this action is defined on every $x \in \mathcal{H}_\Gamma$. The series $A^* = \sum_\gamma A_\gamma^* t^\gamma$ satisfies $\langle Ax, y \rangle = \langle x, A^*y \rangle$, by finite coefficientwise rearrangement.

One might suspect that $\mathcal{A}_\Gamma(H)$ is only a convenient restricted operator class. The next theorem says that the existence of an everywhere-defined adjoint forces exactly this class, without any advance assumption that an operator preserves Hahn sums.

Theorem 6.1 (Automatic adjoint structure). *Let $A, B : \mathcal{H}_\Gamma \rightarrow \mathcal{H}_\Gamma$ be everywhere-defined K -linear maps such that*

$$\langle Ax, y \rangle = \langle x, By \rangle \quad (x, y \in \mathcal{H}_\Gamma). \quad (6.2)$$

There are unique bounded operators $A_\gamma \in \mathcal{B}(H)$ with well-ordered nonzero support such that A acts by (6.1). Moreover $B_\gamma = A_\gamma^$. Consequently*

$$\text{End}_{\text{adj}}(\mathcal{H}_\Gamma) \cong \mathcal{B}(H)((t^\Gamma))$$

as $$ -algebras. No separability of H is assumed.*

Proof. Step 1: bounded coefficient operators. For constant vectors $u, w \in H$, define $A_\gamma u = [t^\gamma]Au$ and $B_\gamma w = [t^\gamma]Bw$. Extracting the coefficient at γ in (6.2) gives

$$\langle A_\gamma u, w \rangle_H = \langle u, B_\gamma w \rangle_H. \quad (6.3)$$

Both maps are defined on all of H . To see that A_γ has closed graph, let $u_n \rightarrow u$ and $A_\gamma u_n \rightarrow z$ in the ordinary Hilbert norm. Testing against any fixed w in (6.3) gives $\langle z, w \rangle = \langle u, B_\gamma w \rangle = \langle A_\gamma u, w \rangle$. Hence $z = A_\gamma u$. The ordinary closed graph theorem makes A_γ bounded. The same argument applies to B_γ , and (6.3) identifies the two as adjoints.

Step 2: one well-ordered support. Let $S = \{\gamma : A_\gamma \neq 0\}$. If S were not well ordered, ordinary choice permits a strictly decreasing sequence $\gamma_1 > \gamma_2 > \dots$ in S . Each $\ker A_{\gamma_n}$ is a proper closed subspace of H , and hence nowhere dense. Baire's theorem gives $u \in H$ outside their countable union. Then the support of Au contains every γ_n , contradicting its being a Hahn series. Thus S is well ordered. Because an ordinary bounded operator is zero exactly when its adjoint is zero, the coefficient family of B has the same support.

Step 3: constants determine the full operator. Let $\tilde{A}, \tilde{B}$ be the convolution operators built from these coefficient families. They are adjoints by lemma 2.1 and agree with A, B on constants. For arbitrary $x \in \mathcal{H}_\Gamma$ and constant $w \in H$,

$$\langle Ax, w \rangle = \langle x, Bw \rangle = \langle x, \tilde{B}w \rangle = \langle \tilde{A}x, w \rangle.$$

If a Hahn vector pairs to zero with every constant w , then each of its ordinary coefficients pairs to zero with all of H , so every coefficient vanishes. Hence $Ax = \tilde{A}x$. The analogous conclusion holds for B . Uniqueness follows by evaluating on constant vectors and extracting coefficients. Composition and the involution are the finite Hahn convolution operations, proving the algebra assertion. $\square$

Corollary 6.2 (Automatic summability and boundedness). *Every everywhere-defined adjointable operator preserves Hahn-summable families. It also admits a bound*

$$\|Ax\| \leq M\|x\| \quad (x \in \mathcal{H}_\Gamma) \quad (6.4)$$

for some $M \in F_{\geq 0}$. If $A \neq 0$ and $\gamma_0 = \min \text{supp } A$, any ordinary real $c > \|A_{\gamma_0}\|_H$ gives the valid bound $M = ct^{\gamma_0}$.

Proof. For a Hahn-summable family, the finiteness in lemma 2.1(i), applied with $\text{supp } A$, allows the two sums to be interchanged, and their union of supports remains well ordered. For the bound, write the leading term of x as $t^\delta u$. At exponent $2(\gamma_0 + \delta)$, the coefficient of $M^2\|x\|^2 - \|Ax\|^2$ is $c^2\|u\|_H^2 - \|A_{\gamma_0}u\|_H^2 > 0$. This remains true when $A_{\gamma_0}u = 0$, since then the leading output moves to a higher exponent. Thus the difference is positive for nonzero x . $\square$

Remark 6.3. The Baire argument is a uniform-support theorem, not a uniform bound on $\|A_\gamma\|_H$. The proof also explains why unrestricted ordinary algebraic endomorphisms, used later for choosing range complements, are not automatically adjointable.

Endomorphisms preserving strong sums, without adjointability, are classified as $\text{End}_\mathbb{C}(H)((t^\Gamma))$ in `duals:thm:strong of docs/surcomplex/three-duals-of-hahn-vector-spaces/`.

7 An injective leading term preserves the whole range defect

7.1 A theorem independent of Hilbert structure

The next result applies to any complex vector space V . Write $V_\Gamma = V((t^\Gamma))$. An ordinary linear map on V extends coefficientwise, even if V also carries a topology and the map is discontinuous in that topology.

Theorem 7.1 (Positive-order defect rigidity). *Let $S : V \rightarrow V$ be injective. Let $E \in \text{End}_\mathbb{C}(V)((t^\Gamma))$ have strictly positive order, or be zero, and put $A = S + E$. Choose an algebraic complement*

$$V = \text{Ran } S \oplus W. \quad (7.1)$$

Then

$$V_\Gamma = A(V_\Gamma) \oplus W((t^\Gamma)). \quad (7.2)$$

In particular, A is injective, $v(Ax) = v(x)$ for $x \neq 0$, and $\text{coker } A \cong W((t^\Gamma))$ as K -vector spaces. The isomorphism depends on the chosen complement, but the nonvanishing of the defect does not.

Proof. At the least exponent δ of a nonzero x , the coefficient of Ax is $Sx_\delta \neq 0$. All contributions from E have greater exponent. This proves injectivity and the valuation assertion.

Define $U : V \rightarrow V$ by $U(Sv + w) = v$ for $v \in V$, $w \in W$. Then $US = I$ and $\ker U = W$. Extend U coefficientwise and use lemma 2.3 to set

$$R = (I + UE)^{-1}, \quad L = RU, \quad \Pi = I - AL.$$

The coefficients of U need not be bounded in any additional topology; all these formulas are legitimate in the ordinary algebraic endomorphism Hahn algebra. The identities

$$LA = R(US + UE) = R(I + UE) = I$$

and

$$U\Pi = U - UARU = U - (I + UE)RU = 0$$

show that Π kills the range of A and takes values in $\ker U = W((t^\Gamma))$. On that latter space L vanishes, so Π is the identity. Finally, every x decomposes as $ALx + \Pi x$; the intersection of the two summands is zero because $LA = I$ and $L|_{W((t^\Gamma))} = 0$. $\square$

Corollary 7.2 (No positive-order repair). *An injective nonsurjective ordinary operator cannot become surjective on V_Γ by adding any globally supported positive-order Hahn perturbation. This includes perturbations that do not commute with it and perturbations with infinitely many positive exponents.*

Corollary 7.3 (Explicit independent obstruction classes). *Suppose $(f_j)_{j \in J} \subset V$ have linearly independent classes in $V/\text{Ran } S$ over $\mathbb{C}$. Their constant Hahn extensions have linearly independent classes in $V_\Gamma/(S + E)V_\Gamma$ over K . Consequently*

$$\dim_K \text{coker}(S + E) \geq \dim_\mathbb{C}(V/\text{Ran } S).$$

Proof. In a nonzero finite sum $f = \sum_j c_j f_j$ with $c_j \in K$, take the least exponent among the nonzero c_j . The coefficient at that exponent is a nonzero finite complex combination of the f_j , and is not in $\text{Ran } S$. But the leading coefficient of $(S + E)x$ is always in $\text{Ran } S$. $\square$

Warning 7.4. Theorem 7.1 gives the actual vector-space quotient, not only a first-order approximation to it. It does *not* assert $\dim_K W((t^\Gamma)) = \dim_\mathbb{C} W$ for arbitrary infinite-dimensional W . The safe dimension statement is the lower bound just proved in corollary 7.3.

7.2 Why the result is more informative than a determinant test

There is no finite determinant for an arbitrary infinite-dimensional S . An entrywise infinitesimal perturbation may even make every finite section invertible, while the leading ordinary coefficient of a proposed inverse vector is excluded from H . The exact splitting (7.2) shows that this is not an accidental choice of data: the full algebraic cokernel persists. On the other hand, if S is an ordinary bijection, $S + E = S(I + S^{-1}E)$ is invertible by the same support-certified Neumann formula. Thus the injective case has a sharp dichotomy, not merely a negative example.

8 The exact spectrum of an ordinary normal operator

8.1 Two spectral definitions and one ordinary reduction

For a K -linear endomorphism A of $\mathcal{H}_\Gamma$, define

$$\sigma_\Gamma^{\text{alg}}(A) = \{\lambda \in K : A - \lambda I \text{ is not bijective on } \mathcal{H}_\Gamma\}.$$

For $A \in \mathcal{A}_\Gamma(H)$, define $\sigma_\Gamma^{\text{adj}}(A)$ by failure of invertibility in the unital algebra $\mathcal{A}_\Gamma(H)$. These definitions need not be identified in advance. In the theorem below, explicit inverses and explicit range obstructions will prove their equality. Neither is the algebra-relative spectrum $\sigma_{\mathcal{A}_I}$ of Part II, which is taken in a different algebra acting on a different vector space. For a constant A , $\sigma_\Gamma^{\text{adj}}(A)$ is the case $\mathfrak{A} = \mathcal{B}(H)$ of Part III's spectrum $\Sigma_\Gamma^{\mathfrak{A}}$ (proposition 3.1).

An ordinary bounded operator T is regarded as a constant member of $\mathcal{A}_\Gamma(H)$. We often suppress the extension subscript on T .

Lemma 8.1 (Normal isolated-point decomposition). *Let $T \in \mathcal{B}(H)$ be normal and $a \in \sigma_\mathbb{C}(T)$. Put $N = \ker(T - aI)$, $M = N^\perp$, and $S = (T - aI)|_M$. Then M, N reduce T and S is injective. If a is isolated in $\sigma_\mathbb{C}(T)$, then $N \neq 0$ and S is boundedly invertible on M . If a is an accumulation point, then S is not surjective.*

Proof. These facts follow from the ordinary normal spectral theorem [8]; here is the precise use of it. The spectral projection of $\{a\}$ is the orthogonal projection onto N . For isolated a , the function $(z - a)^{-1}$ on the rest of the compact spectrum is bounded, so spectral functional calculus gives an inverse on M . The projection cannot be zero, since otherwise a would not belong to the spectrum. For accumulation a , if S were onto, its injectivity and the bounded inverse theorem would give a bounded inverse. The spectrum of T would then be contained in $\{a\}$ together with a compact set separated from a , contradicting accumulation. This also covers $N = 0$. $\square$

8.2 Classification and all inverse formulas

Theorem 8.2 (Accumulation-monad spectrum). *For every ordinary bounded normal $T \in \mathcal{B}(H)$,*

$$\sigma_\Gamma^{\text{alg}}(T) = \sigma_\Gamma^{\text{adj}}(T) = \sigma_\mathbb{C}(T) \cup (\sigma_\mathbb{C}(T)' + \mathfrak{m}_K). \quad (8.1)$$

No scalar of negative valuation is in this spectrum. A nonzero infinitesimal displacement of an isolated ordinary spectral point is in the resolvent, whereas every infinitesimal displacement of an accumulation point remains in the spectrum.

Proof. We partition all scalars $\lambda \in K$ into exhaustive cases.

Infinite scalar. If $v(\lambda) < 0$, then

$$(T - \lambda I)^{-1} = - \sum_{n \geq 0} \lambda^{-n-1} T^n. \quad (8.2)$$

This is a support-certified Neumann inverse: the scalar λ^{-1} has positive valuation. It belongs to $\mathcal{A}_\Gamma(H)$, irrespective of the ordinary growth of $\|T^n\|$.

Finite scalar with nonspectral residue. Write $\lambda = a + \varepsilon$, where $a \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_K$. If $a \notin \sigma_{\mathbb{C}}(T)$, let $B = (T - aI)^{-1} \in \mathcal{B}(H)$. Then

$$(T - aI - \varepsilon I)^{-1} = \sum_{n \geq 0} \varepsilon^n B^{n+1}. \quad (8.3)$$

The formula includes $\varepsilon = 0$. For nonzero ε , use lemma 2.3 on εB ; coefficients need not satisfy any ordinary norm-convergence estimate.

Undisplaced ordinary spectral point. If $\varepsilon = 0$ and $a \in \sigma_{\mathbb{C}}(T)$, then $T - aI$ is not an ordinary bijection, by the bounded inverse theorem. A nonzero ordinary kernel vector remains a Hahn kernel vector. If there is no kernel but surjectivity fails, choose a constant f outside its ordinary range. The coefficient at zero in any putative solution would have to solve $(T - aI)y_0 = f$, which is impossible. Thus the Hahn operator is not bijective.

Displaced isolated point. Suppose a is isolated and $\varepsilon \neq 0$. Use lemma 8.1; let P, Q project onto N, M . The inverse is

$$(T - aI - \varepsilon I)^{-1} = -\varepsilon^{-1}P + \sum_{n \geq 0} \varepsilon^n S^{-n-1}Q. \quad (8.4)$$

The first term is a single scalar Hahn series times a bounded projection; the second is a positive-order Neumann series on M . Both have well-ordered operator support and lie in $\mathcal{A}_\Gamma(H)$. The finite orthogonal splitting verifies the inverse on both summands. If $M = 0$, the second summand is simply zero.

Displaced accumulation point. Suppose $a \in \sigma_{\mathbb{C}}(T)'$ and $\varepsilon \neq 0$. On M in lemma 8.1, the ordinary S is injective and not onto. The perturbation $-\varepsilon I$ has strictly positive order. By theorem 7.1, $S - \varepsilon I$ is not onto $M((t^\Gamma))$. The constant decomposition $H = N \oplus M$ extends to a decomposition of Hahn spaces, and the full operator is block diagonal. It is therefore not onto either. Explicitly, a constant $f \in M \setminus \text{Ran } S$ cannot be its M -component output.

These cases prove exactly the stated set formula. In every nonspectral case we constructed an inverse in $\mathcal{A}_\Gamma(H)$; in every spectral case we proved failure of algebraic bijectivity. This proves equality of the two spectral definitions as well. $\square$

8.3 Point spectrum is rigid, despite spectral thickening

Theorem 8.3 (No new normal eigenvalues). *For ordinary bounded normal T ,*

$$\sigma_{p,\Gamma}(T) = \sigma_{p,\mathbb{C}}(T), \quad \ker(T - aI) = \ker_H(T - aI)((t^\Gamma)) \quad (a \in \mathbb{C}).$$

In particular, the new nonreal spectral points of a self-adjoint operator are not eigenvalues.

Proof. Infinite scalars are in the resolvent by theorem 8.2. Suppose $(T - aI - \varepsilon I)x = 0$ with $x \neq 0$ and $\varepsilon \in \mathfrak{m}_K$. At its leading exponent δ , one obtains $(T - aI)x_\delta = 0$. Thus a is an ordinary eigenvalue and $x_\delta \in N = \ker_H(T - aI)$. If $\varepsilon \neq 0$, apply the ordinary orthogonal projection P onto N . Normality implies $P(T - aI) = 0$, so the full equation gives $-\varepsilon Px = 0$. Hence $Px = 0$, contradicting $Px_\delta = x_\delta \neq 0$. Therefore $\varepsilon = 0$. For a constant a , the kernel equation is coefficientwise, yielding the claimed eigenspace identity. $\square$

Example 8.4 (Normality is indispensable). Let B be the backward shift on $\ell^2(\mathbb{N}_0)$: $Be_0 = 0$, $Be_n = e_{n-1}$ for $n \geq 1$. For $s = t^\eta$ with $\eta > 0$,

$$x = \sum_{n \geq 0} s^n e_n \in \mathcal{H}_\Gamma, \quad Bx = sx.$$

The support $\{n\eta : n \geq 0\}$ is well ordered. This produces a new nonconstant eigenvalue for the nonnormal operator B , in direct contrast to theorem 8.3. No ordinary Hilbert convergence of the vector series is asserted or needed.

Remark 8.5 (The nonnormal constant case). Part III computes $\sigma_\Gamma^{\text{adj}}(T)$ for *every* ordinary bounded T , normal or not, and for constant elements of any unital algebra: the accumulation set is replaced by the finite-index Drazin spectrum (theorem 32.1 and corollary 33.1). Its proof is a second route to the σ^{adj} half of (8.1) (remark 33.2). It does not replace theorem 8.2, which also determines $\sigma_\Gamma^{\text{alg}}$, nor theorem 8.3; and it concerns the spectral formula, not the new eigenvalue of the example above.

9 Consequences that separate finite and infinite spectral geometry

9.1 Self-adjoint spectra are real exactly in the finite-spectrum case

Corollary 9.1 (A sharp spectral-reality criterion). *Let T be ordinary bounded self-adjoint. The following are equivalent:*

- (i) $\sigma_\Gamma(T) \subseteq \mathbb{R}$;
- (ii) $\sigma_\mathbb{C}(T)$ is finite;
- (iii) T has a finite ordinary orthogonal spectral resolution $T = \sum_{j=1}^r a_j P_j$ with $a_j \in \mathbb{R}$.

There is no finite-rank hypothesis on the projections P_j .

Proof. The usual self-adjoint spectrum is a nonempty compact subset of $\mathbb{R}$. A finite set has no accumulation points, so theorem 8.2 proves (ii) $\Rightarrow$ (i). An infinite compact subset has an accumulation point a . For any $s = t^\eta$, $\eta > 0$, the scalar $a + is$ belongs to $\sigma_\Gamma(T)$ and does not belong to $\mathbb{R}$, proving the converse. The equivalence with (iii) is the ordinary spectral theorem applied to a finite spectral set. $\square$

The criterion is finite *spectrum*, not finite dimension. For example, an infinite-rank orthogonal projection with infinite-dimensional kernel has spectrum exactly $\{0, 1\}$ even after Hahn extension. Infinite multiplicity at an isolated point does not create a monad. The accumulation set in (8.1) therefore cannot be replaced by an essential spectrum that includes such points.

9.2 Ordinary compact operators and a continuous-spectrum example

Corollary 9.2 (Ordinary compact normal extension). *If T is an ordinary compact normal operator of infinite rank, then*

$$\sigma_\Gamma(T) = \mathfrak{m}_K \cup \{\text{nonzero ordinary eigenvalues of } T\}.$$

If its ordinary spectrum is finite, there is no thickening.

Proof. The ordinary compact normal spectral theorem says that nonzero spectral points are isolated eigenvalues of finite multiplicity and zero is the only possible accumulation point. Infinite rank forces zero to be an accumulation point. Apply theorem 8.2. $\square$

For an injective compact self-adjoint T , Part IV adds a self-adjoint Hahn perturbation of positive order that need not commute with T : the monad $\mathfrak{m}_K$ stays in the spectrum, and each nonzero eigenvalue is replaced by a finite cluster of real Hahn eigenvalues (theorem 42.1, under a divisibility hypothesis).

Here “compact” describes the original operator on the ordinary Hilbert space. It makes no claim of compactness or compactoidness in the Hahn valuation topology. The sequence (De_n) for $De_n = n^{-1}e_n$, for instance, satisfies $v(De_n - De_m) = 0$ for $n \neq m$, despite $\|De_n\|_H \rightarrow 0$ in the ordinary norm. Thus the valuation ball $\{x : v(x) > 0\}$ separates these outputs at every rank; in the rank-one metric below their pairwise distance is one.

For another concrete example, let T be multiplication by the coordinate on $L^2([0, 1], \mathbb{C})$. Its ordinary spectrum is $[0, 1]$, it has no eigenvalues, and every point of its spectrum is an accumulation point. Thus

$$\sigma_\Gamma(T) = [0, 1] + \mathfrak{m}_K, \quad \sigma_{p,\Gamma}(T) = \emptyset.$$

The usual facts here can be seen directly: away from $[0, 1]$ the reciprocal multiplier is bounded; near any point of $[0, 1]$, vectors supported in arbitrarily small intervals prevent a bounded inverse; a singleton has measure zero and cannot support a nonzero eigenvector. This gives full infinitesimal fibers above a continuous real interval, without adding a single eigenvector.

9.3 Enlarging the scalar workspace does not repair the defect

Corollary 9.3 (Persistence under ordered-group extension). *Let $\Gamma \subseteq \Delta$ be nonzero ordered abelian groups and use the natural coefficient-preserving embedding $K_\Gamma \subseteq K_\Delta$. For constant normal T ,*

$$\sigma_\Delta(T) \cap K_\Gamma = \sigma_\Gamma(T).$$

Moreover, the nonsurjectivity in corollary 7.2 persists after such an extension for the same S and E .

Proof. An element of K_Γ has the same leading valuation, residue and infinitesimal status in the larger group. Apply the exact spectral formula. For the second assertion use the same complement W in theorem 7.1, now forming $W((t^\Delta))$. $\square$

This is a genuine surcomplex-workspace statement. Adding further surreal scales does not turn a prohibited leading ordinary Hilbert coefficient into an allowed one. It does not assert that an unspecified larger class-sized notion of vector could never behave differently. Proposition 25.2 is the corresponding naturality statement for Part II; it is about a different construction and is proved separately.

10 Which constant data have an infinitesimal resolvent?

10.1 The inverse-smooth domain

Even when $S - sI$ is not onto, some constant right-hand sides have a unique solution. The exact criterion is stronger than belonging merely to $\text{Ran } S$.

Theorem 10.1 (All inverse moments are necessary and sufficient). *Let $S : V \rightarrow V$ be injective and $s = t^\eta$ with $\eta > 0$. For constant $f \in V$, the equation*

$$(S - sI)y = f, \quad y \in V((t^\Gamma)), \tag{10.1}$$

has a solution if and only if

$$f \in \bigcap_{k \geq 1} \text{Ran } S^k. \tag{10.2}$$

In that case the solution is unique and is

$$y = \sum_{n \geq 0} s^n S^{-n-1} f. \quad (10.3)$$

The inverse powers denote the unique preimages, not an everywhere-defined inverse operator on V .

Proof. Sufficiency follows by coefficientwise telescoping of (10.3); its support is contained in the well-ordered set $\{n\eta : n \geq 0\}$. Injectivity follows from theorem 7.1.

For necessity, partition the support of a proposed solution according to cosets of $\mathbb{Z}\eta$ in Γ . Restricting a Hahn series to a coset preserves well-ordering, and both S and multiplication by s preserve that coset. On a coset other than $\mathbb{Z}\eta$, the restricted equation is homogeneous. Its least nonzero coefficient would be killed by S , which is impossible. Thus the support lies in $\mathbb{Z}\eta$.

A well-ordered subset of $\mathbb{Z}\eta$ is bounded below in its integer index. For $f \neq 0$, the valuation-preservation assertion of theorem 7.1 forces the least index to be zero. Write $y = \sum_{n \geq 0} s^n y_n$, allowing zero coefficients. The equation gives

$$Sy_0 = f, \quad Sy_n = y_{n-1} \quad (n \geq 1).$$

Consequently $S^{n+1}y_n = f$ for every n , proving (10.2) and the formula. The zero-data case is immediate. $\square$

Example 10.2 (The diagonal inverse-smooth space). For $De_n = n^{-1}e_n$ on $H = \ell^2(\mathbb{N}_{\geq 1})$, the admissible constant data are exactly

$$H_\infty = \left\{ f = (f_n) : \sum_{n \geq 1} n^{2k} |f_n|^2 < \infty \text{ for every integer } k \geq 1 \right\}. \quad (10.4)$$

The formula follows because $D^{-k}f = (n^k f_n)$. Finitely supported vectors belong to H_∞ , so this subspace is dense in the *ordinary* Hilbert topology. Yet $f_n = n^{-2}$ does not belong to it, although $f \in \text{Ran } D$. Thus even some data with an ordinary first preimage lack an infinitesimal resolvent.

The same statement for $D - isI$ follows by replacing the recurrence with $Dy_n = iy_{n-1}$. Its solution, when it exists, is $\sum_{n \geq 0} i^n s^n D^{-n-1} f$. No uniform bound on the ordinary norms of these coefficients is required, only that each one actually belongs to H .

10.2 Why individual scalar inverses are insufficient

Every diagonal scalar $n^{-1} - s$ is invertible in K and has expansion

$$(n^{-1} - s)^{-1} = \sum_{k \geq 0} n^{k+1} s^k.$$

But applying these inverses coordinatewise to f requires the entire coefficient vector $(n^{k+1} f_n)_n$ to lie in ℓ^2 at each k . The Hahn vector space imposes this condition before any comparison of higher scales. There is therefore no contradiction between invertibility of every diagonal entry and failure of surjectivity of the infinite operator. A putative coordinatewise inverse has left the chosen vector space, rather than found a missing scalar inverse. In Part II, where the vector space and the operator algebra are different, the corresponding coordinatewise inverse is admissible; section 3.2 compares the two situations directly.

11 Coercive operators with continuum-dimensional defects

11.1 An explicit large ordinary cokernel

Continue with $H = \ell^2(\mathbb{N}_{\geq 1})$ and $De_n = n^{-1}e_n$. The operator D is bounded, positive, self-adjoint and injective. It is not onto because $(n^{-1})_n \in \ell^2$ has no preimage in ℓ^2 .

Lemma 11.1 (Continuum many independent defect vectors). *For $r \in (1/2, 3/2]$, let $f^{(r)} = (n^{-r})_{n \geq 1} \in H$. Their classes in $H/\text{Ran } D$ are linearly independent over $\mathbb{C}$.*

Proof. Take distinct $r_1 < \dots < r_k$ and complex coefficients with $c_1 \neq 0$. For $g = \sum_{j=1}^k c_j f^{(r_j)}$, one has

$$ng_n = c_1 n^{1-r_1} (1 + o(1)).$$

Its squared magnitude is bounded below by a positive real multiple of n^{2-2r_1} for sufficiently large n . Since $r_1 \leq 3/2$, the exponent is at least -1 , and the series of those squared magnitudes diverges. Thus $(ng_n) \notin \ell^2$, which is exactly $g \notin \text{Ran } D$. There are continuum many choices of r . $\square$

Theorem 11.2 (Every positive-order perturbation retains a large defect). *For every $E \in \mathcal{B}(H)((t^\Gamma))$ of positive order, including zero,*

$$\dim_K \text{coker}(D + E) \geq 2^{\aleph_0}.$$

The same lower bound holds with leading operator D^2 .

Proof. Apply corollary 7.3 and lemma 11.1. Since $\text{Ran } D^2 \subseteq \text{Ran } D$, the same vectors remain independent modulo $\text{Ran } D^2$. $\square$

11.2 A nonreal spectral point with a strong lower bound

For $s = t^\eta$, $\eta > 0$, set $A = D - isI$. Since $D = D^*$,

$$A^*A = D^2 + s^2I, \quad \|Ax\|^2 = \|Dx\|^2 + s^2\|x\|^2 \geq s^2\|x\|^2. \quad (11.1)$$

Nevertheless, A is not onto by theorem 7.1. Thus the usual lower-bound argument for excluding nonreal self-adjoint spectrum stops precisely at surjectivity, not at injectivity.

The operator even has an attained upper norm bound:

$$\|Ax\| \leq \sqrt{1 + s^2} \|x\|, \quad \|Ae_1\| = \sqrt{1 + s^2}. \quad (11.2)$$

Indeed D is an ordinary contraction and its extension remains a contraction; for this diagonal operator the inequality also follows by writing $I - D^2 = R^*R$ with ordinary bounded diagonal $Re_n = \sqrt{1 - n^{-2}}e_n$. Equations (11.1)–(11.2) show that neither lack of a bound nor lack of an attained norm explains the failure of inversion.

11.3 Positive coercivity does not imply surjectivity

Theorem 11.3 (A coercive noninvertible self-adjoint operator). *Let*

$$C = D^2 + s^2I. \quad (11.3)$$

Then C is everywhere-defined, adjointable and self-adjoint, with

$$\langle Cx, x \rangle \geq s^2 \langle x, x \rangle, \quad (11.4)$$

$$\|Cx\| \leq (1 + s^2) \|x\|. \quad (11.5)$$

The upper bound is attained at e_1 . The operator is injective but not surjective, and its cokernel has dimension at least $2^{\aleph_0}$ over K . Specifically, the constant vector $f = (n^{-2})_{n \geq 1}$ is not in its range.

Proof. Self-adjointness and (11.4) follow from $\langle D^2x, x \rangle = \langle Dx, Dx \rangle \geq 0$. The contraction property of D^2 , the triangle inequality and $Ce_1 = (1 + s^2)e_1$ give (11.5) and its attainment. The leading coefficient operator D^2 is injective, so theorem 7.1 gives injectivity, and theorem 11.2 gives the cokernel bound.

For a direct witness, if $Cy = f$, valuation preservation forces $v(y) = 0$. The coefficient at zero must solve $D^2y_0 = f$, that is, $(y_0)_n = 1$ for every n . This is not an ℓ^2 vector. Thus no such y exists. $\square$

For clarity, coercivity is a bound by a strictly positive element s^2 of the ordered scalar field. It is not a bound by a positive *ordinary real* independent of the infinitesimal scale. If instead $C \geq cI$ with ordinary $c > 0$ and an ordinary bounded positive leading operator with this bound, the classical inverse and the positive-order Neumann construction would apply. The infinitesimal scale is exactly where the distinction becomes mathematically significant.

Corollary 11.4 (Failure of a specified Lax–Milgram analogue). *On this Hahn inner-product space, a bounded coercive sesquilinear form need not solve all variational problems, even when the right-hand side is represented by an actual vector. In particular,*

$$b(u, v) = \langle Cu, v \rangle, \quad \ell(v) = \langle f, v \rangle, \quad f = (n^{-2})_n,$$

defines a bounded coercive form and a bounded conjugate-linear functional, but there is no u satisfying $b(u, v) = \ell(v)$ for all $v \in \mathcal{H}_\Gamma$.

Proof. Cauchy–Schwarz and (11.5) bound the form; (11.4) is coercivity, and $|\ell(v)| \leq \|f\| \|v\|$. If the identity held for all v , nondegeneracy of the inner product would give $Cu = f$, contradicting theorem 11.3. $\square$

This does not contradict the ordinary complex Lax–Milgram theorem: the scalar field, the topology and the duality structure have changed. It is an explicit obstruction to transferring that theorem using only its familiar-looking algebraic inequalities.

12 Completeness does not restore orthogonal projection

12.1 The rank-one topology and its completeness

For this subsection only, assume Γ is a nonzero ordered subgroup of $\mathbb{R}$. For $x \neq 0$ define

$$\rho(x) = e^{-v(x)}, \quad \rho(0) = 0, \quad d(x, y) = \rho(x - y).$$

The scalar absolute value here is $|a|_v = e^{-v(a)}$, not the F -valued modulus. The metric is ultrametric. Its topology agrees with the topology of the field-valued norm balls on $\mathcal{H}_\Gamma$: a sufficiently higher valuation makes a vector smaller than any prescribed positive radius, and choosing a still higher monomial radius gives the converse inclusion of neighborhood bases.

It is not the ordinary Hilbert topology on the constant copy of H . Every nonzero constant vector has ρ -size one. Nor is it the fine subspace topology from the full surcomplex class, where the set-sized scalar workspace is discrete [1].

Proposition 12.1 (Rank-one valuation completeness). *For any complex vector space V , the metric space $V((t^\Gamma))$ with metric d is complete. In particular, $\mathcal{H}_\Gamma$ is complete in this topology.*

Proof. Let (x_n) be Cauchy. For each real cutoff R , all sufficiently late x_n have exactly the same coefficients at exponents $\gamma \leq R$, since eventually $v(x_n - x_m) > R$. Define x_γ to be the eventually stable coefficient.

The support of this coefficient family is well ordered. Otherwise it would contain a strictly decreasing sequence $\gamma_1 > \gamma_2 > \dots$. Choose a cutoff at least γ_1 . Below that cutoff the entire stable family agrees with one sufficiently late Hahn vector x_N , whose well-ordered support cannot contain that sequence. This is a contradiction. Thus $x = \sum x_\gamma t^\gamma$ belongs to $V((t^\Gamma))$. By the same stabilization statement, $v(x_n - x) > R$ eventually for every real R , proving convergence. Notice that no coefficient-space topology was used: stabilization is exact, not approximate. $\square$

12.2 A proper closed subspace with zero orthogonal complement

Return to an arbitrary nonzero ordered abelian group Γ . The valuation topology has neighborhoods $\{x : v(x) > \gamma\}$, $\gamma \in \Gamma$. These define the same topology as field-valued norm balls by the leading norm formula. No real-valued metric or completeness hypothesis is needed for the next two results.

Theorem 12.2 (Closed-range orthogonality failure). *For the coercive $C = D^2 + s^2I$, its range $M = C\mathcal{H}_\Gamma$ is closed and proper in the valuation topology at every rank, but*

$$M^\perp = \{0\}, \quad M^{\perp\perp} = \mathcal{H}_\Gamma \neq M. \quad (12.1)$$

In particular, M is not the range of a self-adjoint idempotent.

Proof. Use the splitting in theorem 7.1 with $S = D^2$ and $E = s^2I$. Its projection $\Pi = I - C(I + UE)^{-1}U$ has kernel exactly M : it kills M , and $\Pi x = 0$ gives $x = C(I + UE)^{-1}Ux$. Every coefficient of Π has nonnegative exponent, since U, S are constant and E has positive order. Therefore $v(\Pi x) \geq v(x)$, so Π is valuation-continuous. The valuation topology is Hausdorff; hence $M = \Pi^{-1}(\{0\})$ is closed. It is proper by theorem 11.3.

In rank one there is also the original metric proof: $v(Cx) = v(x)$ makes C an isometry, and its range is complete and closed by proposition 12.1. The projection proof explains why rank one is unnecessary for closedness.

If $z \in M^\perp$, then $\langle Cx, z \rangle = 0$ for every x . Since $C = C^*$, $\langle x, Cz \rangle = 0$ for every x , so $Cz = 0$. Injectivity gives $z = 0$. This proves (12.1). A self-adjoint idempotent P has $\mathcal{H}_\Gamma = \text{Ran } P \oplus \ker P$ and $\ker P = (\text{Ran } P)^\perp$, by the finite algebraic projection identities. If its range were M , the complement would be zero and the range all of $\mathcal{H}_\Gamma$, a contradiction. $\square$

The displayed range is not orthogonally closed, although it is closed in the valuation topology. Thus the usual Hilbert identification of these two closure notions fails. Known non-Archimedean failures of orthomodularity also warn against unconditional projection theorems [10, 11]. The theorem above gives a particularly rigid witness: it is the closed range of a bounded, coercive, self-adjoint injection and retains a continuum-sized algebraic deficit.

12.3 A continuous functional outside inner-product duality

Proposition 12.3 (Explicit failure of valuation Riesz representation). *There is a nonzero K -linear, valuation-continuous functional $F : \mathcal{H}_\Gamma \rightarrow K$ that vanishes on $M = C\mathcal{H}_\Gamma$ but is not of the form $F(x) = \langle x, z \rangle$ for any $z \in \mathcal{H}_\Gamma$.*

Proof. Choose an algebraic complement W of $\text{Ran } D^2$ in H , and take the projection Π in theorem 7.1 for $S = D^2$, $E = s^2I$. Its coefficients have nonnegative exponents: U and S are constant, E has positive order, and $(I + UE)^{-1}$ has nonnegative order. Consequently $v(\Pi x) \geq v(x)$.

Choose a nonzero complex linear functional $\psi : W \rightarrow \mathbb{C}$, and extend it coefficientwise to $\psi_\Gamma : W((t^\Gamma)) \rightarrow K$. Set $F = \psi_\Gamma \Pi$. Coefficient extraction gives $v(Fx) \geq v(x)$, so F is valuation-continuous. It is nonzero because Π is the identity on $W((t^\Gamma))$, and it kills M . If $F(x) = \langle x, z \rangle$, then vanishing on M implies $z \in M^\perp = 0$, which would force $F = 0$. $\square$

This functional is strong in the sense of the report `docs/surcomplex/three-duals-of-hahn-vector-spaces/` (`duals:thm:strong`); its leading coefficient is discontinuous, so it lies in the first gap of `duals:thm:riesz` (`duals:rem:rhs-riesz`).

The algebraic maps U and ψ need not be continuous for the ordinary Hilbert norm. That does not prevent the displayed Hahn maps from being valuation-continuous. Confusing those topologies would incorrectly apply an ordinary closed graph or Riesz argument. Conversely, this proposition does not contradict theorem 6.1: the functional just constructed has no adjoint represented by a Hahn vector.

13 When does an operator have a least field-valued norm?

13.1 Bounds versus the existence of their least element

For $A \in \mathcal{A}_\Gamma(H)$, let

$$\mathcal{U}(A) = \{M \in F_{\geq 0} : \|Ax\| \leq M\|x\| \text{ for all } x \in \mathcal{H}_\Gamma\}.$$

Corollary 6.2 makes this set nonempty. By a field-valued *operator norm* we mean a least element of this set, not an arbitrary chosen bound and not the valuation norm of the operator.

Lemma 13.1 (Ordinary contraction bounds survive). *If $T \in \mathcal{B}(H)$ and $n = \|T\|_H$, then $\|Tx\| \leq n\|x\|$ for every $x \in \mathcal{H}_\Gamma$. For nonzero x with leading coefficient u , the finite scalar $\|Tx\|/\|x\|$ has standard part*

$$\text{st}\left(\frac{\|Tx\|}{\|x\|}\right) = \frac{\|Tu\|_H}{\|u\|_H}.$$

The right side is zero if $Tu = 0$.

Proof. The ordinary positive operator $n^2I - T^*T$ has a bounded positive square root R by ordinary Hilbert spectral theory. The coefficientwise identity $R^*R = n^2I - T^*T$ implies $n^2\|x\|^2 - \|Tx\|^2 = \|Rx\|^2 \geq 0$. For the standard part, use the leading norm formula in proposition 5.1. If $Tu = 0$, the output has strictly larger valuation or is zero, so the quotient has residue zero. $\square$

Theorem 13.2 (Norm-attainment criterion). *For an ordinary bounded operator T on nonzero H , the set $\mathcal{U}(T)$ has a least element if and only if T attains its ordinary operator norm on a unit vector. When it does, the least element is $\|T\|_H$ and is attained by a constant unit vector in $\mathcal{H}_\Gamma$.*

Proof. Put $n = \|T\|_H$. If T attains n , the constant maximizing vector forces every upper bound to be at least n , while lemma 13.1 proves that n is a bound. This includes $T = 0$, which attains its norm on every unit vector.

Now suppose T does not attain its norm; then $n > 0$. For each nonzero $x \in \mathcal{H}_\Gamma$, lemma 13.1 gives

$$\text{st}\left(\frac{\|Tx\|}{\|x\|}\right) < n.$$

Therefore every positive finite M with $\text{st}(M) = n$ is an upper bound, including $n - \varepsilon$ for every positive infinitesimal ε . Every finite M with residue greater than n and every positive infinite M is a bound as well.

No finite nonnegative M with residue less than n can be a bound: choose an ordinary unit vector u with $\|Tu\|_H > \text{st}(M)$ using the definition of the ordinary supremum. Its constant extension violates that bound. We have thus described all upper bounds. A bound with residue greater than n , or an infinite bound, can be decreased to n . A bound with residue n can be decreased by any fixed positive infinitesimal and remains positive with the same residue. There is no least upper bound. $\square$

Example 13.3 (A bounded self-adjoint operator without a field-valued norm). Let $Te_n = (1 - 1/(n+1))e_n$ on $\ell^2(\mathbb{N}_{\geq 1})$. Its ordinary norm is one but is not attained. Indeed, for every nonzero u ,

$$\|u\|_H^2 - \|Tu\|_H^2 = \sum_{n \geq 1} \left(1 - \left(1 - \frac{1}{n+1}\right)^2\right) |u_n|^2 > 0,$$

where this is an ordinary convergent real sum. The diagonal values approach one, proving the supremum is one. Hence its Hahn extension has many field-valued bounds, including $1 - s$ and $1 - 2s$, but no least one.

13.2 The two mechanisms are independent

The operator $D^2 + s^2 I$ in theorem 11.3 has an attained least upper norm bound $1 + s^2$ but is not invertible. The operator in example 13.3 fails norm attainment for a different reason: every individual Hahn vector has a strictly submaximal *ordinary leading ratio*, so infinitesimally smaller bounds cannot be distinguished by any one vector. Neither phenomenon should be used as an explanation for the other.

14 Why finite invertibility is misleading

Let $D_N = \text{diag}(1, 1/2, \dots, 1/N)$ and $C_N = D_N^2 + s^2 I$. Every C_N is invertible over K . For the truncated right-hand side $f_N = (n^{-2})_{1 \leq n \leq N}$,

$$(C_N^{-1} f_N)_n = \frac{1}{1 + n^2 s^2} = \sum_{k \geq 0} (-1)^k n^{2k} s^{2k}. \quad (14.1)$$

The constant coefficient of this finite solution is the N -vector of ones, whose ordinary Hilbert norm is $\sqrt{N}$. No single ordinary ℓ^2 vector can serve as the stable constant coefficient as $N \rightarrow \infty$. This directly identifies the obstruction missed by every finite matrix determinant.

For $A_N = D_N - isI$, all diagonal resolvents likewise exist, with

$$(n^{-1} - is)^{-1} = \sum_{k \geq 0} i^k n^{k+1} s^k.$$

The leading inverse coefficient has ordinary norm N . Infinite extension would require the unbounded ordinary operator $\text{diag}(n)$ as a bounded coefficient, contradicting theorem 6.1. More strongly, particular constant data fail the vector-space condition of theorem 10.1; thus the problem is actual nonsurjectivity, not just exclusion from a preferred inverse algebra.

Remark 14.1 (The same matrix in the other part). The operator $\text{diag}(n)$ excluded here is admitted in Part II, where the coefficient algebra is $\text{RCF}_I(\mathbb{C})$ rather than $\mathcal{B}(H)$, and where the finite-section question has a positive answer of a different kind: theorem 21.4 gives an exact finite spatial certificate for each individual coefficient. The two statements are about different algebras and do not conflict; see section 3.2.

15 Part I: interpretation and remaining boundaries

15.1 What has been settled in this construction

The questions raised by this particular scalar-extension model now have sharp answers. Adjointability is not an opaque class condition: it is exactly global Hahn support with bounded ordinary coefficients. For constant normal operators, infinitesimal resolvent behavior is completely classified by the ordinary spectral accumulation set. Infinitesimal perturbations of injective leading operators cannot remove any ordinary range defect. The leading ordinary coefficient gives a direct and robust witness, even where every finite section is invertible.

These statements distinguish spectral membership from eigenvalue existence, compactness in the original Hilbert topology from compactness in the valuation topology, upper norm bounds from a least such bound, and metric completeness from orthogonal completeness. None of those pairs can be silently identified after extending the scalars.

15.2 What would require a different theorem

The formula (8.1) concerns *constant normal* operators. General Hahn operators with multiple coefficient scales can have more complicated spectra; example 8.4 already shows why the point-spectrum rigidity fails without normality. The perturbation theorem handles injective leading coefficients and proves a large class of obstructions, but is not a classification of all spectra of all Hahn operator series.

Two later parts of this report supply such different theorems for two classes, and leave the general classification open.

- *Constant nonnormal operators.* Part III computes the algebra-relative spectrum $\sigma_\Gamma^{\text{adj}}(T)$ of every constant bounded T , with the Drazin spectrum in place of the accumulation set (theorem 32.1). The Volterra operator has no ordinary accumulation point and still acquires the whole monad (section 35.2), so the accumulation-only formula is false without normality. The bijectivity spectrum $\sigma_\Gamma^{\text{alg}}$ of nonnormal constant operators is determined there only in examples.
- *A nonconstant class.* Part IV treats $T + E$ with T injective compact self-adjoint and E a self-adjoint Hahn series of positive order that need not commute with T , over a divisible Γ : both spectra agree and equal the monad $\mathfrak{m}_K$ together with finite Hermitian clusters (theorem 42.1). The eigenvalues move, unlike those of a constant normal operator (theorem 8.3), but only within their clusters. This local answer does *not* assemble into a global spectral theorem with bounded coefficients: an explicit trace-class example has no Hahn-unitary diagonalizer in $\mathcal{A}_\Gamma(H)$ and no coefficientwise eigenvalue trace sum or determinant product (theorems 43.1, 45.1 and 45.2).

Spectra of general Hahn operator series with several coefficient scales, including nonnormal nonconstant ones, remain unclassified.

One could change the vector construction to allow coefficientwise objects outside H , choose a different sequence-space completion, or impose a stronger orthogonal-completeness axiom. Such changes must be specified and analyzed rather than treated as harmless notational variations. Enlarging only the exponent group is not enough, by corollary 9.3.

15.3 Why this matters beyond a counterexample

In finite dimension, replacing $\mathbb{R}/\mathbb{C}$ by a real-closed field and its complexification preserves much of spectral algebra. In infinite dimension, that replacement cannot by itself preserve the analytic

role of Hilbert space. An infinitesimal imaginary displacement may fail to produce a resolvent, even when it gives a strict lower norm bound. An infinitesimal positive regularizer may leave a continuum of unsolved linear equations. These are precise obstructions to using the model as an automatic replacement for complex Hilbert space, not a rejection of surcomplex analysis in every possible formulation.

The constructive side is equally useful. The automatic adjoint theorem gives a well-defined algebra for exact symbolic manipulation. The inverse formulas identify every ordinary normal resolvent that actually exists. The inverse-smooth criterion identifies exactly which constant data remain solvable at a monomial infinitesimal shift. The defect splitting gives explicit noncanonical projections and an exact algebraic quotient. Together these form a workable theory with both positive structure and sharp limits.

Part II. Row- and column-finite Hahn matrices: exact synthesis and algebra-relative spectra

Standing hypotheses for Part II. Γ is a set-sized ordered abelian group, *not assumed divisible*; I is a nonempty index set. “Spectrum” in this part means $\sigma_{\mathcal{A}_I}$, failure of two-sided invertibility inside the named unital K -algebra $\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$, acting on $\mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma))$. It is not a Banach-algebra spectrum, and it is not the failure-of-bijectivity spectrum of Part I, which lives on a different vector space.

16 The question, the answer, and its scope

Passing from finitely many to infinitely many coordinates changes the problem: products can cease to be defined, formal eigenvectors can escape the chosen vector space, and infinitely many small denominators need not define a bounded operator.

We address the following precise question. Suppose an infinite diagonal matrix has distinct ordinary real entries and is perturbed by genuinely infinitesimal surcomplex coefficients. Is there a natural associative operator algebra, together with a vector space carrying a positive surreal-valued inner product, in which *all* eigenvectors can be constructed coherently, summed against arbitrary vectors, and used to recover exact spectral projections and resolvents?

The answer is affirmative in a coefficientwise locally finite Hahn category. The category is part of the theorem, not an incidental convention. The operators need not be bounded on an ordinary Hilbert space; the vector space is not ordinary ℓ^2 ; and Hahn summation is not shorthand for a sequence of topological limits. Those distinctions allow accumulation of real energy levels while retaining coefficientwise exactness.

The repository was checked for this part at commit `a3124af`; the complete commit identifier appears in the bibliography [3] and in appendix B. In particular, a still-unfinished Lean implementation of a normal-form bridge is not being treated as an open mathematical conjecture. The surcomplex interpretation in section 25 uses the established normal-form theorem [5], not an unproved declaration in a software project.

17 The coefficient algebra and its Hahn vector space

17.1 Row- and column-finite coefficients

Let I be a nonempty set. A matrix $M = (M_{ij})_{i,j \in I}$ is *row- and column-finite* if each of its rows and each of its columns has finite support. The number of nonzero entries need not be uniformly bounded as the row or column varies. Set

$$R_I = \text{RCF}_I(\mathbb{C}), \quad V_I = \mathbb{C}^{(I)}.$$

Every diagonal matrix with entries in $\mathbb{C}$ belongs to R_I , even when its entries are unbounded in the ordinary absolute value.

Lemma 17.1. *The usual finite row-by-column multiplication makes R_I an associative unital $\mathbb{C}$ -algebra. Conjugate transpose is an involution, and R_I acts faithfully on V_I . With the ordinary finite-support inner product, $\langle Mx, y \rangle = \langle x, M^*y \rangle$.*

Proof. For a fixed row of MN , only the finitely many rows of N selected by that row of M can contribute. Their supports have finite union. The analogous column argument proves column-finiteness. Every entry in an iterated product involves only finitely many intermediate indices, so associativity follows from distributivity of finite sums. The identity matrix belongs to R_I , and conjugate transpose interchanges the two finiteness conditions. A finite-support vector is taken to a finite-support vector because the relevant columns have finite union of supports. The adjoint identity is a finite computation. Finally, an operator acting as zero on every standard vector e_i has every column zero. $\square$

The requirement is deliberately imposed on *coefficients*, rather than on the complete matrix of Hahn entries. The eventual diagonalizer will generally have infinitely many nonzero entries in a row or column, spread over increasingly complicated exponents.

17.2 The Hahn algebra and module

With the conventions of section 2.1, define

$$K = \mathbb{C}((t^\Gamma)), \quad F = \mathbb{R}((t^\Gamma)), \quad \mathcal{A}_I = R_I((t^\Gamma)), \quad \mathcal{H}_I = V_I((t^\Gamma)). \quad (17.1)$$

The usual Hahn convolution makes K a field and F an ordered subfield; a nonzero real series is positive exactly when its leading coefficient is positive. The field property does not require Γ to be divisible: a nonzero series factors as a nonzero constant times a monomial times a unit of the form $1 + \text{positive support}$, whose inverse is given by lemma 2.3. Write $\mathfrak{m}_F = \{f \in F : \text{supp } f \subset \Gamma_{>0}\}$; it includes zero and is not the cone of positive-valued elements.

Proposition 17.2. *The space $\mathcal{A}_I$ is an associative unital K -algebra with involution*

$$\left(\sum M_\gamma t^\gamma \right)^* = \sum M_\gamma^* t^\gamma.$$

It acts faithfully on the K -vector space $\mathcal{H}_I$ by convolution. For products in this algebra and its module,

$$v(XY) \geq v(X) + v(Y), \quad v(Xx) \geq v(X) + v(x).$$

The order function on $\mathcal{A}_I$ need not be a field valuation: if I contains two distinct indices, $\mathcal{A}_I$ has zero divisors. For a singleton I , $\mathcal{A}_I = K$ is a field and its order function is the field valuation.

Proof. At a prescribed exponent, lemma 2.1(i) leaves finitely many coefficient products; lemma 17.1 puts each in R_I or V_I . The same finite-factor support argument permits reassociation of every triple product. Scalar Hahn series act through scalar matrices. The support of a product is contained in the sum of the supports, proving the displayed inequalities. Faithfulness follows by evaluating on each e_i and then comparing coefficients. The adjoint identity follows coefficientwise. For distinct $i, j \in I$, the nonzero constant diagonal matrix units E_{ii} and E_{jj} belong to R_I and satisfy $E_{ii}E_{jj} = 0$. If I is a singleton, identifying a matrix with its sole entry gives $R_I = \mathbb{C}$ and hence $\mathcal{A}_I = K$. $\square$

17.3 The inner product and its positivity

Proposition 17.3. *The formula*

$$\langle x, y \rangle = \sum_{\alpha, \beta} \langle x_\alpha, y_\beta \rangle_{\mathbb{C}(I)} t^{\alpha+\beta} = \sum_{i \in I}^H \bar{x}_i y_i \quad (17.2)$$

defines a conjugate-linear/linear K -valued inner product on $\mathcal{H}_I$. For $x \neq 0$,

$$\langle x, x \rangle \in F_{>0}, \quad v(\langle x, x \rangle) = 2v(x).$$

*The adjoint relation $\langle Xx, y \rangle = \langle x, X^*y \rangle$ holds for $X \in \mathcal{A}_I$.*

Proof. For each total exponent there are finitely many pairs (α, β) , and each pair involves finite-support vectors. Thus the family over i in (17.2) satisfies both conditions for strong Hahn summability. Sesquilinearity and the adjoint relation follow from finite coefficient calculations. If $\gamma = v(x)$, the coefficient at 2γ in $\langle x, x \rangle$ is

$$\sum_i |(x_\gamma)_i|^2 > 0.$$

There can be no smaller exponent, and no other pair of support exponents sums to 2γ . This proves positivity and the valuation formula. $\square$

Remark 17.4 (No divisibility, and no norm). Proposition 17.3 is positive without assuming that every positive element of F has a square root. Indeed, for arbitrary Γ the field F need not be real closed. The only roots used in the main construction are roots of units with residue 1, obtained by binomial Hahn series through corollary 2.6. In particular no analogue of $\|x\| = \sqrt{\langle x, x \rangle}$ is introduced here; the norm of proposition 5.1 is a Part I object, but its construction does not require divisibility. The even-leading-exponent argument also supplies a square root of the squared length in Part II; we simply do not use a norm in this part's algebraic construction.

Lemma 17.5 (Strong sums are respected). *If (x_a) is strongly summable in $\mathcal{H}_I$ and $X \in \mathcal{A}_I$, then (Xx_a) is strongly summable and $X(\sum_a^H x_a) = \sum_a^H Xx_a$. The analogous statement holds for multiplication by a fixed element in $\mathcal{A}_I$ and for taking the inner product with a fixed vector.*

Proof. The union of resulting supports lies in the sum of two well-ordered sets. At any exponent there are finitely many pairs of coefficient exponents, and for each such pair only finitely many family members can contribute. The claimed identities are therefore identities between finite sums at each coefficient. $\square$

18 Exact infinite diagonalization

18.1 The divided commutator

Let $(d_i)_{i \in I}$ be pairwise distinct elements of $\mathbb{C}$, and put $D = \text{diag}(d_i)$. Define two coefficientwise linear maps on R_I and hence on $\mathcal{A}_I$:

$$\pi(M) = \text{diag}(M_{ii}), \quad \mathcal{R}_D(M)_{ij} = \begin{cases} M_{ij}/(d_i - d_j), & i \neq j, \\ 0, & i = j. \end{cases} \quad (18.1)$$

They do not enlarge Hahn support, and they preserve the row- and column-finiteness of every coefficient. On off-diagonal matrices, $\mathcal{R}_D$ is the inverse of $X \mapsto [D, X]$.

No uniform bound for $\mathcal{R}_D$ in a real operator norm is claimed. Every individual denominator is a nonzero ordinary complex number and therefore has Hahn valuation zero. Arbitrarily small ordinary gaps do not change supports. This is exactly the point at which the pairwise-distinct hypothesis does its work, and remark 18.4 records what happens when it is absent.

18.2 A similarity theorem without self-adjointness

Theorem 18.1 (Coherent simple-residue diagonalization). *Let $D = \text{diag}(d_i) \in R_I$ have pairwise distinct complex entries, and let $B \in \mathcal{A}_I$ have positive support. There exist unique*

$$V = \mathbf{1} + W \in \mathbf{1} + (\mathcal{A}_I)_{>0}, \quad \pi(W) = 0, \quad \Lambda = D + \Delta,$$

with Δ diagonal of positive support, such that

$$(D + B)V = V\Lambda. \quad (18.2)$$

The matrix V is invertible in $\mathcal{A}_I$. If $S = \text{supp } B$, then

$$\text{supp } W, \text{ sup } \Delta \subseteq S^* \setminus \{0\}.$$

The construction uses only finite coefficient operations, the nonzero numbers $d_i - d_j$, and strong Hahn summation.

Proof. Introduce a formal bookkeeping variable h , distinct from the Hahn monomials, and seek

$$V(h) = \sum_{n \geq 0} h^n V_n, \quad \Lambda(h) = D + \sum_{n \geq 1} h^n \Delta_n, \quad V_0 = \mathbf{1}, \quad \pi(V_n) = 0 \ (n \geq 1).$$

At degree $n \geq 1$, the equation $(D + hB)V(h) = V(h)\Lambda(h)$ is

$$[D, V_n] + BV_{n-1} = \Delta_n + \sum_{k=1}^{n-1} V_{n-k} \Delta_k.$$

Thus the recursion is forced:

$$T_n = BV_{n-1} - \sum_{k=1}^{n-1} V_{n-k} \Delta_k, \tag{18.3}$$

$$\Delta_n = \pi(T_n) = \pi(BV_{n-1}), \tag{18.4}$$

$$V_n = -\mathcal{R}_D(T_n). \tag{18.5}$$

The second equality in (18.4) follows because multiplying an off-diagonal V_{n-k} by a diagonal matrix does not create diagonal entries. These formulas define every coefficient and prove the formal eigenidentity.

Inductively, V_n and Δ_n are homogeneous of degree n in B , with support contained in nS . Each coefficient is in R_I , by lemma 17.1 and the support preservation of (18.1). Corollary 2.5 therefore licenses the substitutions

$$V = \mathbf{1} + \sum_{n \geq 1}^H V_n, \quad \Delta = \sum_{n \geq 1}^H \Delta_n.$$

It also licenses substitution in the products in the formal equation, proving (18.2). The support claim follows at once. The inverse of V exists by corollary 2.6.

For uniqueness, suppose (V', Λ') is another pair with the stated normalizations. Then $T = V^{-1}V'$ belongs to $\mathbf{1} + (\mathcal{A}_I)_{>0}$ and satisfies $\Lambda T = T \Lambda'$. The diagonal equation is $(\lambda_i - \lambda'_i)T_{ii} = 0$. Since T_{ii} has residue 1, it is nonzero, so $\lambda_i = \lambda'_i$. For $i \neq j$, the element $\lambda_i - \lambda_j$ has nonzero residue $d_i - d_j$, and hence $T_{ij} = 0$. Thus T is diagonal. Finally $V_{ii} = V'_{ii} = 1$ gives $T_{ii} = 1$, proving $T = \mathbf{1}$ and the asserted uniqueness. $\square$

The bookkeeping construction does not assert that $h = 1$ is a permissible substitution in arbitrary formal power series. It is permissible here because the degree- n coefficient is supported in nS , and lemma 2.1(ii) supplies the missing summability proof.

18.3 Positivity turns the similarity into a unitary

Theorem 18.2 (Normalized infinite Hahn spectral theorem). *Assume now that every d_i is real, that the d_i are pairwise distinct, and that $B = B^* \in \mathcal{A}_I$ has positive support. With $A = D + B$, there is a unique pair (U, Λ) satisfying*

$$AU = U\Lambda, \quad U^*U = UU^* = \mathbf{1}, \quad U \in \mathbf{1} + (\mathcal{A}_I)_{>0}, \tag{18.6}$$

where $\Lambda = \text{diag}(\lambda_i)$, $\lambda_i \in F$, $\lambda_i - d_i \in \mathfrak{m}_F$, and every diagonal entry U_{ii} is real and strictly positive. Both $U - \mathbf{1}$ and $\Lambda - D$ are supported in $S^* \setminus \{0\}$, where $S = \text{supp } B$.

Here $\mathfrak{m}_F$ includes zero and is not the cone of positive-valued elements. In particular, the correction to an eigenvalue may have either sign.

Proof. Use theorem 18.1 and write $v_i = Ve_i$. Since $Av_i = \lambda_i v_i$ and $A = A^*$,

$$\lambda_i \langle v_i, v_i \rangle = \langle v_i, Av_i \rangle = \overline{\langle v_i, Av_i \rangle}.$$

The norm square is positive and nonzero by proposition 17.3; therefore $\lambda_i = \overline{\lambda_i}$. For distinct i, j , self-adjointness similarly gives

$$(\lambda_j - \lambda_i) \langle v_i, v_j \rangle = 0.$$

The eigenvalue difference has nonzero real residue, so the columns are orthogonal.

Consequently $N = V^*V$ is diagonal. Because $V = \mathbf{1} + W$ and $W_{ii} = 0$,

$$N_{ii} = 1 + \langle We_i, We_i \rangle, \quad N = \mathbf{1} + \pi(W^*W). \quad (18.7)$$

The diagonal entries are positive with residue 1. The binomial series

$$H = N^{-1/2} = \sum_{m \geq 0}^H \binom{-1/2}{m} (N - \mathbf{1})^m$$

exists by corollary 2.6, is real diagonal, and satisfies $HNH = \mathbf{1}$. Its diagonal entries have leading coefficient 1, hence are positive. Define $U = VH$. Since H commutes with Λ , $AU = UA$ and $U^*U = HNH = \mathbf{1}$. Both V and H are invertible in $\mathcal{A}_I$, so U is invertible and $U^* = U^{-1}$; this proves the other unitary identity as well. Notice that in an infinite algebra a left inverse alone would not have sufficed.

Because $V_{ii} = 1$, one has $U_{ii} = H_{ii} > 0$. The support claims follow from lemma 2.1(ii) applied to the homogeneous expansions; no exponent division is involved in the binomial normalization.

For uniqueness, let (U', Λ') satisfy the same requirements. The intertwining matrix $T = U^*U'$ has residue $\mathbf{1}$. As in the uniqueness proof of theorem 18.1, first $\Lambda' = \Lambda$ and then T is diagonal. It is unitary. From $U'_{ii} = U_{ii}T_{ii}$ and positivity of both diagonal entries, T_{ii} is positive real. The identity $\overline{T_{ii}}T_{ii} = 1$ then forces $T_{ii} = 1$. $\square$

Corollary 18.3 (No new scales). *Assume the hypotheses of theorem 18.2: in particular that $D = \text{diag}(d_i)$ has pairwise distinct ordinary real entries, so that every divisor $d_i - d_j$ used in (18.1) is a nonzero ordinary complex constant of valuation zero, and that $B = B^*$ has globally well-ordered positive support S . Then, under those hypotheses, all eigenvalues λ_i and all entries of the normalized diagonalizer U lie in the original Hahn field K , and their correction supports lie in the additive monoid S^* generated by S ; consequently the result holds over an arbitrary ordered exponent group, with no passage to its divisible hull. If in addition the coefficients of A are real Hahn, the diagonalizer can be taken, and by uniqueness is, real Hahn orthogonal.*

Proof. The support assertion was proved in theorems 18.1 and 18.2. When all coefficients are real, the recursion and the binomial series use real operations throughout. $\square$

Remark 18.4 (The monoid statement is not true without the distinct-residue hypothesis). The support conclusion of corollary 18.3 must not be lifted out of its hypothesis. The collection's finite-dimensional spectral report contains an example, its `spec:ex:monoid`, which shows that the corresponding unconditional statement is false: with $\Gamma = \mathbb{Z}$ and

$$A = \begin{pmatrix} t^3 & t^4 \\ t^4 & 0 \end{pmatrix}, \quad \lambda_+ = t^3 + t^5 - t^7 + 2t^9 - \dots,$$

the exponent 5 is not in the nonnegative monoid generated by the raw input exponents 3 and 4, and that report concludes that a theorem asserting that eigenvalue supports stay in the raw input monoid “would therefore be false” [2].

There is no conflict. Writing that matrix as $D + B$ with B of positive support forces $D = \text{diag}(0, 0)$: both diagonal entries, t^3 and 0, have residue zero. The residues are therefore not pairwise distinct, the divided commutator $\mathcal{R}_D$ of (18.1) does not exist, and theorems 18.1 and 18.2 and corollary 18.3 do not apply. Corollary 18.3 divides only by differences $d_i - d_j$ of pairwise distinct ordinary complex constants, each of valuation zero; that is precisely what keeps every correction exponent inside S^* . The report cited above states the correct unconditional invariant for the general finite case — the input *subgroup*, supplemented by the positive monoids of the normalized perturbations at each node — and that is a different statement from the one proved here.

18.4 Initial coefficients and their meaning

For a one-scale perturbation $B = t^\eta B_1 + t^{2\eta} B_2 + \dots$, $\eta > 0$, the first terms are

$$\lambda_i = d_i + t^\eta (B_1)_{ii} + t^{2\eta} \left((B_2)_{ii} + \sum_{j \neq i} \frac{|(B_1)_{ji}|^2}{d_i - d_j} \right) + \mathcal{O}(t^{3\eta}), \quad (18.8)$$

$$U_{ji} = t^\eta \frac{(B_1)_{ji}}{d_i - d_j} + \mathcal{O}(t^{2\eta}) \quad (j \neq i), \quad (18.9)$$

$$U_{ii} = 1 - \frac{t^{2\eta}}{2} \sum_{j \neq i} \frac{|(B_1)_{ji}|^2}{(d_i - d_j)^2} + \mathcal{O}(t^{3\eta}). \quad (18.10)$$

Here the remainder notation is a valuation statement within the one-scale series under discussion. Every displayed sum is finite for a fixed i . Equations (18.8)–(18.10) follow directly from (18.3)–(18.7); they are the familiar perturbation coefficients, not independent novelty claims.

What changes in the present theorem is that all orders and all coordinates fit into one operator with globally controlled Hahn support. A uniform bound for the real numbers in (18.9) is unnecessary, but a finite coefficient support in every row and column remains essential to the chosen algebra.

19 Spectral synthesis, commutants, and projections

Throughout this section the hypotheses of theorem 18.2 hold, and $u_i = Ue_i$ denotes its normalized eigenvector.

Theorem 19.1 (Exact Hahn spectral synthesis). *The family $(u_i)_{i \in I}$ is orthonormal. Every $x \in \mathcal{H}_I$ has the unique strongly summable expansion*

$$x = \sum_{i \in I}^H u_i \langle u_i, x \rangle. \quad (19.1)$$

*More precisely, the coordinate map $x \mapsto (\langle u_i, x \rangle)_{i \in I}$ identifies $\mathcal{H}_I$ with itself when the right side is interpreted as the Hahn vector U^*x . One has the identities*

$$\langle x, x \rangle = \sum_{i \in I}^H |\langle u_i, x \rangle|^2, \quad (19.2)$$

$$Ax = \sum_{i \in I}^H \lambda_i u_i \langle u_i, x \rangle. \quad (19.3)$$

The eigenvalues of A on $\mathcal{H}_I$ are exactly the λ_i , and each corresponding eigenspace is Ku_i .

Proof. Orthonormality is the entrywise meaning of $U^*U = \mathbf{1}$. Put $y = U^*x$. At each exponent, y has finite coordinate support. Therefore the family $(e_i y_i)_{i \in I}$ is strongly summable in $\mathcal{H}_I$ and sums to y . Lemma 17.5 gives

$$x = Uy = \sum_i^H Ue_i y_i = \sum_i^H u_i \langle u_i, x \rangle,$$

proving existence and summability. Taking inner products with each u_i proves uniqueness whenever such a strong expansion is given. The diagonal action of Λ takes $\mathcal{H}_I$ to itself because $\Lambda \in \mathcal{A}_I$. Applying U to the coordinate expansion of Λy proves (19.3). The norm identity follows from $\langle x, x \rangle = \langle y, y \rangle$ and proposition 17.3.

Finally, if $Ax = zx$ with $x \neq 0$, then $(\lambda_i - z)y_i = 0$ for all i . At least one y_i is nonzero, so $z = \lambda_i$ for that index. Distinct residues make the λ_i pairwise distinct; all other coordinates of y vanish. This proves the eigenspace statement. $\square$

The expansion need not be finite over K , so (u_i) is generally not a Hamel basis. Nor does (19.2) involve the ordinary convergence of a series of nonnegative real numbers: its terms and sum lie in F , and the assertion is strong Hahn summability.

19.1 The coherent diagonal algebra

Let

$$\mathcal{D}_I = (\mathbb{C}^I)((t^\Gamma)) \subseteq \mathcal{A}_I \quad (19.4)$$

be the algebra of diagonal Hahn operators. Thus a family $(a_i)_{i \in I}$ of elements of K defines an element of $\mathcal{D}_I$ precisely when $\bigcup_i \text{supp } a_i$ is well ordered. There is no finite-incidence condition on the indices i here: an arbitrary diagonal matrix of complex numbers is one legitimate coefficient in R_I .

For finite I , every family is coherent, so $\mathcal{D}_I = K^I$; the same equality holds for the trivial value group. If I is infinite and $\Gamma \neq 0$, the inclusion is strict. Choose distinct indices i_n and $\eta > 0$, and set $a_{i_n} = t^{-n\eta}$ for $n \geq 1$, with all other entries zero. Their support union contains the decreasing sequence $-\eta, -2\eta, \dots$, so this family does not belong to $\mathcal{D}_I$.

Coherence places no ordinary bound on a fixed diagonal coefficient: the constant matrix with entries n at i_n and zero elsewhere belongs to $\mathcal{D}_I$ when I is infinite. These are separate conditions, and the resolvent proof below uses precisely global support coherence.

Theorem 19.2 (The complete commutant). *Within the specified Hahn operator algebra,*

$$\text{Cent}_{\mathcal{A}_I}(A) = U\mathcal{D}_IU^*. \quad (19.5)$$

In particular this commutant is commutative. Its idempotents are exactly

$$P_J = UE_JU^*, \quad J \subseteq I, \quad (19.6)$$

where $E_J = \text{diag}(\mathbf{1}_{i \in J})$. Every such idempotent is self-adjoint. The correspondence $J \mapsto P_J$ is a complete Boolean algebra isomorphism, with complement $\mathbf{1} - P_J$ and product $P_J P_L = P_{J \cap L}$.

Proof. If X commutes with A , then $Y = U^* X U$ commutes with Λ . At entry (i, j) this says $(\lambda_i - \lambda_j)Y_{ij} = 0$. For $i \neq j$ the first factor is nonzero in the field K , so Y is diagonal. Its membership in $\mathcal{A}_I$ is precisely the support coherence in (19.4). The converse is immediate, proving (19.5).

An idempotent of $\mathcal{D}_I$ has entries satisfying $a_i^2 = a_i$ in a field, so each is 0 or 1. Every choice is allowed, since its complete operator support is contained in $\{0\}$. These are exactly the E_J . Conjugating proves the projection statement and the finite Boolean identities. Arbitrary unions and intersections of subsets give the joins and meets for the partial order on idempotents defined by $P \leq Q$ when $PQ = P$; no assertion about an infinite operator-valued series is needed for completeness of this Boolean algebra. $\square$

Corollary 19.3 (Projection sums are pointwise Hahn sums). *For every subset $J \subseteq I$ and every $x \in \mathcal{H}_I$,*

$$P_J x = \sum_{i \in J}^H u_i \langle u_i, x \rangle.$$

If $(J_a)_{a \in L}$ are pairwise disjoint, then $(P_{J_a}x)_{a \in L}$ is strongly summable in $\mathcal{H}_I$ and its sum is $P_{\bigcup_a J_a}x$.

Proof. In the coordinates $y = U^*x$, these are the sums of disjoint coordinate pieces of a Hahn vector. At any exponent, only finitely many of the pieces can be nonzero. Apply lemma 17.5 to transport the sums by U . $\square$

Remark 19.4 (Operator sums and vectorwise additivity). For infinite I , the operator family $(P_{\{i\}})_{i \in I}$ is *not* strongly Hahn summable in $\mathcal{A}_I$: its coefficient at exponent zero is $E_{\{i\}} \neq 0$ for every i . Thus $\sum_i^H P_{\{i\}}$ is undefined in that algebra, whereas $\sum_i^H P_{\{i\}}x = x$ holds for each $x \in \mathcal{H}_I$. This distinction already occurs for $B = 0$.

It does not by itself exclude a spectral-measure interpretation. Classical projection-valued measures require countable additivity in the strong operator topology, which tests convergence on each Hilbert vector, rather than an operator-norm sum; see [9, Definition 5.1 and footnote 18]. Here $J \mapsto P_J$ on the power set of I is a projection-valued set function with vectorwise *Hahn* additivity by corollary 19.3. Such additivity does not imply convergence in the intrinsic valuation topology.

For an explicit failure, take $\Gamma = \mathbb{Q}$, $I = \mathbb{N}_{>0}$, $D = \text{diag}(n)$, and $B = 0$, so $U = \mathbf{1}$ and $P_{\{n\}} = E_{\{n\}}$. Put

$$\gamma_n = 1 - \frac{1}{n+1}, \quad x = \sum_{n \geq 1}^H t^{\gamma_n} e_n.$$

The exponents increase strictly, hence form a well-ordered set, and each coefficient has one coordinate. Thus $x \in \mathcal{H}_I$, but

$$v\left(x - \sum_{n=1}^N P_{\{n\}}x\right) = \gamma_{N+1} < 1$$

for every N . The partial sums never enter the neighborhood of x with error valuation greater than 1. Consequently even this rank-one example lacks countable additivity in the topology of pointwise valuation convergence. The established object is the Hahn-additive projection calculus; no general topologically countably additive spectral-measure or Borel integration theorem is asserted.

19.2 An unrestricted ordinary label calculus

Every function $f : I \rightarrow \mathbb{C}$ defines the operator

$$T_f = U \text{diag}(f(i))U^* \in \text{Cent}_{\mathcal{A}_I}(A).$$

This is a unital involution-preserving homomorphism from $\mathbb{C}^I$ into $\mathcal{A}_I$. No continuity or ordinary boundedness of f is required. It is a *label multiplier calculus*: it does not assign values to an ordinary function at arbitrary surreal arguments. More generally, a K -valued label family is allowed exactly when it belongs to the coherent diagonal algebra $\mathcal{D}_I$. Polynomial expressions $p(A)$ have labels $p(\lambda_i)$ because finite polynomial operations preserve coherence.

20 The exact algebra-relative spectrum and all resolvents

Definition 20.1. For $X \in \mathcal{A}_I$, define

$$\sigma_{\mathcal{A}_I}(X) = \{z \in K : X - z\mathbf{1} \text{ has no two-sided inverse in } \mathcal{A}_I\}.$$

This is a spectrum relative to a unital algebra, with no norm or compactness assumption. It is the second of the two senses recorded in table 1.

Diagonalization alone does not automatically identify this spectrum. Entrywise invertibility of infinitely many diagonal entries does not, in general, guarantee that their inverses have globally well-ordered support. The simple-residue hypothesis gives the additional coherence. For example, on $I = \mathbb{N}_0$ and with $\eta > 0$, the diagonal $Q = \text{diag}(t^{n\eta})$ belongs to $\mathcal{D}_I$, but its entrywise inverse has support $\{-n\eta : n \geq 0\}$ and is not coherent. Any matrix inverse would have to have exactly those diagonal entries and zero off-diagonal entries, by multiplying entrywise against Q . Thus Q is not invertible in $\mathcal{A}_I$. It is not onto $\mathcal{H}_I$ either: the Hahn vector $\sum_{n \geq 0} t^{n\eta} e_n$ would require the preimage $(1, 1, \dots)$, which has an inadmissible infinite-support coefficient at zero.

Theorem 20.2 (Coherent resolvents at every noneigenvalue). *Under the hypotheses of theorem 18.2,*

$$\sigma_{\mathcal{A}_I}(A) = \{\lambda_i : i \in I\}. \quad (20.1)$$

For $z \in K$ outside that set,

$$(A - z\mathbf{1})^{-1} = U \text{diag}((\lambda_i - z)^{-1}) U^* \in \mathcal{A}_I. \quad (20.2)$$

The conclusion and the proof also hold for the non-Hermitian simple-complex-residue setting of theorem 18.1, with U, U^* replaced by V, V^{-1} .

Proof. The eigenvector u_i prevents $A - \lambda_i \mathbf{1}$ from being invertible. It remains to prove coherence of the diagonal inverse when every $\lambda_i - z$ is nonzero. Write $\Lambda = D + \Delta$ with $\text{supp } \Delta \subset \Gamma_{>0}$.

First suppose $v(z) < 0$. Then $\Lambda - z\mathbf{1} = -z(\mathbf{1} - z^{-1}\Lambda)$. Since $v(\Lambda) \geq 0$, the operator $z^{-1}\Lambda$ has positive support. The inverse therefore belongs to $\mathcal{D}_I$ by lemma 2.3.

Now suppose $v(z) \geq 0$, also allowing $z = 0$. Write $z = c + \zeta$, where $c \in \mathbb{C}$ is the coefficient at exponent zero and ζ has positive support. If $c \neq d_i$ for every i , the ordinary diagonal matrix $C = \text{diag}((d_i - c)^{-1})$ belongs to R_I . It need not have bounded complex entries, and none are required. The factorization

$$\Lambda - z\mathbf{1} = (D - c\mathbf{1})(\mathbf{1} + C(\Delta - \zeta\mathbf{1}))$$

has a positive-support second factor and an invertible constant first factor. Its inverse is again in $\mathcal{D}_I$.

Finally, if $c = d_j$, then there is exactly one such index j . Split the diagonal algebra into its $E_{\{j\}}$ and $\mathbf{1} - E_{\{j\}}$ parts. On the second part the preceding argument applies, using arbitrary harmless values at the omitted index to write the factorization. On the first part, the inverse is the single scalar Hahn series $(\lambda_j - z)^{-1}$ placed at index j . It exists by hypothesis. The union of the supports of these two pieces is a finite union of well-ordered sets. Adding them gives the desired diagonal inverse. Conjugation by U proves (20.2).

Nothing in the coherence argument used reality of the distinct constants d_i ; the last assertion follows in the same way. $\square$

Example 20.3 (An accumulation point need not be in this spectrum). Take $I = \mathbb{N}_{>0}$, $D = \text{diag}(1/n)$, and $B = 0$. Then 0 is an ordinary accumulation point of the labels, but

$$D^{-1} = \text{diag}(n) \in R_I \subset \mathcal{A}_I.$$

Consequently $0 \notin \sigma_{\mathcal{A}_I}(D)$. This does not contradict the ordinary bounded-operator spectrum on ℓ^2 , where the displayed inverse is unbounded. It records the different algebra in which invertibility is being tested. It also does not contradict corollary 9.2, which puts the whole infinitesimal monad $\mathbf{m}_K$ into the failure-of-bijectivity spectrum of the same diagonal data as a compact operator on $\ell^2((t^\Gamma))$; section 3.2 sets the two side by side, and section 3.3 derives both from Part III's single formula.

Corollary 20.4 (The inhomogeneous equation). *If $z \notin \{\lambda_i : i \in I\}$, every equation $(A - z\mathbf{1})x = f$ with $f \in \mathcal{H}_I$ has the unique solution*

$$x = \sum_i^H u_i \frac{\langle u_i, f \rangle}{\lambda_i - z}.$$

The series is strongly summable in $\mathcal{H}_I$.

Proof. The coherent inverse diagonal operator takes U^*f to a Hahn vector. Apply U and theorem 19.1. Invertibility gives uniqueness. $\square$

21 Weighted walks control every coefficient

21.1 The graph attached to a Hahn perturbation

Let $S = \text{supp } B \subset \Gamma_{>0}$ and write $B = \sum_{s \in S} B_s t^s$. Construct an edge-labeled directed graph on I by putting an edge

$$j \xrightarrow{s} i \quad \text{when} \quad (B_s)_{ij} \neq 0.$$

Loops are allowed. Since $B_s = B_s^*$, every nonloop edge has a reverse edge with the same label. For a fixed s , every vertex has finitely many incoming and outgoing edges. The graph obtained by forgetting all labels need not be locally finite when S is infinite.

A walk has a finite sequence of edges; its weight is the sum of the positive labels. The empty walk has weight zero. Although an edge label can be infinitesimal relative to another in Γ , no edge has zero or negative weight.

Lemma 21.1 (Finite walks at a prescribed weight). *For fixed starting vertex i and fixed $\gamma \in \Gamma$, there are only finitely many walks beginning at i and having total weight γ . In particular their visited vertices form a finite set.*

Proof. Lemma 2.1(ii) gives finitely many ordered label words with total weight γ . For any one such word, there are finitely many choices of the first edge and, recursively, finitely many choices at each subsequent step. The word length is finite, so the complete set of walks is finite. Taking a finite union over words proves the assertion. $\square$

Theorem 21.2 (Coefficientwise walk dependence). *For the intermediate-normalized V and the normalized unitary U , the coefficient at a positive exponent γ in V_{ji} or U_{ji} is a finite sum of expressions supported on walks from i to j of weight γ . The coefficient of $\lambda_i - d_i$ is a finite sum of expressions supported on closed walks at i of weight γ .*

Each expression is a product of edge coefficients and their conjugates, multiplied by a rational number and by inverse powers of nonzero real label differences at visited vertices. Thus those coefficients are completely determined by the finitely many relevant walks and the ordinary labels at their vertices.

This is a dependence statement, not a claim that every permitted walk contributes a nonzero term, or that its contribution is just its raw edge product.

Proof. Work first with the homogeneous recursion (18.3)–(18.5). A coefficient of $V_0 = \mathbf{1}$ is represented by an empty walk. Multiplication by B attaches an edge to the end of a walk, with matrix multiplication summing over the intermediate vertex. The projection π retains closed walks. In the term $(V_{n-k}\Delta_k)_{ji}$, the scalar $(\Delta_k)_{ii}$ is represented by a closed walk at i ; traverse that closed walk and then a walk from i to j representing $(V_{n-k})_{ji}$. Division by $d_j - d_i$ changes no walk or exponent. This proves the assertion for V and Λ by induction on the homogeneous degree.

For normalization, (18.7) expresses $N_{ii} - 1$ through products $\overline{W_{ki}}W_{ki}$. More generally, at a fixed coefficient one factor can come from one walk from i to k and the other from another. Reverse the conjugated walk, using Hermitian symmetry of the labeled edges. Their concatenation is a closed walk at i . Powers of $N - \mathbf{1}$ concatenate such closed walks; the binomial coefficients contribute only rational scalars. Finally $U_{ji} = V_{ji}H_{ii}$ concatenates those closed walks with a walk from i to j .

Lemma 2.1(ii) bounds the homogeneous degrees at any prescribed exponent, and lemma 21.1 bounds every index sum within them. All coefficient formulas are therefore genuinely finite. $\square$

21.2 Marked-edge stability

Suppose B and $\tilde{B}$ satisfy the same hypotheses with the same D . Their support union S is well ordered and positive. Use the union of the two edge graphs, and mark an edge of label s whenever $(\tilde{B}_s - B_s)_{ji} \neq 0$. For a vertex i , let κ_i be the minimum weight of a closed walk at i containing a marked edge, or $+\infty$ if no such walk exists. The minimum exists whenever the set is nonempty, since its weights form a subset of the well-ordered monoid S^* .

Theorem 21.3 (Exact marked-walk stability). *The eigenvalues labeled by i satisfy*

$$v(\tilde{\lambda}_i - \lambda_i) \geq \kappa_i. \quad (21.1)$$

More generally, the coefficient of $\tilde{U}_{ji} - U_{ji}$ at exponent γ vanishes unless there is a walk from i to j of weight γ containing a marked edge.

In particular, set $E = \tilde{B} - B$. Then

$$v(\tilde{U} - U) \geq v(E), \quad v(\tilde{\Lambda} - \Lambda) \geq v(E).$$

If E has zero diagonal and $S \neq \emptyset$, the second estimate sharpens to

$$v(\tilde{\Lambda} - \Lambda) \geq v(E) + \min S. \quad (21.2)$$

Zero differences are interpreted as having valuation $+\infty$.

Proof. The recursions and the normalization use the same rational expressions in the edge coefficients for both inputs, with identical denominators because D is fixed. Subtract a product of coefficients from its counterpart by successively replacing its factors. Each resulting term contains a factor from E or its conjugate. It therefore has a marked edge in its associated walk. This argument is finite at every exponent by theorem 21.2, so it proves both dependence assertions and (21.1).

A term containing a factor from E has weight at least $v(E)$, since all its other edges have positive weights. If E has zero diagonal, a closed walk containing a marked edge cannot have length one. It has at least one further edge with weight at least $\min S$. This proves (21.2). The same bounds hold uniformly for every matrix coefficient and hence for the global operator order function. $\square$

These statements are valuation precision bounds, not analytic continuity in a real operator norm. They are also stronger than merely saying that finite jets depend on finite jets: they identify *which spatial changes* can first become visible at a given Hahn scale.

21.3 Finite sections with exact coefficient certificates

Let $F_0 \subset I$ be finite and contain i , and let A_{F_0} denote the principal compression of A to F_0 . Its diagonalizer and eigenvalues are normalized by the same rules. Write $\lambda_i^{F_0}$ for its eigenvalue with

residue d_i . Define the escape weight

$$\kappa_i(F_0) = \min\{\text{wt}(w) : w \text{ is a closed walk at } i \text{ and visits } I \setminus F_0\},$$

again using $+\infty$ for an empty set.

Theorem 21.4 (Finite-section exactness). *For every finite $F_0 \ni i$,*

$$v(\lambda_i - \lambda_i^{F_0}) \geq \kappa_i(F_0). \quad (21.3)$$

For a fixed exponent γ , the coefficient of λ_i is exactly that of a finite principal section containing all vertices of closed walks at i of weight γ . The analogous statement for U_{ji} holds with walks from i to j and with both endpoints included. Consequently every single requested coefficient has a finite spatial certificate, even when I and $\text{supp } B$ are infinite.

Proof. For the coefficient assertion, all relevant walks are finite in number by lemma 21.1. If their vertices lie in F_0 , every finite expression in theorem 21.2 is unchanged by compression. In particular, every coefficient strictly below $\kappa_i(F_0)$ agrees, which proves (21.3).

Alternatively, remove all edges crossing between F_0 and its complement while keeping the diagonal and all internal edges. The resulting infinite operator is a direct sum on those two index sets. Its recursion restricted to F_0 is exactly the recursion for A_{F_0} . A closed walk at i contains a removed crossing edge exactly when it leaves F_0 . Thus theorem 21.3 gives the same inequality. $\square$

The finite section for one coefficient need not work for *all* coefficients below a prescribed arbitrary-rank exponent. Such an initial segment can be infinite and can involve infinitely many spatial vertices. The theorem guarantees finiteness at each exact weight; it does not replace an infinite Hahn support by a finite truncation without additional data.

Corollary 21.5 (One-scale radius bound). *Let $A = D + t^\eta C$, where $\eta > 0$ and $C = C^* \in R_I$, and let F_0 contain the graph ball of radius r about i in the undirected support graph of C . Then*

$$v(\lambda_i - \lambda_i^{F_0}) \geq (2r + 2)\eta. \quad (21.4)$$

The coefficient of $t^{n\eta}$ in λ_i depends only on the ball of radius $\lfloor n/2 \rfloor$ about i . The coefficient of $t^{n\eta}$ in U_{ji} vanishes when $\text{dist}(i, j) > n$.

Proof. A walk that leaves the radius- r ball requires at least $r + 1$ edges to reach the outside and at least $r + 1$ more to return. Every edge has weight η . A closed walk of length n cannot visit a vertex farther than $\lfloor n/2 \rfloor$ from its starting point, while a walk from i to j has length at least their graph distance. Apply theorems 21.2 and 21.4. $\square$

21.4 The threshold is sharp

Theorem 21.6 (A nonzero first boundary coefficient). *Let $L \geq 1$ and let $A_L(t)$ be the tridiagonal matrix on vertices $0, \dots, L$ with pairwise distinct real diagonal entries $d_0, \dots, d_L$ and off-diagonal entries tb_k and $\overline{tb_k}$ between k and $k + 1$, where $b_k \in \mathbb{C} \setminus \{0\}$. Let $\lambda_0^{[L]}$ be its branch with residue d_0 , and compare it with the corresponding branch for the section $0, \dots, L - 1$. Then*

$$\lambda_0^{[L]} - \lambda_0^{[L-1]} = \frac{\prod_{k=0}^{L-1} |b_k|^2}{(d_0 - d_L) \prod_{k=1}^{L-1} (d_0 - d_k)^2} t^{2L} + \mathcal{O}(t^{2L+2}). \quad (21.5)$$

In particular the leading coefficient is nonzero, and the exponent in (21.4) cannot be increased in general.

Proof. Use $q = t^2$ and put $c_k = |b_k|^2$. Schur elimination on the tail gives the scalar equation

$$x - d_0 = qc_0 H_1^{[L]}(x, q), \quad H_k^{[L]}(x, q) = \frac{1}{x - d_k - qc_k H_{k+1}^{[L]}(x, q)}, \quad (21.6)$$

with $H_L^{[L]} = (x - d_L)^{-1}$. For x near d_0 , all constant terms of the denominators are nonzero, so these are formal power series in q with coefficients rational in x , regular at d_0 .

For $L \geq 2$, subtract the full and truncated tail fractions. At the last common vertex one obtains

$$H_{L-1}^{[L]} - H_{L-1}^{[L-1]} = \frac{qc_{L-1}}{(x - d_{L-1})^2(x - d_L)} + \mathcal{O}(q^2).$$

At each preceding vertex, the identity

$$\frac{1}{a - qcH} - \frac{1}{a - qc\tilde{H}} = \frac{qc(H - \tilde{H})}{(a - qcH)(a - qc\tilde{H})}$$

adds one power of q , one factor c , and a squared leading denominator. Thus

$$qc_0(H_1^{[L]} - H_1^{[L-1]}) = \frac{\prod_{k=0}^{L-1} c_k}{(x - d_L) \prod_{k=1}^{L-1} (x - d_k)^2} q^L + \mathcal{O}(q^{L+1}).$$

To justify passage from the tail difference to the root difference, let

$$G_L(x, q) = x - d_0 - qc_0 H_1^{[L]}(x, q), \quad x_L = \lambda_0^{[L]}, \quad x_{L-1} = \lambda_0^{[L-1]}.$$

Both roots are $d_0 + \mathcal{O}(q)$. The formal divided difference of G_L between these two roots is a unit $1 + \mathcal{O}(q)$, since $G_L(x, 0) = x - d_0$. If $C(x)$ denotes the displayed coefficient of q^L , then

$$G_L(x_{L-1}, q) = -C(d_0)q^L + \mathcal{O}(q^{L+1}), \quad G_L(x_L, q) = 0.$$

Subtracting and dividing by that unit gives $x_L - x_{L-1} = C(d_0)q^L + \mathcal{O}(q^{L+1})$, as claimed. For $L = 1$ the truncated branch is exactly d_0 , and the original Schur equation gives the same formula directly. This proves (21.5). The nonzero denominator and the nonzero positive numerator prove sharpness. $\square$

22 An accumulating-level chain with an exact eigenpair

Consider $I = \mathbb{N}_0$, $\Gamma = \mathbb{Z}$, and the real symmetric operator

$$(Ax)_n = \frac{x_n}{n+1} + t(x_{n-1} + x_{n+1}), \quad x_{-1} = 0. \quad (22.1)$$

Its diagonal labels $d_n = 1/(n+1)$ are pairwise distinct but accumulate at zero, and the neighboring gaps

$$d_n - d_{n+1} = \frac{1}{(n+1)(n+2)}$$

have no uniform positive real lower bound. Theorem 18.2 nevertheless constructs all its normalized eigenvectors simultaneously in $\mathbb{C}^{(\mathbb{N}_0)}((t))$.

22.1 The first eigenvalue can be written in closed form

Proposition 22.1 (An exact branch and its eigenvector). *For the operator (22.1), the eigenvalue with residue $d_0 = 1$ is*

$$\lambda_0 = \sqrt{1 + 4t^2}, \quad (22.2)$$

where the square root has residue 1. Define

$$q = \frac{\lambda_0 - 1}{2t} = \frac{2t}{\lambda_0 + 1}. \quad (22.3)$$

Then $v(q) = 1$, the intermediate-normalized eigenvector has coordinates $(v_0)_n = (n + 1)q^n$, and the normalized eigenvector is

$$(u_0)_n = (n + 1)q^n \left(\frac{(1 - q^2)^3}{1 + q^2} \right)^{1/2}. \quad (22.4)$$

All square roots in these formulas are near-one binomial Hahn roots.

Proof. The binomial series defines (22.2); it gives $q = t + \mathcal{O}(t^3)$ and the exact identities

$$t = \frac{q}{1 - q^2}, \quad \lambda_0 = \frac{1 + q^2}{1 - q^2} = t(q + q^{-1}). \quad (22.5)$$

For $n \geq 0$, direct substitution gives

$$\begin{aligned} \frac{(n + 1)q^n}{n + 1} + t(nq^{n-1} + (n + 2)q^{n+1}) &= q^n(1 + t(nq^{-1} + (n + 2)q)) \\ &= \lambda_0(n + 1)q^n. \end{aligned}$$

For $n = 0$, the term with the factor n is zero, agreeing with the boundary condition. Because $v(q^n) = n$, at any exponent only finitely many vector coordinates can contribute. The displayed sequence is therefore a Hahn vector, not merely an arbitrary sequence of elements of K . It has zeroth coordinate 1 and residue e_0 . The eigenspace statement in theorem 19.1, together with its residue and zeroth coordinate, identifies it with the unique intermediate-normalized branch constructed in theorem 18.1.

The elementary formal identity

$$\sum_{n \geq 0} (n + 1)^2 z^n = \frac{1 + z}{(1 - z)^3}$$

can be evaluated at $z = q^2$ by corollary 2.6. Hence

$$\langle v_0, v_0 \rangle = \frac{1 + q^2}{(1 - q^2)^3}.$$

Taking its inverse near-one square root gives (22.4). □

Proposition 22.1 is a directly proved explicit example; no separate priority claim is made for this solvable tridiagonal model. It also checks the perturbation construction by a method independent of the matrix recursion. Its eigenvalue begins

$$\lambda_0 = 1 + 2t^2 - 2t^4 + 4t^6 - 10t^8 + 28t^{10} + \mathcal{O}(t^{12}). \quad (22.6)$$

The included exact computation obtains these same coefficients from the six-vertex principal section. Corollary 21.5 shows that the section $0, \dots, 5$ gives the infinite branch correctly through degree 10; its first possible error is degree 12.

22.2 The first finite-section error is explicit

For the path section $0, \dots, L$ with unit edge coefficients, the coefficient in theorem 21.6 simplifies to

$$\frac{1}{(1 - 1/(L+1)) \prod_{k=1}^{L-1} (1 - 1/(k+1))^2} = L(L+1).$$

Indeed, $1 - 1/(k+1) = k/(k+1)$ and the product telescopes. Therefore

$$\lambda_0^{[L]} - \lambda_0^{[L-1]} = L(L+1)t^{2L} + \mathcal{O}(t^{2L+2}). \quad (22.7)$$

The infinite branch and the section $0, \dots, L$ agree through degree $2L$, so the same first coefficient describes the error made by truncating the infinite chain at $L-1$.

This is stronger than a qualitative finite-section limit: it specifies the first scale at which the omitted boundary is visible and gives its nonzero coefficient. It makes sense without any ordinary numerical choice of a small real t .

23 A higher-rank example: exact sums without cofinal powers

Let $\Gamma = \mathbb{Z}^2$ with lexicographic order, and set $a = (0, 1)$, $b = (1, 0)$. Both are positive, but $na < b$ for every positive integer n . Consider

$$A = \begin{pmatrix} 0 & t^a & 0 \\ t^a & 1 & t^b \\ 0 & t^b & 3 \end{pmatrix}. \quad (23.1)$$

The normalized branch at vertex 0 has the coefficient identities

$$[t^{2a}]\lambda_0 = -1, \quad [t^{a+b}]U_{20} = \frac{1}{3}, \quad [t^{2a+2b}]\lambda_0 = -\frac{1}{3}. \quad (23.2)$$

The first uses the edge $0 \leftrightarrow 1$. The distant eigenvector coordinate uses the path $0 \rightarrow 1 \rightarrow 2$, and the last eigenvalue coefficient is the first return through the distant edge $1 \leftrightarrow 2$. These coefficients follow from the recursion, or from the shortest corresponding walk expressions; the included bivariate calculation checks them exactly.

There can be infinitely many powers of t^a before the scale t^b . Accordingly, a finite total-degree computation in two formal variables is *not* the same as computing every coefficient below b in the lexicographic valuation. The verification program checks specified bidegrees and explicitly does not conflate these truncations.

Proposition 23.1 (Why geometric partial sums need not converge). *In this Hahn field, the family $((t^a)^n)_{n \geq 0}$ is strongly summable with sum $(1 - t^a)^{-1}$, but its finite partial sums do not converge to that sum in the valuation topology of K .*

Proof. Lemma 2.3 gives the sum and its algebraic identity. The exact remainder after degree N is

$$\frac{t^{(N+1)a}}{1 - t^a},$$

of valuation $(N+1)a < b$. Thus it never belongs to the neighborhood of zero defined by $v(x) > b$. This prevents convergence to the Hahn sum. $\square$

In a surreal realization sending a to 1 and b to ω , the monomials become ω^{-1} and $\omega^{-\omega}$. The same exact remainder remains larger in magnitude than the latter infinitesimal scale. The spectral theorem uses the word-finiteness proof, so it remains valid despite this failure of sequential convergence.

24 Three boundaries that cannot be omitted

24.1 No bounded Hilbert-space diagonalizer with these jets

Proposition 24.1 (The inverse-gap obstruction). *On ordinary $\ell^2(\mathbb{N}_{>0})$, let $D = \text{diag}(1/n)$ and let C be the nearest-neighbor adjacency operator. Both D and C are bounded and self-adjoint. There is no formal diagonalization of $D + tC$ of the form*

$$(D + tC)U(t) = U(t)\Lambda(t), \quad U(t) = \mathbf{1} + tX + \cdots,$$

with X a bounded operator and $\Lambda(t)$ diagonal with constant term D . In particular no ordinary operator-norm analytic near-identity unitary diagonalizer can have that property.

Proof. The order- t off-diagonal equation is $[D, X] + C = \text{diagonal}$. At the neighboring entry $(n, n+1)$ it gives

$$X_{n,n+1} = -\frac{1}{1/n - 1/(n+1)} = -n(n+1).$$

Every matrix entry of a bounded operator has absolute value at most its operator norm. The displayed sequence is unbounded, a contradiction. An operator-norm analytic family would have a bounded derivative at zero, so it is ruled out as well. $\square$

The diagonalizer of theorem 18.2 does have this first coefficient, and that coefficient belongs to R_I : it is supported on neighboring diagonals, even though its real entries are unbounded. Thus the theorem does not secretly solve the ordinary small-divisor or bounded-operator problem. It solves a different, explicitly algebraic problem with exact infinitesimal scales. The same contrast appears from the other side in section 14, where the excluded object is an unbounded diagonal coefficient of an adjointable operator.

24.2 Repeated residues can destroy every eigenvector

Let T be the bilateral adjacency operator on finite-support vectors indexed by $\mathbb{Z}$:

$$(Tx)_n = x_{n-1} + x_{n+1}.$$

It is real symmetric and row- and column-finite. A nonzero finite-support vector cannot be an eigenvector of T : at the index immediately above the largest nonzero coordinate, Tx is nonzero while the vector itself is zero. The same observation shows that T is injective on $\mathbb{C}^{(\mathbb{Z})}$.

Proposition 24.2 (A positive Hahn operator with no eigenvectors). *For any $\eta \in \Gamma_{>0}$, the operator*

$$A = \mathbf{1} + t^\eta T \in \text{RCF}_{\mathbb{Z}}(\mathbb{C})((t^\Gamma))$$

is self-adjoint and satisfies $\langle x, Ax \rangle > 0$ for every nonzero $x \in \mathbb{C}^{(\mathbb{Z})}((t^\Gamma))$. Nevertheless it has no eigenvalue in K for its action on that Hahn vector space.

Proof. If $\gamma = v(x)$, the leading term of $\langle x, x \rangle$ has exponent 2γ and positive coefficient. The term $t^\eta \langle x, Tx \rangle$ has valuation at least $2\gamma + \eta$. Thus $\langle x, Ax \rangle$ has the same positive leading term as the norm square.

Suppose $Ax = zx$ with $x \neq 0$. Injectivity of T on the leading finite-support coefficient gives $v(Tx) = v(x) = \gamma$. Hence $z \neq 1$ and $v(z - 1) = \eta$ in the equation $t^\eta Tx = (z - 1)x$. If c is the nonzero leading coefficient of $t^{-\eta}(z - 1)$, comparison at exponent $\gamma + \eta$ gives $Tx_\gamma = cx_\gamma$. That is a forbidden finite-support eigenvector of the bilateral adjacency operator. The contradiction proves the assertion. $\square$

All residue labels in this example are 1. Thus replacing “pairwise distinct ordinary residues” in theorem 18.2 merely by “positive self-adjoint” would be false. The general phenomenon of positive infinite row- and column-finite matrices without eigenvalues is already emphasized in [18]; the proof here shows explicitly how it survives the chosen Hahn extension.

24.3 Entrywise Hahn matrices are not enough

Proposition 24.3 (Failure without global support coherence). *Let $I = \mathbb{N}_0$ and $\Gamma = \mathbb{Z}$. The entrywise Hahn diagonal matrix $M = \text{diag}(t^{-n})_{n \geq 0}$ does not define an operator from $\mathcal{H}_I$ to itself by coordinatewise multiplication, although every entry belongs to K and every individual exponent coefficient is row- and column-finite.*

Proof. Its union of supports is $\{0, -1, -2, \dots\}$, which is not well ordered; thus $M \notin \mathcal{A}_I$. More concretely,

$$x = \sum_{n \geq 0} t^n e_n \in \mathcal{H}_I$$

would be sent to the sequence $(1, 1, 1, \dots)$. This would require an infinite-support coefficient at exponent zero, so it is not an element of $\mathcal{H}_I$. $\square$

Likewise, a coefficient matrix with an infinite nonzero column can send one e_i outside $\mathbb{C}^{(I)}$; requiring row-finiteness as well makes the adjoint remain in the same algebra. These hypotheses are simple sufficient conditions for the whole framework. The counterexamples do not claim that no larger useful operator categories exist.

25 Transport to surreal numbers and extension of scales

Corollary 25.1 (An infinite surcomplex spectral system). *Whenever the input is given in a workspace embedded by (2.2), every scalar entry of U , every eigenvalue λ_i , and every scalar coordinate of a vector in $\mathcal{H}_I$ is an actual surreal or surcomplex number. All identities in sections 17 to 21 remain valid after this embedding, with sums interpreted as the transported strong sums. The eigenvalues of a Hermitian input are surreal, and its inner products have positive surreal norm squares.*

Proof. The embedding preserves every finite coefficient operation and every strong sum used in the construction. It preserves order and conjugation, so it preserves positivity, adjoints, and the normalization conditions. $\square$

The matrices and vectors remain set-indexed structured objects; an infinite matrix is not being identified with one surcomplex scalar. Nor is this a theorem about every conceivable class-sized operator over $\mathbf{No}[i]$. It provides a rigorous infinite-dimensional theory in a fixed, explicitly embedded workspace.

Proposition 25.2 (Naturality and absence of new spectral points). *Let $\iota : \Gamma \hookrightarrow \Gamma'$ be a strictly increasing additive embedding. Apply it to all Hahn exponents of A , keeping the same index set and complex coefficients. The normalized diagonalizer and eigenvalues in the Γ' -workspace are exactly the images of U and λ_i . Moreover,*

$$\sigma_{\mathcal{A}_{I, \Gamma'}}(\iota_* A) = \{\iota_* \lambda_i : i \in I\}.$$

In particular, enlargement of the value group creates neither new labeled eigenvalues nor additional algebraic spectral points for this simple-residue class.

Proof. An order embedding takes well-ordered supports to well-ordered supports and preserves the finiteness of coefficient incidences. Hence it commutes with all strong sums, products, divided commutators, and near-one binomial operations in the proof of theorem 18.2. The transported pair satisfies the same normalization, and uniqueness identifies it with the pair constructed in the larger workspace. Theorem 20.2 applied there, including to parameters $z \in K_{\Gamma'}$ outside the smaller field, proves the displayed equality. $\square$

This is an algebraic invariance statement. A scale extension can change whether a sequence of partial sums converges in the intrinsic valuation topology, as proposition 23.1 illustrates. The exact Hahn identities are stable even when such sequential behavior changes. Corollary 9.3 is the corresponding statement in Part I; it concerns a different construction and is proved separately, also without divisibility.

26 Effective coefficient calculation

26.1 What a certified computation needs

The proof gives an exact coefficient algorithm once the input presentation supplies the required finite data. To compute a coefficient at a weight γ , one needs a finite list of all ordered input-weight words summing to γ , the coefficient matrices on the finitely many vertices those words can visit, and exact arithmetic for the ordinary labels and edge coefficients. Theorem 21.2 then reduces the problem to finite rational operations and finite matrix jets.

For a one-scale locally finite graph and an integer degree n , the spatial part is particularly explicit: use the radius- $\lfloor n/2 \rfloor$ ball for an eigenvalue coefficient, and a sufficiently large radius- n ball for an eigenvector coefficient. Equations (18.3)–(18.5) compute the intermediate normalization, after which (18.7) computes the unitary normalization.

For an arbitrary abstract ordered group, the existence of finitely many weight decompositions does *not* provide an algorithm that finds them from an order-comparison oracle. Likewise, arbitrary complex coefficients need not have a decidable exact equality test. The theorem is therefore an exact mathematical construction with a stated computation contract, not a universal effective representation of all surreal numbers.

26.2 A coefficient-level description of the implementation

The Part II program uses exact SymPy matrices with rational and Gaussian rational entries. It implements finite multivariate Taylor jets, indexed by tuples $\alpha \in \mathbb{N}^r$ of bounded total degree. With $B = \sum_{\beta \neq 0} B_{\beta} z^{\beta}$, it computes

$$\begin{aligned} T_{\alpha} &= \sum_{0 < \beta \leq \alpha} B_{\beta} V_{\alpha-\beta} - \sum_{0 < \beta < \alpha} V_{\alpha-\beta} \Delta_{\beta}, \\ \Delta_{\alpha} &= \pi(T_{\alpha}), \quad V_{\alpha} = -\mathcal{R}_D(T_{\alpha}). \end{aligned}$$

The second sum means componentwise $\beta \leq \alpha$ with both β and $\alpha - \beta$ nonzero. It next forms $N = V^*V$ and evaluates the truncated binomial series for $N^{-1/2}$. All coefficients retained are checked against the full eigenidentity and both unitary identities.

The program does not numerically approximate eigenvalues. It also does not pretend to enumerate arbitrary Hahn supports or implement infinite matrices. The infinite claims rest on the proofs in this part; the code tests finite algebraic instances and the advertised coefficient formulas.

27 Part II: further questions and a formalization route

27.1 Beyond simple ordinary residue labels

The repeated-residue counterexample proposition 24.2 shows that infinite degeneracy cannot be ignored. A natural next class consists of diagonal residue blocks that are finite, perhaps with unbounded block sizes, together with an explicit hierarchy of positive splitting scales. A useful theorem would need a global support certificate ensuring that the block refinements and their unitary transformations remain coherent across all blocks. Finite-dimensional splitting by itself does not provide that uniform certificate.

Another direction replaces ordinary gaps by Hahn-valued gaps. Then dividing by $d_i - d_j$ can lower valuations by index-dependent amounts. A possible hypothesis would control the total gap losses along weighted walks so that the resulting effective weights are still well ordered and have finite word multiplicity. The present proof does not establish that extension: its divided commutator is support preserving exactly because the nonzero gaps are ordinary constants. Remark 18.4 is the concrete warning attached to this point.

27.2 Larger vector and operator categories

The vector space $\mathcal{H}_I$ admits only finite-support coefficients. Allowing ordinary ℓ^2 coefficients would include many more states, but real small divisors can then invalidate the recurrence, as proposition 24.1 shows. A weighted Hilbert-coefficient or Banach-coefficient Hahn category would have to replace row/column finiteness by analytic estimates compatible with the inverse commutator. Such an extension could connect the exact algebraic results to ordinary spectral approximation, but it is not a consequence of the current construction. Part I is a Hilbert-coefficient theory in that spirit, but for a different class of operators, and it does not supply the missing diagonalization theorem either. Part IV shows that in Part I's bounded-coefficient algebra the corresponding global statement is false even for a trace-class tridiagonal perturbation with simple residues (theorem 43.1), so an extension of this kind would need hypotheses beyond simple residues and small coefficient ideals.

Similarly, the marked-walk bounds identify first possible errors, not their noncancellation in every graph. It is worthwhile to classify graphs and signs for which the minimum escaping-walk weight always survives in the coefficient difference. The path formula (21.5) proves this only for a particular family.

27.3 A proof-assistant decomposition

A formalization can be organized without initially constructing the entire surreal class. The foundational objects are the set-sized algebra R_I , its involution and finite-support module, followed by their Hahn extensions. The essential reusable ingredients are: finite row/column support under products; the inner-product leading-term lemma; ordered-word summability; coefficientwise divided commutators; the homogeneous diagonalization recursion; and the uniqueness argument.

After those, normalization, synthesis, and commutant classification are short algebraic developments. The resolvent theorem needs a separate coherence lemma for infinitely many inverse diagonal entries, while locality requires an inductive relation between coefficient expressions and finite labeled walks. Only the final transport step needs a proved normal-form embedding into actual surreal numbers. No claim is made that these new infinite-dimensional statements have been added to, compiled in, or checked by the repository.

27.4 Conclusion of Part II

The principal result is not that every infinite Hermitian surcomplex matrix has a spectral theorem. That assertion is false in the category defined here, by proposition 24.2. The result is that simple ordinary residues and global positive Hahn support provide a complete, exact, infinite-dimensional spectral mechanism with unusually strong locality and extension properties. Its infinitesimal series are actual algebraic objects, its vector expansions are licensed sums, its resolvents are coherent operators, and its finite-section errors have verifiable scale thresholds.

Part III. Drazin halos: constant elements of an arbitrary Hahn operator algebra

Standing hypotheses for Part III. $\mathfrak{A}$ is a nonzero associative unital complex algebra, with the complex scalars central, and Γ is a nonzero set-sized ordered abelian group. Divisibility, a norm on $\mathfrak{A}$, normality and rank one are *not* assumed; only section 33 and proposition 34.8 add a Banach norm. “Spectrum” in this part means $\Sigma_{\Gamma}^{\mathfrak{A}}(T)$, failure of two-sided invertibility of $T - \lambda 1$ inside the named algebra $\mathfrak{A}_{\Gamma}$, for a *constant* $T \in \mathfrak{A}$ (definition 29.1). It is the third sense recorded in table 1. It is a spectrum relative to an algebra, like $\sigma_{\Gamma}^{\text{adj}}$ of Part I and $\sigma_{\mathcal{A}_r}$ of Part II, and section 3.3 proves that it contains both of them as special cases for constant elements. It is not a failure-of-bijectivity spectrum on a chosen vector space.

Notation. The source manuscript of this part wrote A for the algebra, D for a Drazin inverse, P, Q for its idempotents, B for the Drazin spectrum, V for the Volterra operator, h for a positive-order correction and r for a ramification index. Each of these letters already has a different meaning in Parts I and II, so this part writes instead $\mathfrak{A}$, S^D , P_{nil} , P_{inv} , Z , $\mathcal{V}$, κ and e_a . Its K_{Γ} and $\mathfrak{m}_{\Gamma}$ are the K and $\mathfrak{m}_K$ of this report.

28 The question, the answer, and its scope

28.1 Why nonnormal constant elements are the next step

The distinction between a nilpotent operator and a quasinilpotent but nonnilpotent operator is easy to blur at infinitesimal scales. Over $\mathbb{C}$, every nonzero scalar displacement of either operator can be invertible. Over a Hahn field, an exact inverse must satisfy a support condition. The nilpotent case requires only finitely many negative powers of the displacement. A nonnilpotent singularity can demand an infinite descending sequence of exponents, which is not a Hahn support. Small numerical size is not a replacement for this support requirement.

For constant *normal* operators, theorem 8.2 gives the ordinary spectrum together with infinitesimal halos of the ordinary accumulation points. This part asks what replaces accumulation points when normality is dropped and the Hilbert operator algebra is replaced by an arbitrary unital algebra, and exactly how a nondivisible value group affects polynomial spectral mapping. The answers are finite-index Drazin invertibility and a ramification test. In a Banach algebra, an isolated spectral point can acquire an entire infinitesimal halo when the ordinary resolvent singularity is not a finite pole. Conversely, every finite pole permits an inverse at every nonzero infinitesimal displacement, with an exact valuation determined by its order. These questions are formulated and answered here; they are not represented as named open conjectures already posed in published literature.

28.2 The theorem package at a glance

Write $\sigma_{\mathfrak{A}}(T)$ for the ordinary algebra-relative spectrum of $T \in \mathfrak{A}$ and $\sigma_{D, \mathfrak{A}}(T)$ for the set of $a \in \mathbb{C}$ such that $T - a1$ is not Drazin invertible (definition 30.1). The first main conclusion, theorem 32.1, is

$$\Sigma_{\Gamma}^{\mathfrak{A}}(T) = \sigma_{\mathfrak{A}}(T) \cup (\sigma_{D, \mathfrak{A}}(T) + \mathfrak{m}_K). \quad (28.1)$$

The added halos contain all infinitesimals, including nonmonomial ones with arbitrary well-ordered support. The inverse criterion is unchanged by enlarging the value group.

For a nonconstant $p \in \mathbb{C}[X]$, set $Z = \sigma_{D, \mathfrak{A}}(T)$. For $b \in p(Z)$ define the finite nonempty fiber and the ramification indices

$$Z_b = \{a \in Z : p(a) = b\}, \quad e_a = \text{ord}_{X=a}(p(X) - b).$$

The second conclusion, theorem 34.5, is the exact local image

$$p(\Sigma_{\Gamma}^{\mathfrak{A}}(T)) \cap (b + \mathfrak{m}_K) = \{b\} \cup \left\{ b + \zeta : 0 \neq \zeta \in \mathfrak{m}_K, v(\zeta) \in \bigcup_{a \in Z_b} e_a \Gamma \right\}. \quad (28.2)$$

In contrast, $\Sigma_{\Gamma}^{\mathfrak{A}}(p(T))$ contains the whole halo $b + \mathfrak{m}_K$. Spectral mapping therefore holds exactly when, in each relevant fiber, at least one of the integers e_a acts surjectively on Γ (theorem 34.7). This includes divisible groups but gives a sharp answer for every group.

28.3 Classical ingredients and the proposed novelty

Drazin inverses and their finite-index algebraic identities are classical [19]. Their relation to resolvent poles, their Banach-algebra spectra, and ordinary polynomial spectral mapping are also classical [20, 13]. Generalized Drazin invertibility, which permits a quasinilpotent singular part, is a different condition [21]; only the finite-index notion is used here.

The candidate contribution is the combined arbitrary-rank classification: straightening and descent for every infinitesimal in the original Hahn field, the resulting nonnormal halo formula, and the complete ramification and value-group description of the polynomial image. Independent coefficient proofs are given, but re-proving a classical Laurent identity does not make that identity new. The constant-normal case is an explicit overlap with Part I (remark 33.2), not a new claim. The Banach algebraicity criterion (corollary 33.3) is a Hahn interpretation of a classical Drazin-spectrum characterization. Appendix C records the source's repository and literature audit, and corrects the statements of it that are stale.

29 Hahn algebras and the third spectrum

29.1 The coefficient algebra is part of the statement

Set $\mathfrak{A}_{\Gamma} = \mathfrak{A}((t^{\Gamma}))$, with coefficientwise addition and convolution product; lemma 2.1(i) makes it a unital K -algebra with K central. No topology on $\mathfrak{A}$ is needed. For nonzero $x \in \mathfrak{A}_{\Gamma}$ write $v_{\mathfrak{A}}(x) = \min \text{supp } x$ and $\text{lc}_{\mathfrak{A}}(x) = x_{v_{\mathfrak{A}}(x)}$, and put $v_{\mathfrak{A}}(0) = +\infty$; on scalar series write v and lc . Since $\mathfrak{A}$ may have zero divisors, in general only $v_{\mathfrak{A}}(xy) \geq v_{\mathfrak{A}}(x) + v_{\mathfrak{A}}(y)$. For a nonzero central scalar $s \in K$ and nonzero $x \in \mathfrak{A}_{\Gamma}$, however,

$$v_{\mathfrak{A}}(sx) = v(s) + v_{\mathfrak{A}}(x), \quad \text{lc}_{\mathfrak{A}}(sx) = \text{lc}(s) \text{lc}_{\mathfrak{A}}(x), \quad (29.1)$$

because a nonzero complex number cannot annihilate a nonzero element of $\mathfrak{A}$. Every $s \in \mathcal{O}_K$ is uniquely $a + \varepsilon$ with $a = \text{st}(s) \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_K$. A scalar of negative valuation is called infinite; the *halo* of $a \in \mathbb{C}$ is $a + \mathfrak{m}_K$.

Definition 29.1 (The spectrum relative to a Hahn operator algebra). For a constant $T \in \mathfrak{A}$ put

$$\Sigma_{\Gamma}^{\mathfrak{A}}(T) = \{\lambda \in K : T - \lambda 1 \text{ has no two-sided inverse in } \mathfrak{A}_{\Gamma}\}.$$

The ordinary spectrum $\sigma_{\mathfrak{A}}(T)$ is defined by the same condition for $\lambda \in \mathbb{C}$ and inverses in $\mathfrak{A}$.

For $\mathfrak{A} = \mathcal{B}(H)$, convolution gives an action on $H((t^{\Gamma}))$, and $\mathfrak{A}_{\Gamma}$ is exactly the adjointable algebra $\mathcal{A}_{\Gamma}(H)$ of Part I. The definition is *not* silently replaced by failure of bijectivity in the algebra of all K -linear endomorphisms, or by the spectrum of row- and column-finite matrices on a finite-coordinate module; section 3.3 states exactly which identifications hold. In the examples of Volterra and unbounded Jordan lengths below, actual vector nonsurjectivity is proved directly.

29.2 The support calculus used

All infinite sums in this part are strong Hahn sums in the sense of section 2.1. The positive-support lemma is lemma 2.1; the source manuscript used it in the combined form: *if $S \subseteq \Gamma_{>0}$ and $B \subseteq \Gamma$ are well ordered, then $B + S^*$ is well ordered and each exponent has only finitely many representations as an element of B plus an ordered word in S* . This is lemma 2.1(i) applied to B and S^* , followed by lemma 2.1(ii) inside S^* ; it is classical, not a new result here. If $v_{\mathfrak{A}}(h) > 0$ then $\sum_{n \geq 0} h^n$ inverts $1 - h$ (lemma 2.3), and for a scalar h of positive order and $q \in \mathbb{Q}$ the binomial series $(1 + h)^q = \sum_n \binom{q}{n} h^n$ is strongly defined and obeys the exponent law (corollary 2.6).

Warning 29.2 (No hidden rank-one assumption). This part never assumes that $n\eta$ eventually exceeds every element of Γ for $\eta > 0$; that fails at higher rank. All infinite sums are justified by well ordering and coefficientwise local finiteness, not by convergence of finite subsums in a valuation topology.

30 Drazin decomposition and finite Laurent tails

Definition 30.1 (Finite-index Drazin invertibility). An element $S \in \mathfrak{A}$ is *Drazin invertible* if there exist $S^D \in \mathfrak{A}$ and an integer $\nu \geq 0$ with

$$SS^D = S^D S, \quad S^D S S^D = S^D, \quad S^{\nu+1} S^D = S^\nu. \quad (30.1)$$

The least such ν is the Drazin index; index zero means ordinary invertibility. The *Drazin spectrum* is $\sigma_{D,\mathfrak{A}}(T) = \{a \in \mathbb{C} : T - a1 \text{ is not Drazin invertible}\}$; it is contained in $\sigma_{\mathfrak{A}}(T)$.

Given the identities, put $P_{\text{nil}} = 1 - SS^D$ and $P_{\text{inv}} = SS^D$. Indeed, $(SS^D)^2 = S(S^D SS^D) = SS^D$, so these are complementary idempotents commuting with S . The second Drazin identity gives $S^D P_{\text{nil}} = P_{\text{nil}} S^D = 0$, and the third gives $S^\nu P_{\text{nil}} = 0$. In the corner algebra $P_{\text{inv}} \mathfrak{A} P_{\text{inv}}$, with identity P_{inv} , the element $S P_{\text{inv}}$ is invertible with inverse S^D . Thus P_{nil} selects the nilpotent part and P_{inv} the invertible part; the corresponding elements are $S P_{\text{nil}}$ and $S P_{\text{inv}}$. Either corner may be zero. If $\nu \geq 1$ is least, then $S^{\nu-1} P_{\text{nil}} \neq 0$: otherwise $S^\nu S^D = S^{\nu-1}$ would lower the index by one. No centrality of P_{nil} in the whole algebra is required.

Let u be a central formal variable; the ordinary Laurent algebra $\mathfrak{A}((u))$ permits only finitely many negative integer exponents.

Lemma 30.2 (Laurent inverse criterion). *For $S \in \mathfrak{A}$, the pencil $S - u1$ is invertible in $\mathfrak{A}((u))$ if and only if S is Drazin invertible. If S^D and $P_{\text{nil}} = 1 - SS^D$ are its data of index ν , then*

$$(S - u1)^{-1} = \sum_{n \geq 0} u^n (S^D)^{n+1} - \sum_{j=0}^{\nu-1} u^{-j-1} S^j P_{\text{nil}}. \quad (30.2)$$

The second sum is empty for $\nu = 0$; otherwise its exact pole order is ν .

Proof. Assume the Drazin identities. Multiplying the first sum by $S - u1$ on either side gives P_{inv} : its constant term is SS^D and the higher coefficients cancel. Multiplying the negative sum gives $P_{\text{nil}} - u^{-\nu} S^\nu P_{\text{nil}} = P_{\text{nil}}$. The combined result is 1. Its coefficient at $u^{-\nu}$ is $-S^{\nu-1} P_{\text{nil}} \neq 0$ when $\nu \geq 1$ is least.

Conversely, suppose $B(u) = \sum_{n \geq -m} B_n u^n$ is a two-sided inverse, with $m \geq 0$ and absent coefficients set to zero. Coefficient comparison in both products gives

$$S B_n - B_{n-1} = \delta_{n0} 1 = B_n S - B_{n-1} \quad (n \in \mathbb{Z}). \quad (30.3)$$

In particular each B_n commutes with S . Set $S^D = B_0$ and $P_{\text{nil}} = -B_{-1} = 1 - SS^D$. If $m = 0$, then $SS^D = S^D S = 1$. If $m \geq 1$, the negative recurrences, including the finite lower bound,

give $S^m P_{\text{nil}} = 0$, and the positive recurrences give $S^D = S^m B_m$. Since P_{nil} commutes with S , this gives $P_{\text{nil}} S^D = 0$; the definition of P_{nil} and commutation give $S^D P_{\text{nil}} = 0$. More explicitly, $P_{\text{nil}} B_0 = P_{\text{nil}} S^m B_m = 0$ and $B_0 P_{\text{nil}} = P_{\text{nil}} B_0$, since B_0 commutes with S . It follows that $P_{\text{nil}}^2 = P_{\text{nil}}$, $S^D S S^D = S^D$ and $S^{m+1} S^D = S^m$, which are Drazin identities. Any Drazin data give (30.2); uniqueness of a two-sided inverse therefore identifies its constant coefficient with S^D and proves uniqueness of the Drazin inverse. $\square$

This is the classical finite-Laurent mechanism, written out to expose the exact obstruction that arbitrary Hahn supports must not be allowed to conceal. No complex-analytic convergence was used.

31 Straightening and inverse descent at arbitrary rank

Lemma 31.1 (Strong parameter straightening). *Given $0 \neq \varepsilon \in \mathfrak{m}_K$ and $\eta = v(\varepsilon)$, there is a strong, valuation-preserving $\mathbb{C}$ -algebra automorphism Φ of K with $\Phi(t^\eta) = \varepsilon$. The same construction is an $\mathfrak{A}$ -fixing automorphism of $\mathfrak{A}_\Gamma$. No divisibility or enlargement of Γ is required.*

Proof. Write $\varepsilon = c t^\eta (1 + \kappa)$ with $c \in \mathbb{C}^*$ and $v(\kappa) > 0$. Since Γ is torsion-free, it embeds in $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$. Choose a rational linear functional ℓ on that vector space with $\ell(\eta) = 1$; its restriction to Γ is a homomorphism to $\mathbb{Q}$, not necessarily order-preserving. Choose $L \in \mathbb{C}$ with $e^L = c$ and set $c^q = e^{qL}$ for $q \in \mathbb{Q}$; these are compatible rational powers of the one constant c . Define on monomials

$$\Phi(t^\gamma) = c^{\ell(\gamma)} t^\gamma (1 + \kappa)^{\ell(\gamma)}, \quad (31.1)$$

and extend coefficientwise using strong sums. Let $S = \text{supp } \kappa$. For an input support B , every resulting exponent lies in $B + S^*$, so by the combined support lemma of section 29.2 each output coefficient has finitely many contributors. The same bound works simultaneously for any strongly summable family of inputs, so the map is well defined and strong, including for uncountable supports. Additivity of ℓ and the binomial exponent law make it multiplicative on monomials, and the same support bounds justify all rearrangements, hence multiplication of arbitrary series. This remains true for noncommuting coefficients in $\mathfrak{A}$, since every binomial factor is central. The leading coefficient of a nonzero input x is multiplied by $c^{\ell(v_{\mathfrak{A}}(x))} \neq 0$, so valuation is preserved.

Surjectivity needs an argument: a triangular-looking map is not surjective merely because its leading terms are nonzero. Write $\Phi = \Psi \circ C_c$, where $C_c(t^\gamma) = c^{\ell(\gamma)} t^\gamma$ has an evident inverse and $\Psi(t^\gamma) = t^\gamma (1 + \kappa)^{\ell(\gamma)}$. Put $\Xi = \Psi - \text{id}$. One application of Ξ shifts a starting exponent by a sum of at least one member of S , so a term of $\Xi^n(x)$ is shifted by a word of length at least n in S . Hence

$$\Theta(x) = \sum_{n \geq 0} (-\Xi)^n(x) \quad (31.2)$$

is a strong sum: at a fixed final exponent there are finitely many starting exponents and words, and each word has only finitely many partitions into nonempty blocks, so only finitely many iterations and intermediate terms contribute. The same reasoning shows that Θ preserves strongly summable families. Coefficientwise telescoping gives $(\text{id} + \Xi)\Theta = \Theta(\text{id} + \Xi) = \text{id}$. Hence Ψ is bijective, and its inverse is an algebra homomorphism. If $\kappa = 0$, take $\Xi = 0$. Finally (31.1) sends t^η to $c t^\eta (1 + \kappa) = \varepsilon$ and fixes every coefficient in $\mathfrak{A}$. $\square$

The rational functional chooses a coordinate on the rational span of the exponents; it does not adjoin any fractional exponent to the field. The binomial corrections use only exponents already generated by $\text{supp } \kappa$. This is why nondivisibility causes no obstruction to straightening a parameter that already exists. Related strong Hahn-automorphism constructions belong to the surrounding literature [23]; the general idea is not claimed as new here.

Lemma 31.2 (Cyclic-support inverse descent). *For $S \in \mathfrak{A}$ and $0 < \eta \in \Gamma$, put $u = t^\eta$. The pencil $S - u1$ is invertible in $\mathfrak{A}_\Gamma$ if and only if it is invertible in $\mathfrak{A}((u))$. If it is invertible, its inverse in $\mathfrak{A}_\Gamma$ already belongs to the embedded Laurent subalgebra.*

Proof. Retain the coefficients on the cyclic subgroup $\mathbb{Z}\eta$: $\varpi_\eta(\sum B_\gamma t^\gamma) = \sum_{n \in \mathbb{Z}} B_{n\eta} u^n$. The intersection of a well-ordered support with $\mathbb{Z}\eta$ is well ordered and becomes, under $n \mapsto n\eta$, a subset of $\mathbb{Z}$ with a finite lower bound; thus $\varpi_\eta B \in \mathfrak{A}((u))$. Because the pencil is supported on $\mathbb{Z}\eta$, coefficient comparison gives $\varpi_\eta((S - u1)B) = (S - u1)\varpi_\eta B$ and $\varpi_\eta(B(S - u1)) = (\varpi_\eta B)(S - u1)$. A two-sided inverse therefore projects to a two-sided Laurent inverse. Conversely a Laurent inverse embeds into $\mathfrak{A}_\Gamma$, since a lower-bounded set of integer multiples of a positive element is well ordered. Uniqueness of a two-sided inverse identifies it with its projection. $\square$

The map ϖ_η is not generally a ring homomorphism: if $\eta/2 \in \Gamma$, it sends $t^{\eta/2}$ to zero but its square to u . The proof used only left and right compatibility with the particular cyclically supported pencil, and no claim that $n\eta$ is cofinal in Γ . The restriction to $\mathbb{Z}\eta$ is essential to the finite negative Laurent tail: in lexicographic $\mathbb{Z} \times \mathbb{Z}$, for example, the well-ordered set $\{(-1, n) : n \geq 0\}$ contains infinitely many negative exponents. General Hahn supports need not have only finitely many negative terms.

Theorem 31.3 (Arbitrary-parameter inverse descent). *For $S \in \mathfrak{A}$ and $0 \neq \varepsilon \in \mathfrak{m}_K$, the following are equivalent:*

- (i) $S - \varepsilon 1$ has a two-sided inverse in $\mathfrak{A}_\Gamma$;
- (ii) its inverse is a Laurent series in ε with coefficients in $\mathfrak{A}$ and only finitely many negative powers;
- (iii) S is Drazin invertible in $\mathfrak{A}$.

When they hold, the unique inverse is

$$(S - \varepsilon 1)^{-1} = \sum_{n \geq 0} \varepsilon^n (S^D)^{n+1} - \sum_{j=0}^{\nu-1} \varepsilon^{-j-1} S^j P_{\text{nil}}. \quad (31.3)$$

Proof. Apply lemma 31.1 with $\eta = v(\varepsilon)$. It fixes S and takes t^η to ε , so it preserves the invertibility question. Lemma 31.2 descends an inverse of $S - t^\eta 1$ to its Laurent subalgebra, and lemma 30.2 identifies the condition as Drazin invertibility and supplies the inverse. Applying the strong automorphism to that formula gives (31.3); its nonnegative part is a strong sum by the support lemma and its negative part is finite. $\square$

The conclusion rules out more than one unsuccessful expansion: no inverse with exotic extra exponents can exist when the finite-Laurent obstruction remains. Although the straightening automorphism involved choices, existence and the inverse formula are intrinsic.

32 Spectral halos and exact resolvent scales

Theorem 32.1 (Constant-element spectrum). *For every $T \in \mathfrak{A}$,*

$$\Sigma_\Gamma^\mathfrak{A}(T) = \sigma_\mathfrak{A}(T) \cup (\sigma_{D,\mathfrak{A}}(T) + \mathfrak{m}_K). \quad (32.1)$$

No infinite scalar is spectral. A constant $a \in \mathbb{C}$ is spectral exactly when $a \in \sigma_\mathfrak{A}(T)$. For $0 \neq \varepsilon \in \mathfrak{m}_K$, the scalar $a + \varepsilon$ is spectral exactly when $a \in \sigma_{D,\mathfrak{A}}(T)$.

Proof. If $v(\lambda) < 0$, then $\lambda^{-1}T$ has positive order unless $T = 0$, and the strong geometric series gives

$$(T - \lambda 1)^{-1} = -\lambda^{-1} \sum_{n \geq 0} \lambda^{-n} T^n; \quad (32.2)$$

for $T = 0$ the formula has just its first term. If $\lambda = a \in \mathbb{C}$ and B is a two-sided Hahn inverse, its constant coefficient satisfies $(T - a1)B_0 = B_0(T - a1) = 1$, because $T - a1$ is constant; so the ordinary invertibility question is unchanged by Hahn extension. Every other scalar is uniquely $a + \varepsilon$ with $0 \neq \varepsilon \in \mathfrak{m}_K$, and theorem 31.3 applied to $S = T - a1$ decides exactly this case. Since $\sigma_{D, \mathfrak{A}}(T) \subseteq \sigma_{\mathfrak{A}}(T)$, the halo centres are included correctly. $\square$

Theorem 32.2 (Exact valuation and leading coefficient). *Suppose $S = T - a1$ is Drazin invertible of index $\nu \geq 1$. For $0 \neq \varepsilon \in \mathfrak{m}_K$ put $\eta = v(\varepsilon)$ and $c = \text{lc}(\varepsilon)$. Then*

$$v_{\mathfrak{A}}((T - a1 - \varepsilon 1)^{-1}) = -\nu\eta, \quad \text{lc}_{\mathfrak{A}}((T - a1 - \varepsilon 1)^{-1}) = -c^{-\nu} S^{\nu-1} P_{\text{nil}}. \quad (32.3)$$

For index zero the inverse has valuation zero and leading coefficient S^{-1} . For infinite λ the inverse has valuation $-v(\lambda)$ and leading coefficient $-\text{lc}(\lambda)^{-1}1$.

Proof. In (31.3) the last nonzero negative term is $-\varepsilon^{-\nu} S^{\nu-1} P_{\text{nil}}$, of valuation $-\nu\eta$ and with the stated nonzero leading coefficient; every other term has strictly larger valuation, so no cancellation can affect it. The other cases follow from the initial term of the corresponding geometric series. $\square$

This is a valuation of an operator-valued Hahn series, not a statement about the least field-valued operator-norm bound of section 13. The integer ν measures the largest nilpotent chain in the singular part. Isolation of a spectral point does not by itself determine it.

Corollary 32.3 (Workspace invariance). *Let $\Gamma \hookrightarrow \Lambda$ be an order-preserving group embedding. For every constant $T \in \mathfrak{A}$ and every $\lambda \in K_{\Gamma}$, invertibility of $T - \lambda 1$ in $\mathfrak{A}((t^{\Gamma}))$ is equivalent to invertibility of its image in $\mathfrak{A}((t^{\Lambda}))$; when an inverse exists, the latter is the image of the former.*

Proof. The embedding preserves the sign of the valuation, the constant coefficient, and nonzeroness of the infinitesimal part, hence every case of theorem 32.1. The explicit inverse formulas already exist in the smaller workspace; uniqueness identifies them after extension. $\square$

Adjoining fractional exponents cannot repair a non-Drazin constant pencil. It can repair a scalar root problem; section 34 shows why these two facts are compatible with a change in the image of a spectrum under a polynomial. Corollary 9.3 and proposition 25.2 are the corresponding statements of Parts I and II. Invariance concerns the status of each old scalar: the halos themselves grow with the field, and corollary 42.3 of Part IV makes this explicit for a nonconstant operator, whose eigenvalues do not change while its non-eigenvalue monad grows.

33 Banach algebras: poles, nonnormality and algebraicity

This section adds a Banach norm on $\mathfrak{A}$; the preceding results did not need one. In a Banach algebra, finite-index Drazin invertibility has a familiar analytic meaning: a spectral point is Drazin invertible exactly when it is a finite pole of the ordinary resolvent [20, 13].

To see the mechanism, take $S = T - a1$ with Drazin inverse S^D and use the local resolvent convention $R(u) = (S - u1)^{-1}$. The positive part of (30.2) converges in norm for sufficiently small complex u , being a geometric series in uS^D ; its negative part is finite. It supplies the ordinary resolvent on a punctured disk, with pole order equal to the Drazin index if a is spectral. Conversely, the coefficients of a finite-pole Laurent expansion of the analytic resolvent satisfy

(30.3), so its constant coefficient is a Drazin inverse. With the opposite convention $(u1 - S)^{-1}$ that coefficient would instead be $-S^D$. No accumulation point of the ordinary spectrum can be a finite pole, and an isolated spectral singularity cannot be removable, since continuity of the inverse identity would then invert $T - a1$. Consequently

$$\sigma_{D,\mathfrak{A}}(T) = \text{acc } \sigma_{\mathfrak{A}}(T) \cup \{a \in \sigma_{\mathfrak{A}}(T) : a \text{ is isolated and the resolvent has no finite pole at } a\}. \quad (33.1)$$

Here acc is the ordinary accumulation set, written $\sigma_{\mathbb{C}}(T)'$ in Part I. For a normal operator an isolated spectral point is a simple pole. Without normality, an isolated essential resolvent singularity is an additional source of a whole infinitesimal halo. Generalized Drazin invertibility is not an adequate substitute: it permits a quasinilpotent singular corner [21], whereas theorem 31.3 requires that corner to be nilpotent of *finite* index. The Volterra example of section 35.2 distinguishes them.

Corollary 33.1 (Recovery of the normal case). *For a bounded normal operator T on a nonzero ordinary complex Hilbert space H , with $\mathfrak{A} = \mathcal{B}(H)$,*

$$\Sigma_{\Gamma}^{\mathcal{B}(H)}(T) = \sigma_{\mathbb{C}}(T) \cup (\sigma_{\mathbb{C}}(T)' + \mathfrak{m}_K).$$

Every isolated ordinary spectral point has Drazin index one, regardless of the dimension of its eigenspace.

Proof. The ordinary spectral theorem decomposes the space at an isolated normal spectral point into its eigenspace and orthogonal complement; on the former $T - aI$ is zero, on the latter boundedly invertible (lemma 8.1). Apply (33.1) and theorem 32.1. $\square$

Remark 33.2 (What is recovered from Part I, and which hypotheses are dropped). Since $\mathcal{B}(H)((t^{\Gamma})) = \mathcal{A}_{\Gamma}(H)$, the left side of corollary 33.1 is exactly $\sigma_{\Gamma}^{\text{adj}}(T)$ of section 8.1. The corollary is therefore the σ^{adj} half of theorem 8.2, obtained by a different route: straightening and Laurent descent instead of the defect splitting. At a Drazin point of index one, (31.3) with $S = T - aI$ reads $-\varepsilon^{-1}P_{\text{nil}} + \sum_{n \geq 0} \varepsilon^n (S^D)^{n+1}$; with the notation of lemma 8.1, $P_{\text{nil}} = P$ and S^D is the inverse of $(T - aI)|_M$ composed with Q , and this is the isolated-point inverse (8.4). Relative to that theorem, theorem 32.1 drops *normality* (the accumulation set is replaced by the Drazin spectrum) and the *Hilbert and Banach structure* (any unital complex algebra; a norm enters only to identify $\sigma_{D,\mathfrak{A}}$ through (33.1)). It does *not* drop divisibility relative to the present Part I, which has not assumed divisibility since the revision recorded in section 5.1; the source manuscript's claim to drop it refers to an earlier state of Part I (appendix C.3).

Two parts of theorem 8.2 are *not* recovered. The first is the failure-of-bijectivity spectrum $\sigma_{\Gamma}^{\text{alg}}(T)$ on $\mathcal{H}_{\Gamma}$: Part III is a statement about invertibility in an algebra and says nothing about bijectivity on a vector space, except in its explicitly worked examples (the Volterra operator of section 35.2, and the block operator of section 35.4, both at every infinitesimal displacement). The second is point-spectrum rigidity, theorem 8.3. Theorem 8.2 proves $\sigma_{\Gamma}^{\text{alg}}(T) = \sigma_{\Gamma}^{\text{adj}}(T)$ for constant normal T ; together with corollary 33.1 this gives the full normal formula by two proofs of its algebra half. For nonnormal T one always has $\sigma_{\Gamma}^{\text{alg}}(T) \subseteq \Sigma_{\Gamma}^{\mathcal{B}(H)}(T)$, since an inverse in $\mathcal{A}_{\Gamma}(H)$ is a bijection; equality is not asserted in general (section 37.3).

Corollary 33.3 (Algebraicity criterion). *For an element T of a Banach algebra $\mathfrak{A}$ the following are equivalent:*

- (i) $\Sigma_{\Gamma}^{\mathfrak{A}}(T) \subseteq \mathbb{C}$;
- (ii) $\sigma_{D,\mathfrak{A}}(T) = \emptyset$;
- (iii) T is algebraic over $\mathbb{C}$.

In that case $\Sigma_{\Gamma}^{\mathfrak{A}}(T) = \sigma_{\mathfrak{A}}(T)$ is finite.

Proof. The first equivalence follows from theorem 32.1 and the existence of a positive element of $\Gamma \neq 0$. For completeness, the classical second equivalence can be proved as follows. If $\sigma_{D, \mathfrak{A}}(T)$ is empty, all points of the ordinary spectrum are isolated poles, so the compact spectrum is finite, say $\{a_1, \dots, a_s\}$. Its finite Riesz decomposition gives commuting complementary projections P_j with sum one and $(T - a_j 1)^{\nu_j} P_j = 0$ for the finite pole orders ν_j ; hence $\prod_j (T - a_j 1)^{\nu_j} = 0$. Conversely, factor an annihilating polynomial into powers of distinct linear factors. Polynomial Chinese remaindering gives complementary idempotents polynomial in T ; on each part T is a scalar plus a finite-index nilpotent, so every scalar shift is a direct sum of an invertible and a finite-index nilpotent part, hence Drazin invertible. The empty-Drazin-spectrum characterization itself is classical [13]; its exact Hahn spectral interpretation is the consequence asserted here. $\square$

The Banach assumption is essential for the algebraicity implication. For $\mathfrak{A} = \mathbb{C}(X)$ and $T = X$, every $T - a$ is invertible, both ordinary and Drazin spectra are empty, yet T is transcendental over $\mathbb{C}$. No algebraicity claim is made for arbitrary unital complex algebras. For an ordinary bounded self-adjoint T , being algebraic is the same as having finite spectrum, so corollary 33.3 is consistent with the spectral-reality criterion corollary 9.1 of Part I; the two statements concern different spectra (σ^{adj} here, σ^{alg} there), which agree for constant normal operators by theorem 8.2.

34 Ramified polynomial spectral mapping

We return to an arbitrary nonzero unital complex algebra $\mathfrak{A}$ and a nonconstant $p \in \mathbb{C}[X]$. A nondivisible value group obstructs scalar roots; the following lemmas turn that obstruction into an exact spectral-mapping theorem.

Lemma 34.1 (Power image). *For every integer $e \geq 1$,*

$$\{x^e : x \in \mathfrak{m}_K\} = \{0\} \cup \{\zeta \in \mathfrak{m}_K \setminus \{0\} : v(\zeta) \in e\Gamma\}. \quad (34.1)$$

Proof. Necessity follows from $v(x^e) = ev(x)$. Conversely write $\zeta = dt^{e\gamma}(1+k)$ with $d \in \mathbb{C}^*$ and $v(k) > 0$. Since $e\gamma > 0$, also $\gamma > 0$. Choose a complex root $d^{1/e}$ and put $x = d^{1/e}t^\gamma(1+k)^{1/e}$, by the strong binomial series. Then $x \in \mathfrak{m}_K$ and $x^e = \zeta$. $\square$

Lemma 34.2 (Exact local polynomial image). *Suppose $p(a) = b$ with $a, b \in \mathbb{C}$, and put $e = \text{ord}_{X=a}(p(X) - b)$. There is a bijection $\varphi_a : \mathfrak{m}_K \rightarrow \mathfrak{m}_K$, induced by a formal complex power series with nonzero linear coefficient, such that $p(a+x) - b = \varphi_a(x)^e$ for $x \in \mathfrak{m}_K$. Consequently $p(a + \mathfrak{m}_K) - b = \{x^e : x \in \mathfrak{m}_K\}$.*

Proof. Factor $p(a+X) - b = cX^e(1 + Xg(X))$ with $c \in \mathbb{C}^*$ and $g \in \mathbb{C}[X]$. Choose $c^{1/e}$ and put $\varphi_a(X) = c^{1/e}X(1 + Xg(X))^{1/e} \in X\mathbb{C}[[X]]$. Its nonzero linear coefficient gives a unique formal compositional inverse $\psi_a \in X\mathbb{C}[[X]]$: at each degree the next unknown coefficient is multiplied by that nonzero linear coefficient and all other terms are already determined. Every series and composition in these identities can be evaluated at any element of $\mathfrak{m}_K$ by corollary 2.6; the values stay in $\mathfrak{m}_K$, and the identities make the resulting maps mutually inverse on the whole of $\mathfrak{m}_K$. The defining e th-power identity proves the claim. No ordinary convergence radius was used. For $x \neq 0$ the nonzero linear coefficient is the unique leading term in each substitution, so $v(\varphi_a(x)) = v(x) = v(\psi_a(x))$. In particular, both substitutions preserve nonzeroness as well as positive valuation. $\square$

A ramification index is thus an exact condition on the value group, not an approximate estimate along one selected scale; the leading coefficient causes no additional obstruction because $\mathbb{C}$ contains its roots.

Lemma 34.3 (Commuting factors). *A finite product of commuting elements in a unital algebra is a unit if and only if every factor is a unit.*

Proof. If the product q is a unit, each factor commutes with q and hence with q^{-1} , and the product of all other factors times q^{-1} is both a left and a right inverse of that factor. The converse is immediate. $\square$

Proposition 34.4 (Ordinary polynomial mapping). *For every nonconstant $p \in \mathbb{C}[X]$,*

$$\sigma_{\mathfrak{A}}(p(T)) = p(\sigma_{\mathfrak{A}}(T)), \quad \sigma_{D, \mathfrak{A}}(p(T)) = p(\sigma_{D, \mathfrak{A}}(T)). \quad (34.2)$$

Proof. Factor $p(X) - b$ over $\mathbb{C}$ and apply lemma 34.3 at T for the first equality. For the second, work temporarily in the auxiliary field $\mathbb{C}((u^{\mathbb{Q}}))$; this imposes no divisibility assumption on the workspace of the later theorems. Let $a_1, \dots, a_s$ be the distinct roots of $p(X) - b$, with multiplicities $e_1, \dots, e_s$. For every e_j th root of unity ξ , lemma 34.2 gives a root $\alpha_{j, \xi} = a_j + \psi_{a_j}(\xi u^{1/e_j})$ of $p(X) - b - u$. These roots are distinct: different a_j have different residues, and for the same a_j the series ψ_{a_j} is injective on infinitesimals. Their number is $\sum_j e_j = \deg p$, so with c_p the leading coefficient of p ,

$$p(X) - b - u = c_p \prod_{j=1}^s \prod_{\xi^{e_j}=1} (X - \alpha_{j, \xi}).$$

After evaluation at T all factors commute. By theorem 31.3, a factor is a unit exactly when $T - a_j 1$ is Drazin invertible, and the product is a unit exactly when $p(T) - b 1$ is Drazin invertible. The commuting-factor lemma proves that this happens exactly when every $T - a_j 1$ is Drazin invertible, which is the second equality. $\square$

Theorem 34.5 (Exact ramified spectral image). *Let $Z = \sigma_{D, \mathfrak{A}}(T)$ and, for $b \in p(Z)$, let Z_b and e_a be as in section 28.2; these multiplicities refer to the polynomial germ, not to the multiplicity of an operator eigenvalue. Then for $b \in p(Z)$,*

$$p(\Sigma_{\Gamma}^{\mathfrak{A}}(T)) \cap (b + \mathfrak{m}_K) = \{b\} \cup \left\{ b + \zeta : 0 \neq \zeta \in \mathfrak{m}_K, v(\zeta) \in \bigcup_{a \in Z_b} e_a \Gamma \right\}. \quad (34.3)$$

For $b \in p(\sigma_{\mathfrak{A}}(T)) \setminus p(Z)$ this intersection is $\{b\}$; for $b \notin p(\sigma_{\mathfrak{A}}(T))$ it is empty. In contrast,

$$\Sigma_{\Gamma}^{\mathfrak{A}}(p(T)) = p(\sigma_{\mathfrak{A}}(T)) \cup (p(Z) + \mathfrak{m}_K). \quad (34.4)$$

In particular $p(\Sigma_{\Gamma}^{\mathfrak{A}}(T)) \subseteq \Sigma_{\Gamma}^{\mathfrak{A}}(p(T))$, and these formulas describe every possible omission from the polynomial image.

Proof. By theorem 32.1, a spectral point is either an ordinary point of $\sigma_{\mathfrak{A}}(T)$ or a member of a halo $a + \mathfrak{m}_K$ with $a \in Z$. The standard part of $p(a + \varepsilon)$ is $p(a)$, so contributions to a halo $b + \mathfrak{m}_K$ come only from its ordinary fibre. For each $a \in Z_b$, lemmas 34.1 and 34.2 give exactly the condition $v(\zeta) \in e_a \Gamma$. Their finite union proves (34.3), including the cases with no Drazin-spectral preimage. Finally apply theorem 32.1 to $p(T)$ and substitute proposition 34.4. $\square$

Lemma 34.6 (Finite unions of integer-multiple subgroups). *For any abelian group G and positive integers $e_1, \dots, e_s$ with $s \geq 1$, one has $G = \bigcup_j e_j G$ if and only if $e_j G = G$ for some j . If G is ordered and the union is proper, it omits a positive element.*

Proof. Assume every $e_j G$ is proper. Choose a prime divisor p_j of e_j with $p_j G \neq G$; one exists, since surjectivity of multiplication by all prime factors would give surjectivity of multiplication by e_j . Let $\mathcal{P}$ be the finite set of chosen primes and choose $x_\ell \notin \ell G$ for each $\ell \in \mathcal{P}$. By the integer Chinese remainder theorem choose $c_\ell \equiv 1 \pmod{\ell}$ and $c_\ell \equiv 0 \pmod{q}$ for the other $q \in \mathcal{P}$. Then $x = \sum_{\ell \in \mathcal{P}} c_\ell x_\ell$ has the same class as x_ℓ modulo ℓG , so lies in none of the ℓG and hence in none

of the $e_j G$. The converse is immediate. In an ordered group $x \neq 0$, and one of $x, -x$ is positive and still outside every subgroup in the union. $\square$

Theorem 34.7 (Sharp spectral-mapping criterion). *For a nonconstant $p \in \mathbb{C}[X]$, the equality*

$$\Sigma_\Gamma^\mathfrak{A}(p(T)) = p(\Sigma_\Gamma^\mathfrak{A}(T)) \quad (34.5)$$

holds if and only if, for every $b \in p(\sigma_{\mathbb{D}, \mathfrak{A}}(T))$, at least one $a \in Z_b$ has $e_a \Gamma = \Gamma$. If the condition fails at b , there is a positive $\delta \in \Gamma$ with

$$b + t^\delta \in \Sigma_\Gamma^\mathfrak{A}(p(T)) \setminus p(\Sigma_\Gamma^\mathfrak{A}(T)). \quad (34.6)$$

Proof. Theorem 34.5 says equality holds exactly when $\bigcup_{a \in Z_b} e_a \Gamma$ contains every positive element for every relevant b ; as a union of subgroups it then contains all of Γ . Lemma 34.6 gives exactly the stated condition, and if the union is proper its positive omitted element supplies (34.6). $\square$

For divisible Γ the criterion always holds. More generally, an unramified Drazin-spectral preimage, $e_a = 1$, suffices for its fibre whatever the group. For $\Gamma = \mathbb{Z}$ every relevant fibre must contain such an unramified preimage; for $\Gamma = \mathbb{Z}[1/2]$, ramification of order a power of two creates no obstruction, but an exclusively order-three fibre does. If T is algebraic, its Drazin spectrum is empty and the criterion is vacuous, so finite matrices alone cannot exhibit the nontrivial halo obstruction. The obstruction couples a non-Drazin singularity with insufficient scalar roots in the chosen workspace. It is not a failure of ordinary complex polynomial spectral mapping, and it is not a claimed failure over the full surcomplex field (corollary 36.2).

Proposition 34.8 (Pole-order profile). *Let $\mathfrak{A}$ be a Banach algebra and $b \in p(\sigma_\mathfrak{A}(T)) \setminus p(\sigma_{\mathbb{D}, \mathfrak{A}}(T))$. For each spectral a with $p(a) = b$ let ν_a be its finite pole order and $e_a = \text{ord}_{X=a}(p(X) - b)$. The Drazin index of $p(T) - b1$ is*

$$\nu_b = \max_{a \in \sigma_\mathfrak{A}(T), p(a)=b} \left\lceil \frac{\nu_a}{e_a} \right\rceil, \quad (34.7)$$

and its inverse at a displacement $0 \neq \varepsilon \in \mathfrak{m}_K$ has exact valuation $-\nu_b v(\varepsilon)$.

Proof. The indicated spectral preimages are finite poles. Their finite Riesz decomposition separates them from a part on which $p(T) - b1$ is invertible. More precisely, let P_a be the corresponding Riesz idempotents and work in the corner $P_a \mathfrak{A} P_a$, whose identity is P_a . There $TP_a = aP_a + N_a$ with $N_a = (T - a1)P_a$ of nilpotency index ν_a . Thus $(p(T) - b1)P_a = N_a^{e_a} w_a(N_a)$, where $w_a(0) \neq 0$ and the polynomial w_a is evaluated with corner identity P_a ; the factor $w_a(N_a)$ is invertible and commutes with N_a , so the k th power vanishes exactly when $N_a^{e_a k}$ does, giving index $\lceil \nu_a / e_a \rceil$. The index of a finite direct sum is the maximum. The complementary corner, if nonzero, has index zero; the nonempty spectral fibre ensures $\nu_b \geq 1$. Apply theorem 32.2 for the valuation. $\square$

The ceiling formula is an elementary nilpotent-algebra fact, not claimed as new by itself; here it gives the exact resolvent exponent.

35 Examples separating poles, halos and scalar-root defects

35.1 Finite Jordan blocks and infinite-rank nilpotents

Let J_ν be the nilpotent $\nu \times \nu$ Jordan matrix with ones on the superdiagonal. For $T = aI + J_\nu$,

$$\Sigma_\Gamma^\mathfrak{A}(T) = \{a\}, \quad (T - aI - \varepsilon I)^{-1} = - \sum_{j=0}^{\nu-1} \varepsilon^{-j-1} J_\nu^j.$$

Every nonzero infinitesimal displacement is invertible, although the inverse has valuation $-\nu v(\varepsilon)$: a large inverse and a spectral value are different notions. For example J_5^2 has nilpotency index three, and its inverse at displacement ε has valuation $-3v(\varepsilon)$, not $-5v(\varepsilon)$ or $-\frac{5}{2}v(\varepsilon)$.

Infinite rank alone does not force a halo. On $H \oplus H$ let $N = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}$ with X any bounded operator, possibly compact of infinite rank. Then $N^2 = 0$, so $\Sigma_\Gamma^{\mathcal{B}(H)}(N) = \{0\}$ and its shifted inverse has only finitely many negative powers.

35.2 Volterra: an isolated point that acquires a whole halo

Let $H = L^2([0, 1], \mathbb{C})$ and $(\mathcal{V}f)(x) = \int_0^x f(s) ds$. The operator is injective; its range consists exactly of the L^2 classes with an absolutely continuous representative g satisfying $g(0) = 0$ and $g' \in L^2([0, 1])$. Indeed, an indefinite integral has those properties, and conversely $g = \mathcal{V}g'$ almost everywhere. The constant function 1 has no such representative and is not in the range. Repeated integration gives

$$(\mathcal{V}^n f)(x) = \frac{1}{(n-1)!} \int_0^x (x-s)^{n-1} f(s) ds \quad (n \geq 1), \quad (35.1)$$

and the squared Hilbert–Schmidt norm of this kernel is $1/((n-1)!)^2(2n-1)(2n) \leq 1/(n!)^2$. Thus $\|\mathcal{V}^n\| \leq 1/n!$, and for every ordinary $z \neq 0$ the series $-\sum_{n \geq 0} z^{-n-1} \mathcal{V}^n$ converges in operator norm and inverts $\mathcal{V} - zI$. Since $\mathcal{V}$ is not onto, its ordinary spectrum is exactly $\{0\}$.

Nevertheless $\mathcal{V}^n 1 = x^n/n! \neq 0$ for every n , so $\mathcal{V}$ is not nilpotent. It is not Drazin invertible either: if S^D were a Drazin inverse, then $P_{\text{inv}} = \mathcal{V}^n (S^D)^n$ for every $n \geq 1$, of norm at most $\|S^D\|^n/n! \rightarrow 0$; hence $P_{\text{inv}} = 0$ and $\mathcal{V}$ would be nilpotent. Theorem 32.1 gives

$$\Sigma_\Gamma^{\mathcal{B}(H)}(\mathcal{V}) = \mathfrak{m}_K. \quad (35.2)$$

The ordinary spectrum has no accumulation points at all, so an accumulation-only formula is false without normality; this is the nonnormal counterpart of example 8.4, which exhibits the failure of point-spectrum rigidity rather than of the spectral formula.

In this example the spectral obstruction is also genuine vector nonsurjectivity on $H((t^\Gamma))$. If $x \neq 0$ has leading exponent δ , the coefficient of $(\mathcal{V} - \varepsilon I)x$ at δ is $\mathcal{V}x_\delta \neq 0$, because the infinitesimal term has strictly larger order. The operator is therefore injective and preserves the leading exponent; if its image contained the constant vector 1, then $\delta = 0$ and $\mathcal{V}x_0 = 1$, which is impossible. This is theorem 7.1 with $S = \mathcal{V}$ and $E = -\varepsilon I$, and it does not rely on any equivalence of categories. Thus $\sigma_\Gamma^{\text{alg}}(\mathcal{V}) = \Sigma_\Gamma^{\mathcal{B}(H)}(\mathcal{V}) = \mathfrak{m}_K$: away from $\mathfrak{m}_K$ the algebra inverses are bijections.

35.3 The missing odd valuation levels

Choose $\Gamma = \mathbb{Z}$ and $p(X) = X^e$ with $e \geq 2$. For Volterra, $Z = \{0\}$ and the relevant ramification index is e . Theorem 34.5 gives

$$\Sigma_\Gamma^{\mathcal{B}(H)}(\mathcal{V}^e) = \mathfrak{m}_K, \quad \{\lambda^e : \lambda \in \Sigma_\Gamma^{\mathcal{B}(H)}(\mathcal{V})\} = \{0\} \cup \{\zeta \in \mathfrak{m}_K \setminus \{0\} : v(\zeta) \in e\mathbb{Z}\}. \quad (35.3)$$

For $e = 2$ every odd positive valuation is missing from the image: $t \in \Sigma_\Gamma^{\mathcal{B}(H)}(\mathcal{V}^2)$ but t is not the square of a point of $\Sigma_\Gamma^{\mathcal{B}(H)}(\mathcal{V})$. After passage to $\Gamma = \mathbb{Q}$, $t^{1/2}$ exists and belongs to the spectrum of $\mathcal{V}$, so this image defect disappears; yet $\mathcal{V}^2 - tI$ is still not invertible there, by corollary 32.3. What changed was the existence of a spectral preimage, not the invertibility of the target pencil.

35.4 Norm-convergent finite sections can miss an entire halo

On $H = \bigoplus_{n \geq 1} \mathbb{C}^n$ consider

$$\mathcal{N} = \bigoplus_{n \geq 1} 2^{-n} J_n. \quad (35.4)$$

Let $\mathcal{N}_M$ retain the first M blocks and vanish elsewhere. It has finite rank and is nilpotent, with $\|\mathcal{N} - \mathcal{N}_M\| \leq 2^{-(M+1)} \rightarrow 0$; hence $\mathcal{N}$ is compact. It is not nilpotent, because arbitrarily long blocks occur. For $z \in \mathbb{C}^*$, finitely many initial blocks have finite-dimensional inverses, and on the remaining blocks $\|2^{-n} J_n\| < |z|/2$, so ordinary Neumann series bound their inverse norms by $2/|z|$; the inverses form a bounded direct sum. Thus $\sigma_{\mathbb{C}}(\mathcal{N}) = \{0\}$, as $\mathcal{N}$ has nonzero kernel. It is not Drazin invertible: for $k \geq 1$ only the blocks with $n > k$ contribute to $\mathcal{N}^k$, so $\|\mathcal{N}^k\| = 2^{-k(k+1)}$, and Drazin data would give $\|P_{\text{inv}}\| \leq 2^{-k(k+1)} \|S^D\|^k \rightarrow 0$, making $\mathcal{N}$ nilpotent. Therefore

$$\Sigma_{\Gamma}^{\mathcal{B}(H)}(\mathcal{N}_M) = \{0\} \text{ for every } M, \quad \Sigma_{\Gamma}^{\mathcal{B}(H)}(\mathcal{N}) = \mathfrak{m}_K. \quad (35.5)$$

There is a direct support witness for every $0 \neq \varepsilon \in \mathfrak{m}_K$. Put $\eta = v(\varepsilon) > 0$ and $c = \text{lc}(\varepsilon)$. The constant vector y whose n th block is $2^{-n} e_n$ lies in H , since $\sum_n 4^{-n} < \infty$. A putative solution x of $(\mathcal{N} - \varepsilon I)x = y$ must, in the n th block, equal

$$x^{(n)} = - \sum_{j=0}^{n-1} \varepsilon^{-j-1} (2^{-n} J_n)^j 2^{-n} e_n.$$

Its final term has valuation $-n\eta$ and leading vector $-c^{-n} 2^{-n^2} e_1 \neq 0$; every earlier term has strictly larger valuation. The coefficient of the global vector x at $-n\eta$ is therefore nonzero, since its projection onto the n th block is nonzero. Its support would contain $-\eta, -2\eta, -3\eta, \dots$, which is not well ordered, so no Hahn-vector solution exists. A finite number of block checks can never establish the uniform bound on nilpotent chain lengths that would remove this obstruction. At $\varepsilon = 0$ the operator already has nonzero kernel, whereas outside $\mathfrak{m}_K$ its algebra inverse gives a bijection. Consequently

$$\sigma_{\Gamma}^{\text{alg}}(\mathcal{N}) = \Sigma_{\Gamma}^{\mathcal{B}(H)}(\mathcal{N}) = \mathfrak{m}_K.$$

This extension to nonmonomial displacements follows from the same support argument; the source's explicit block calculation used a monomial. This is the Drazin counterpart of section 14, where every finite section of a normal operator is invertible but the infinite one is not.

35.5 Two ranks and a nonmonomial displacement

Let $\Gamma = \mathbb{Z} \times \mathbb{Z}$ in lexicographic order and $\eta = (0, 1) > 0$. The sequence $n\eta$ never exceeds $(1, 0)$, yet all the preceding inverse formulas remain strongly valid. For example take $\varepsilon = t^{(0,1)}(1 + t^{(0,2)} + t^{(1,0)})$. At a Drazin point of index three, its shifted inverse has exact valuation $(0, -3)$, irrespective of the nonmonomial correction. The straightening lemma applies within this very group and adjoins no fractions. In contrast, the image of the squaring map on infinitesimals consists of zero and the nonzero infinitesimals whose valuation has both coordinates even; inverse descent and the scalar-power criterion answer different questions. Under the ordered embedding $(m, n) \mapsto m\omega + n$ and $t^\gamma = \omega^{-\gamma}$, the displacement is the actual positive surreal number $\varepsilon = \omega^{-1}(1 + \omega^{-2} + \omega^{-\omega})$, and the exact inverse valuation at index three corresponds to the leading monomial ω^3 , with operator coefficient given by theorem 32.2.

36 Actual surreal and surcomplex specialization

The normal-form bridge is the imported classical theorem of section 2.4, with the sign convention $t^\gamma = \omega^{-\gamma}$ of (2.2) [5, 22]; it is not a newly proposed construction of the surreal field. Positive

valuation is exactly infinitesimal size in the embedded fields. Any *set* of surcomplex scalars fits in a common workspace: take the union of the supports of their real and imaginary normal forms, negate the exponents, and form the additive subgroup they generate, adjoining 1 to make it nonzero. This is set-sized, and its rational span is a set-sized divisible workspace when scalar roots are wanted. No single set-sized workspace is asserted to contain all surreal numbers.

For a fixed set-sized complex algebra $\mathfrak{A}$, let $\mathfrak{A}_{\mathbf{No}}$ denote the class of expressions $\sum_{\gamma \in S} a_\gamma \omega^{-\gamma}$ with $a_\gamma \in \mathfrak{A}$ and $S \subset \mathbf{No}$ a well-ordered set. These are *operator-valued normal forms*, not scalar surreal numbers when $\mathfrak{A}$ is noncommutative. Any finite collection of them lies in a common $\mathfrak{A}_\Gamma$, so additions and products are defined there and independent of enlargement. This is a locality construction, not a theorem with a proper class passed as its set-sized group parameter. For constant $T \in \mathfrak{A}$ define $\Sigma_{\mathbf{No}, \mathfrak{A}}(T)$ by failure of a two-sided inverse of $T - \lambda 1$ in $\mathfrak{A}_{\mathbf{No}}$, where $\lambda \in \mathbf{No}[i]$, and write $\mathfrak{m}_{\mathbf{No}[i]}$ for the surcomplex infinitesimals, including zero.

Theorem 36.1 (Surcomplex halo classification). *For every constant $T \in \mathfrak{A}$,*

$$\Sigma_{\mathbf{No}, \mathfrak{A}}(T) = \sigma_{\mathfrak{A}}(T) \cup (\sigma_{\mathbf{D}, \mathfrak{A}}(T) + \mathfrak{m}_{\mathbf{No}[i]}). \quad (36.1)$$

The exact resolvent formulas and leading-scale assertions hold at every actual surcomplex parameter in their indicated cases.

Proof. Place the normal-form support of λ in a set-sized nonzero ordered subgroup Γ . If theorem 32.1 gives an inverse, its formula already lies in $\mathfrak{A}_\Gamma \subseteq \mathfrak{A}_{\mathbf{No}}$. Conversely a putative inverse in $\mathfrak{A}_{\mathbf{No}}$ has set-sized support; enlarge Γ to a set-sized subgroup containing both supports. The set-sized theorem then forces the same ordinary or Drazin invertibility condition, which is independent of the enlargement by corollary 32.3. The exact leading terms follow from theorem 32.2. $\square$

Corollary 36.2 (Polynomial mapping for the full surcomplex scalars). *For nonconstant $p \in \mathbb{C}[X]$, $\Sigma_{\mathbf{No}, \mathfrak{A}}(p(T)) = p(\Sigma_{\mathbf{No}, \mathfrak{A}}(T))$.*

Proof. For any particular parameter and its support choose a set-sized divisible workspace as above. The criterion of theorem 34.7 holds there, and inverses and spectral membership are unchanged after further enlargement. Alternatively, every positive surreal valuation is divisible by each positive integer, and the local germ construction supplies all the required spectral preimages in the full field. $\square$

The nondivisible counterexamples are therefore genuine phenomena of exact Hahn subfields of the surcomplexes, while the non-Drazin halo itself persists over the full surcomplex scalar field. Passing from the integer workspace to the full field fills missing polynomial preimages; it does not remove the target spectral obstruction.

For a real unital algebra $\mathfrak{A}_{\mathbb{R}}$, complexify to $\mathfrak{A} = \mathfrak{A}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$. At real Hahn parameters, invertibility of a real pencil in the complexification is equivalent to invertibility in the real coefficient algebra, because the unique complex inverse is fixed by conjugation and so has real coefficients; likewise uniqueness of the Drazin inverse shows that Drazin data of a real element, when they exist in the complexification, are real. The real-parameter version of theorem 36.1 follows by intersection with $\mathbf{No}$. This avoids choosing rational powers of negative real leading coefficients in the straightening proof: real coefficients descend by conjugation only after the complex theorem is established. For the real Volterra operator, every nonzero real surreal infinitesimal remains a spectral value in the corresponding real operator-normal-form algebra.

37 Part III: interpretation, boundaries and further questions

37.1 What the results establish

There are two independent mechanisms. A constant pencil admits a Hahn inverse at a nonzero infinitesimal displacement exactly when it has a Drazin decomposition, with an invertible corner and a corner nilpotent of finite index, and that condition survives all exponent-workspace enlargements. A polynomial can then map the resulting halos onto only selected valuation levels when the workspace lacks enough roots, and the defect is determined exactly by the ramification indices over the Drazin-spectral fibres. The argument is not a transfer of the analytic theory of complex operators into a non-Archimedean setting: it uses finite algebra on the singular part and exact support calculus on the regular part. The surreal extension localizes all supports in set-sized subfields rather than declaring a proper class to be a Banach field. The Volterra example explains why an ordinary inverse series that converges for every nonzero complex parameter can still be inadmissible at every nonzero infinitesimal: its negative exponent tail descends without a least element. The increasing-block example explains why all finite nilpotent approximants may pass while their norm limit fails. These are structural obstructions, not floating-point problems.

37.2 Non-claims that matter mathematically

The spectra of this part are relative to the named algebras $\mathfrak{A}_\Gamma$ and $\mathfrak{A}_{\mathbf{No}}$, and concern *constant* elements only. No classification of all linear endomorphisms, of all possible non-Archimedean Hilbert completions, or of all Hahn operator series is asserted. Actual vector nonsurjectivity was proved for the two explicit examples, not inferred universally from lack of an inverse in a specified subalgebra. The Banach algebraicity corollary depends on compactness of the ordinary spectrum and the finite Riesz decomposition and is false for arbitrary complex algebras. The pole-order formula concerns a finite integer index; generalized Drazin invertibility permits an infinite-order obstruction and cannot replace that hypothesis. Every infinite operation uses strong support conditions, and no conclusion about convergence of finite subsums in a higher-rank valuation topology is implicit. No least field-valued operator norm, trace-class determinant, spectral measure or physical application is supplied by this part. The main theorems are proved in the text; they have not been independently refereed or formalized in Lean, and the finite program of section 49.3 cannot validate arbitrary support well ordering, proper-class foundations, or the infinite-dimensional spectral formula.

37.3 Natural next questions

The first genuine extension is from a constant pencil to $T_0 + E - \lambda 1$, with E of positive Hahn order and not necessarily commuting with T_0 . The straightening argument no longer removes E , and a singular Drazin corner can interact with its complement. A useful goal is an exact Schur-reduction criterion with a support certificate, particularly when the singular corner is infinite dimensional; no such general classification is claimed. *Status in this report:* Part IV answers one case of this question, for $\mathfrak{A} = \mathcal{B}(H)$, T_0 injective compact self-adjoint and E self-adjoint in $\mathcal{A}_\Gamma(H)$, over a divisible Γ (theorem 42.1); there the halo of the Drazin-spectral point 0 stays the whole of $\mathfrak{m}_K$, and each isolated simple pole $c \neq 0$ is replaced by finitely many Hermitian cluster eigenvalues in $c + \mathfrak{m}_K$. The general question remains open.

A second question concerns the operator category. For which nondivisible value groups and vector modules does bijectivity of an everywhere-defined coefficientwise bounded Hahn operator force its inverse to have a common well-ordered operator support? An answer would relate the present algebra-relative spectrum to larger endomorphism spectra, such as $\sigma_\Gamma^{\text{alg}}$ of Part I; the examples show why assuming that these categories coincide is unsafe. A third direction is

an analytic functional calculus beyond polynomials: ordinary holomorphic germs with nonzero finite ramification can be handled locally by the same formal inverse construction, but a global Hahn-valued calculus must specify common domains and admissible coefficient supports. For formal verification, the clean first target is the finite-Laurent Drazin criterion, which needs only a unital algebra and coefficient recurrences; the next are cyclic-support projection and strong straightening. No checked implementation is included.

Part IV. Fredholm windows: compact self-adjoint residues under noncommuting Hahn perturbations

Standing hypotheses for Part IV. H is an ordinary *infinite-dimensional separable* complex Hilbert space, Γ is a nonzero set-sized ordered abelian group that is **divisible**, and the vector space, inner product and operator algebra are those of Part I: $\mathcal{H}_\Gamma = H((t^\Gamma))$ with the inner product (5.1), linear in the first variable, and $\mathcal{A}_\Gamma(H) = \mathcal{B}(H)((t^\Gamma))$. Divisibility is a hypothesis of this part only; it is used to make K algebraically closed and F real closed, so that finite Hermitian clusters diagonalize inside the field (lemma 41.3). This report has not examined which Part IV statements survive without it. “Spectrum” in this part means both Part I spectra: $\sigma_\Gamma^{\text{alg}}$, failure of bijectivity on $\mathcal{H}_\Gamma$, and $\sigma_\Gamma^{\text{adj}}$, failure of invertibility in $\mathcal{A}_\Gamma(H)$; the main theorem proves that they agree for the operators studied. The source manuscript wrote $\mathcal{B}$, $\sigma_{\mathcal{B}}$ and σ_{bij} for $\mathcal{A}_\Gamma(H)$, $\sigma_\Gamma^{\text{adj}}$ and $\sigma_\Gamma^{\text{alg}}$, res for the standard part st , and W , M , U , V , D , B for objects that are renamed below to avoid Part I and Part II meanings.

38 The question and the answer

A useful surcomplex spectral theory must distinguish three operations: solving each finite spectral cluster, assembling all clusters into one operator, and detecting the spectrum by a scalar analytic determinant. In the coefficientwise Hahn setting these operations have different answers. This part gives a class in which all three answers are proved exactly. The principal example starts with a compact self-adjoint operator whose ordinary eigenvalues accumulate only at zero. An infinitesimal noncommuting perturbation moves the individual eigenvalues. Nevertheless zero does not simply acquire a sequence of displaced eigenvalues: it retains an entire complex infinitesimal neighbourhood of non-eigenvalue spectral points. Every nonzero ordinary spectral window is finite-dimensional, but the accumulation window is not Fredholm in the algebraic sense of section 42.1.

Let T be an ordinary injective compact self-adjoint operator on H and

$$A = T + E, \quad E = E^* \in \mathcal{A}_\Gamma(H), \quad v(E) > 0. \quad (38.1)$$

The coefficient operators of E need only be bounded; they need not be compact or commute with each other or with T . For an ordinary eigenvalue $c \neq 0$ of T write $m_c = \dim_{\mathbb{C}} \ker(T - cI)$. Theorem 42.1 constructs m_c real Hahn eigenvalues $\lambda_{c,1}, \dots, \lambda_{c,m_c}$ with standard part c , counted with multiplicity, and proves

$$\sigma_\Gamma^{\text{adj}}(A) = \sigma_\Gamma^{\text{alg}}(A) = \mathfrak{m}_K \cup \bigcup_{c \in \sigma_{\mathbb{C}}(T) \setminus \{0\}} \{\lambda_{c,1}, \dots, \lambda_{c,m_c}\}. \quad (38.2)$$

The second set is exactly the point spectrum. For $z \in \mathfrak{m}_K$, $A - zI$ is injective but not surjective, with cokernel isomorphic to the Hahn extension of an algebraic complement of $\text{Ran } T$. Thus even nonreal infinitesimals belong to the spectrum of this self-adjoint operator. This is not a claim about the spectrum of an ordinary complex Hilbert-space operator.

Relation to Part I. For $E = 0$, (38.2) is the injective compact self-adjoint case of corollary 9.2, and Part I is more general there (normal, not necessarily injective, any Γ , any H). The new content is the noncommuting perturbation. This is the case that section 15.2 describes as requiring a different theorem. The local answer, however, does not assemble into a global bounded-coefficient spectral theorem: theorem 43.1 gives an explicit tridiagonal perturbation with

trace-class coefficients and individually defined eigenbranches but no Hahn-unitary diagonalizer in $\mathcal{A}_\Gamma(H)$, and theorems 45.1 and 45.2 show that even ordinary coefficientwise eigenvalue sums and products fail for it. A coefficientwise lift of the trace-class Fredholm determinant nevertheless exists and has an exact Fredholm alternative (theorem 44.4); it detects every finite-coupling spectral window, but no nonzero coherent Hahn analytic function can detect the whole spectral monad at the accumulation point (theorem 46.4).

Relation to Part III. In the language of Part III, $\sigma_{\mathbb{D}, \mathfrak{A}}(T) = \{0\}$ for $\mathfrak{A} = \mathcal{B}(H)$: zero is an accumulation point, and each nonzero eigenvalue is an isolated simple pole. Theorem 32.1 then gives $\Sigma_\Gamma^{\mathcal{B}(H)}(T) = \mathfrak{m}_K \cup (\sigma_{\mathbb{C}}(T) \setminus \{0\})$ at $E = 0$. Part IV answers the first question of section 37.3 for this class: under a noncommuting E the halo of the Drazin-spectral point stays all of $\mathfrak{m}_K$, while each pole c is replaced by a finite Hermitian cluster.

What is inherited and what is added. The range-defect factorization used below is theorem 7.1 of Part I, re-proved in the source by a different factorization (remark 40.1); the Hahn support lemma, ordinary compact spectral theory, the Riesz projection construction, finite Hermitian diagonalization, and the classical trace-class determinant identities are likewise inherited or classical, and their transfer alone is not presented as new. Ordinary analytic perturbation theory supplies isolated spectral projections and finite-dimensional reduction [26]; the direct rotation of two nearby orthogonal projections is the classical Sz.-Nagy–Kato construction, with the explicit formula in Simon’s account [27]; ordinary Fredholm determinant identities are standard trace-ideal theory, used in Bornemann’s formulation [28]; non-Archimedean Fredholm theory has a substantial precedent in Serre’s work [29], whose category of completely continuous operators on a non-Archimedean Banach space is not the coefficient algebra used here. The proposed additions are the full formula (38.2) for arbitrary noncommuting self-adjoint positive-order perturbations, the trace-class counterexample to global bounded-coefficient diagonalization and to spectral sums and products, and the exact interface between Hahn Fredholm determinants and the unavoidable accumulation monad. A targeted search did not locate these combined statements; it is not an exhaustive priority determination.

39 The operator category and the identity-lifting device

39.1 The algebra is specified in advance

All objects are those of sections 5 and 6. Part I proves that $\mathcal{A}_\Gamma(H)$ is exactly the algebra of everywhere-defined adjointable operators (theorem 6.1); this part does not need that theorem, because it specifies $\mathcal{A}_\Gamma(H)$ at the outset. Membership in $\mathcal{A}_\Gamma(H)$ requires every coefficient to be an ordinary bounded operator and one well-ordered operator support; it does not require $\sup_\gamma \|A_\gamma\| < \infty$. Conversely, a matrix whose entries are individually valid Hahn series need not define an element of $\mathcal{A}_\Gamma(H)$: its coefficient at a single exponent can be an unbounded ordinary matrix. This is the obstruction used in section 43. For $X \in \mathcal{A}_\Gamma(H)$ also write $\sigma_{p,\Gamma}(X) = \{\lambda : \ker(X - \lambda I) \neq 0\}$. The inclusion $\sigma_\Gamma^{\text{alg}}(X) \subseteq \sigma_\Gamma^{\text{adj}}(X)$ always holds; equality is proved below for the operators studied, not inferred for every $X \in \mathcal{A}_\Gamma(H)$.

Divisibility makes K algebraically closed and F real closed. The complex statement is the standard Hahn-field closedness theorem [25, Corollary 4]; the real statement follows equivalently from positive square roots and complexification, and a positive square root can be built directly from half the leading exponent and a near-one binomial series (section 5.1). The results apply to genuine surcomplex coefficients through (2.2); they posit no set containing the proper class $\mathbf{No}[i]$ and do not identify the intrinsic valuation topology with the fine topology of the surreal class.

The support tools are lemmas 2.1 and 2.3; the correction exponents of $(I + F)^{-1}$ lie in $(\text{supp } F)^* \setminus \{0\}$. If $X = X_0 + X_+$ with X_0 an invertible ordinary bounded operator and

$v(X_+) > 0$, factoring out X_0 gives an inverse in $\mathcal{A}_\Gamma(H)$ with nonnegative support. This criterion is sufficient, not necessary: $t^n I$ is invertible although its standard part is zero.

39.2 Lifting analytic identities coefficientwise

Let X, Y be ordinary complex Banach spaces and f holomorphic near $x_0 \in X$. For a positive-support Hahn perturbation $h = \sum_{s \in S} h_s t^s$ define

$$f^H(x_0 + h) = \sum_{n \geq 0} \frac{1}{n!} d^n f(x_0)[h, \dots, h], \quad (39.1)$$

expanded into the finite coefficient words of lemma 2.1(ii). It is a formal Hahn sum; no inequality involving ordinary norms of the h_s is required.

Lemma 39.1 (Identity lifting). *Ordinary holomorphic identities between maps as above lift to identities of their Hahn Taylor expansions whenever the ordinary base points lie in the domains of the identities. Finite compositions, products and continuous multilinear operations commute with this lifting.*

Proof. Fix an exponent γ . Only finitely many words in S contribute to that coefficient, and only finitely many directions h_s occur in them. Introduce one ordinary complex variable for each such direction and restrict the ordinary identity to the resulting finite-dimensional parameter space near zero. Equality of its Taylor coefficients gives the coefficient equality at γ . For composition, zero-constant inner increments and the same finite-word condition justify all regrouping. $\square$

A real-analytic identity involving adjoints is treated by real auxiliary parameters and compared coefficientwise in the same way; this is used for orthogonal projections of self-adjoint families. An infinite ordinary sum at one exponent must first be evaluated in its ordinary coefficient space: infinitely many nonzero constants are not a strongly Hahn-summable family. Ordinary traces, contour integrals and determinants are taken inside the coefficient space before the Hahn support argument is applied.

40 The inherited range defect and the infinitesimal reservoir

The range-defect mechanism is theorem 7.1: for $S : V \rightarrow V$ injective, $F \in \text{End}_{\mathbb{C}}(V)((t^\Gamma))$ of positive order and an algebraic complement $V = \text{Ran } S \oplus W$, the operator $S + F$ is injective and $\text{coker}(S + F) \cong W((t^\Gamma))$. Its elementary factorization allows completely noncommuting perturbations, and it is printed once, in Part I.

Remark 40.1 (A second proof, from the Part IV source). Define $L : V \rightarrow V$ by $L(Sx + w) = x$, so that $LS = I$ and

$$S + F = (I + FL)S.$$

Since FL has positive support, lemma 2.3 inverts $I + FL$ in the Hahn algebra of all endomorphisms, so multiplication by $I + FL$ is an automorphism of $V((t^\Gamma))$ carrying $S(V((t^\Gamma)))$ onto $\text{Ran}(S + F)$. Applying the two ordinary projections of $V = \text{Ran } S \oplus W$ to every coefficient does not enlarge supports, so $V((t^\Gamma)) = (\text{Ran } S)((t^\Gamma)) \oplus W((t^\Gamma))$, and the coefficientwise inverse of S on its range gives $S(V((t^\Gamma))) = (\text{Ran } S)((t^\Gamma))$. Both conclusions follow. This factorizes on the left of S , whereas the Part I proof builds a left inverse and a complementary projection; both use only the Neumann lemma. For compact injective S with nonclosed range, L is generally unbounded; it is used only in the auxiliary algebra of all endomorphisms, and the factorization produces no bounded inverse, no bounded projection onto W , and no orthogonal decomposition of $\mathcal{H}_\Gamma$.

Proposition 40.2 (The size of a compact range defect). *If T is injective, compact and self-adjoint on an ordinary infinite-dimensional separable H , then $\dim_{\mathbb{C}}(H/\text{Ran } T) = 2^{\aleph_0}$. Consequently $W((t^\Gamma))$ contains at least $2^{\aleph_0}$ vectors that are linearly independent over K .*

Proof. The compact spectral theorem gives an orthonormal eigenbasis with nonzero real eigenvalues $d_n \rightarrow 0$. Choose a subsequence with $0 < |d_{n_k}| \leq 2^{-k^2}$ and, for real $a \in (0, 1)$, put $x_{n_k}^{(a)} = |d_{n_k}|^a$ and $x_j^{(a)} = 0$ off the subsequence; the vector is square summable. For a nonzero combination $x = \sum_{j=1}^r c_j x^{(a_j)}$ with $a_1 < \dots < a_r$ and $c_1 \neq 0$,

$$\frac{x_{n_k}}{d_{n_k}} = \frac{|d_{n_k}|^{a_1}}{d_{n_k}} \left(c_1 + \sum_{j=2}^r c_j |d_{n_k}|^{a_j - a_1} \right);$$

the bracket tends to c_1 and the first factor tends to infinity in absolute value, so these coordinates are not square summable and $x \notin \text{Ran } T$. This gives continuum many independent cosets; the underlying set of a separable Hilbert space has the cardinality of the continuum, which gives the other bound. Independent cosets are represented by independent vectors of W . A finite K -linear dependence among their constant embeddings would, at any exponent with a nonzero scalar coefficient, give an ordinary complex dependence, so they remain independent over K (this is the argument of corollary 7.3). $\square$

The exact K -dimension of $W((t^\Gamma))$ is not asserted for every large Γ ; only the lower bound is used (compare warning 7.4). For the particular operator $De_n = n^{-1}e_n$, lemma 11.1 gives the same lower bound by explicit power vectors.

Corollary 40.3 (The infinitesimal reservoir). *Under (38.1), every $z \in \mathfrak{m}_K$ satisfies*

$$\ker(A - zI) = 0, \quad \text{coker}(A - zI) \cong W((t^\Gamma)), \quad H = \text{Ran } T \oplus W.$$

In particular $\mathfrak{m}_K \subseteq \sigma_\Gamma^{\text{alg}}(A)$, and no point of $\mathfrak{m}_K$ is an eigenvalue.

Proof. Apply theorem 7.1 with $S = T$ and positive-order perturbation $E - zI$ (zero allowed); proposition 40.2 makes the cokernel nonzero. $\square$

41 Isolated spectral windows are exactly finite-dimensional

Fix a nonzero ordinary eigenvalue c of T . Let P_c° be its ordinary orthogonal spectral projection and $H_c = P_c^\circ H$. Compactness makes c isolated and $m_c = \dim H_c < \infty$. Choose a positively oriented ordinary circle $\mathcal{C}_c$ around c containing no other point of $\sigma_{\mathbb{C}}(T)$ in its closed interior and none on its boundary; its radius may shrink as $c \rightarrow 0$, and no common radius is required.

41.1 Coefficientwise Riesz projections and direct rotation

Put $R_0(\zeta) = (\zeta I - T)^{-1}$ for $\zeta \in \mathcal{C}_c$ and define

$$R(\zeta) = \sum_{n \geq 0} (R_0(\zeta)E)^n R_0(\zeta), \quad P_c = \frac{1}{2\pi i} \int_{\mathcal{C}_c} R(\zeta) d\zeta. \quad (41.1)$$

The integral is an ordinary bounded-operator-valued contour integral of each Hahn coefficient, not an integral over a surcomplex contour.

Proposition 41.1. *Let $S = \text{supp } E \subseteq \Gamma_{>0}$. Formula (41.1) defines $P_c \in \mathcal{A}_\Gamma(H)$ with*

$$P_c^2 = P_c = P_c^*, \quad AP_c = P_c A, \quad P_c = P_c^\circ + \text{positive-order terms},$$

and $\text{supp}(P_c - P_c^\circ) \subseteq S^ \setminus \{0\}$.*

Proof. At a fixed exponent the resolvent expansion has finitely many ordered coefficient words, so each coefficient is a finite sum of products of ordinary resolvents and bounded operators; on the compact circle these are norm-continuous, and their contour integrals exist. The constant coefficient is P_c° . For any finite collection of directions E_s , the ordinary Riesz projection of $T + \sum_s z_s E_s$ on the fixed circle, for complex parameters near zero, is holomorphic, idempotent and commutes with the perturbed operator; its Taylor coefficients are the contour coefficients above, and lemma 39.1 proves idempotence and commutation. Each E_s is self-adjoint because $E = E^*$, so for real parameters near zero the projection is orthogonal; comparing real Taylor coefficients gives $P_c^* = P_c$. Lemma 2.1 gives the support containment. $\square$

Lemma 41.2 (Support-controlled direct rotation). *Let $P = P^2 = P^* \in \mathcal{A}_\Gamma(H)$ and let P° be an ordinary orthogonal projection with $v(P - P^\circ) > 0$. Put*

$$\Omega = PP^\circ + (I - P)(I - P^\circ), \quad U = \Omega(I - (P - P^\circ)^2)^{-1/2}, \quad (41.2)$$

*the inverse square root being its binomial expansion at I . Then $U^*U = UU^* = I$, $UP^\circ = PU$ and $v(U - I) > 0$. If $\text{supp}(P - P^\circ) \subseteq S^* \setminus \{0\}$, then $\text{supp}(U - I) \subseteq S^* \setminus \{0\}$.*

Proof. Direct multiplication using $P^2 = P$ and $(P^\circ)^2 = P^\circ$ gives $\Omega^*\Omega = \Omega\Omega^* = I - (P - P^\circ)^2$ and $P\Omega = \Omega P^\circ$. The square $(P - P^\circ)^2$ commutes with both projections and hence with Ω . The binomial series for $(I - (P - P^\circ)^2)^{-1/2}$ exists by corollary 2.6, is self-adjoint, commutes with Ω, P, P° , and squares to the inverse. The unitary and intertwining identities follow, and all nonconstant terms have support in the stated monoid. $\square$

This is the classical two-projection calculation of [27], recorded in the present algebra. Apply it to $P = P_c$ and $P^\circ = P_c^\circ$, and call the unitary U_c . Then $U_c^*AU_c$ commutes with P_c° , so relative to $H = H_c \oplus H_c^\perp$ it has an exact block form

$$U_c^*AU_c = A_c \oplus A_c^\perp, \quad A_c \in M_{m_c}(K), \quad A_c = A_c^*. \quad (41.3)$$

The standard part of A_c is cI_{m_c} and that of $A_c^\perp$ is $T|_{H_c^\perp}$; both blocks have nonnegative support, and their positive exponents lie in S^* .

41.2 Finite Hermitian algebra over the Hahn field

Lemma 41.3 (Finite Hermitian diagonalization). *Every Hermitian matrix over $K = F[i]$ is unitarily diagonalizable, with eigenvalues in F . If its entries are finite and its standard part is cI_m , all its eigenvalues are finite with standard part c .*

Proof. Since K is algebraically closed, the characteristic polynomial has a root and an eigenvector $x \neq 0$. For a Hermitian X , $\lambda\langle x, x \rangle = \langle Xx, x \rangle = \overline{\langle Xx, x \rangle}$ with $\langle x, x \rangle > 0$ in F , so λ is real. The positive square root of $\langle x, x \rangle$ normalizes x ; its orthogonal complement is invariant, and induction gives an orthonormal eigenbasis. For the standard-part assertion, write the monic characteristic polynomial as $X^m + \sum_{j < m} a_j X^j$ with $v(a_j) \geq 0$. A root of negative valuation would make X^m strictly smaller in valuation than every other term, impossible in a zero sum; so every root is finite, and reducing the characteristic equation gives $(\text{st } \lambda - c)^m = 0$. $\square$

Theorem 41.4 (Exact isolated-cluster reduction). *For every nonzero ordinary eigenvalue c of T , the construction above produces a finite Hermitian matrix A_c with exactly m_c real Hahn eigenvalues $\lambda_{c,1}, \dots, \lambda_{c,m_c}$ counted with multiplicity, all with standard part c . For every $\lambda \in c + \mathfrak{m}_K$,*

$$\ker(A - \lambda I) \cong \ker(A_c - \lambda I), \quad A - \lambda I \text{ invertible in } \mathcal{A}_\Gamma(H) \iff \det(\lambda I_{m_c} - A_c) \neq 0. \quad (41.4)$$

The kernel is finite-dimensional, of dimension the multiplicity of λ in the Hermitian eigenvalue list.

Proof. Lemma 41.3 applies to A_c . On the complementary block the standard part of $A_c^\perp - \lambda I$ is $T|_{H_c^\perp} - cI$, an ordinary bounded invertible operator because c has been removed from the compact spectrum; the remaining terms have positive valuation, and lemma 2.3 inverts the block in its bounded-coefficient Hahn algebra. If the finite determinant is nonzero, invert $A_c - \lambda I$ over K , combine with the complementary inverse and conjugate by U_c ; a finite matrix of scalar Hahn entries acts by a bounded coefficient at each exponent on the finite-dimensional block, so the inverse lies in $\mathcal{A}_\Gamma(H)$, even if some exponents are negative. If the determinant is zero, a nonzero finite block eigenvector gives a nonzero vector of $\mathcal{H}_\Gamma$ after conjugation. The complementary inverse also proves the kernel isomorphism. $\square$

Remark 41.5 (Precisely what is support controlled). The entries of $A_c - cI$ and the projection and rotation corrections have support in the original positive monoid S^* . For a *multiple* residue eigenvalue it is not asserted that each root of the finite characteristic polynomial has support in that monoid; the roots belong to the original divisible Hahn field, and section 47.2 shows that their supports can leave S^* . For a *simple* residue eigenvalue, ordinary analytic eigenvalue perturbation and lemma 39.1 give the eigenbranch directly with support in S^* . This matches the Part II boundary remark 18.4: monoid support is a consequence of simple residues, not an unconditional invariant.

42 The complete spectrum of a noncommuting perturbation

Theorem 42.1 (Compact-residue spectral classification). *Under the standing hypotheses of Part IV and (38.1), with T injective, compact and self-adjoint,*

$$\sigma_\Gamma^{\text{adj}}(A) = \sigma_\Gamma^{\text{alg}}(A) = \mathfrak{m}_K \cup \bigcup_{c \in \sigma_{\mathbb{C}}(T) \setminus \{0\}} \{\lambda_{c,1}, \dots, \lambda_{c,m_c}\}, \quad (42.1)$$

$$\sigma_{p,\Gamma}(A) = \bigcup_{c \in \sigma_{\mathbb{C}}(T) \setminus \{0\}} \{\lambda_{c,1}, \dots, \lambda_{c,m_c}\}. \quad (42.2)$$

All eigenvalues are real with finite-dimensional eigenspaces. At every $z \in \mathfrak{m}_K$, $A - zI$ is injective with cokernel $W((t^\Gamma))$, where $H = \text{Ran } T \oplus W$ is any algebraic complement; this cokernel has K -dimension at least $2^{\aleph_0}$.

Proof. Every $\lambda \in K$ falls in one of four disjoint cases. If $v(\lambda) < 0$, then $A - \lambda I = -\lambda(I - \lambda^{-1}A)$ with $v(\lambda^{-1}A) > 0$, so the second factor has a Neumann inverse in $\mathcal{A}_\Gamma(H)$. If λ is finite and $c = \text{st } \lambda \notin \sigma_{\mathbb{C}}(T)$, then $A - \lambda I = (T - cI) + (E - (\lambda - c)I)$ is an ordinary invertible operator plus a positive-order term, again invertible in $\mathcal{A}_\Gamma(H)$. If $c \neq 0$ belongs to $\sigma_{\mathbb{C}}(T)$, it is an isolated eigenvalue of finite multiplicity, and theorem 41.4 gives exactly the listed eigenvalues in its halo; every other point of the halo has an inverse in $\mathcal{A}_\Gamma(H)$. Finally, if $\text{st } \lambda = 0$, including $\lambda = 0$, then $\lambda \in \mathfrak{m}_K$ and corollary 40.3 gives injectivity and non-surjectivity with the stated cokernel. Every excluded point has a constructed algebra inverse and every included point fails bijectivity, proving equality of the two spectra. The point-spectrum and dimension statements follow from the finite Hermitian blocks and proposition 40.2. $\square$

Solving formal characteristic equations near each nonzero c would establish only that the displayed eigenvalues belong to the point spectrum; it would say nothing about infinite scalars, ordinary resolvent halos, or the monad at zero. The four-case proof controls all of these. The hypotheses have identifiable roles: compactness makes every nonzero ordinary spectral point an isolated finite cluster; injectivity makes the whole zero monad a non-eigenvalue reservoir through theorem 7.1; self-adjointness makes the clusters Hermitian and their roots real; divisibility keeps those roots inside the selected field; and infinite dimension ensures that an injective compact operator is not onto. None of these roles is replaced by a claim of completeness of a surreal Hilbert space.

42.1 The algebraic Fredholm spectrum

Call a K -linear endomorphism of $\mathcal{H}_\Gamma$ *algebraically Fredholm* when its kernel and cokernel are finite-dimensional over K , with algebraic index the difference of these dimensions; no topological closed-range condition is included. Put $\sigma_{\text{ess,alg}}(X) = \{\lambda \in K : X - \lambda I \text{ is not algebraically Fredholm}\}$.

Corollary 42.2. *Under the hypotheses of theorem 42.1, $\sigma_{\text{ess,alg}}(A) = \mathfrak{m}_K$, and at every point outside $\mathfrak{m}_K$ the algebraic Fredholm index of $A - \lambda I$ is zero.*

Proof. At a point of $\mathfrak{m}_K$ the cokernel has dimension at least the continuum. Outside that monad the operator is either invertible or conjugate to a finite square matrix plus an invertible complementary block, and a finite square matrix has kernel and cokernel of the same finite dimension. $\square$

Arbitrary positive-order bounded self-adjoint coefficient perturbations therefore preserve the full algebraic Fredholm spectral monad of this compact residue. The terminology is explicitly algebraic; no identification with a Banach-space or compactoid essential spectrum is made, and the prime of Part I is not an essential spectrum either (section 9.1).

42.2 Extension of the exponent group

Corollary 42.3 (Fixed point spectrum, growing spectral monad). *Let $\iota : \Gamma \hookrightarrow \Gamma'$ be an ordered embedding into another nonzero divisible set-sized ordered abelian group, and extend all Hahn series coefficientwise. The point spectrum and the finite cluster multiplicities of A are unchanged under the scalar embedding, whereas*

$$\sigma_{\Gamma'}^{\text{adj}}(A) = \mathfrak{m}_{K_{\Gamma'}} \cup \iota(\sigma_{\Gamma}(A)). \quad (42.3)$$

Enlarging the workspace can add spectral points without adding eigenvalues.

Proof. The constructions of P_c, U_c, A_c commute with an ordered exponent embedding: at every exponent the same finite algebraic and ordinary contour operations are performed. The characteristic polynomial of A_c already splits over K , so it has no new roots in the extension and its multiplicities are unchanged. Apply theorem 42.1 over Γ' ; its non-eigenvalue part is exactly the enlarged infinitesimal ideal. $\square$

For example, extend $\Gamma = \mathbb{Q}$ into $\mathbb{Q}^2$ with lexicographic order by $q \mapsto (0, q)$; then $it^{(1,0)}$ is a new spectral point but not a new eigenvalue. The formula concerns spectra in their respective modules; it does not assert that scalar extension leaves the module unchanged. It is consistent with corollaries 9.3 and 32.3, which say that each old scalar keeps its spectral status.

43 A trace-class counterexample to simultaneous diagonalization

The local finite-dimensional theorem does not imply a global bounded-coefficient diagonalization, even for a one-scale perturbation with exceptionally small ordinary coefficient ideals.

43.1 The operator and its individual eigenvalues

Let $H = \ell^2(\mathbb{N}_{\geq 1})$, fix $\eta > 0$ in Γ , and put

$$D_\star e_n = 2^{-n} e_n, \quad Y = \sum_{n \geq 1} \frac{1}{n^2} (|e_n\rangle\langle e_{n+1}| + |e_{n+1}\rangle\langle e_n|), \quad A_\star = D_\star + t^\eta Y. \quad (43.1)$$

The sum defining Y is an ordinary trace-norm convergent series of rank-one operators, evaluated before it is used as a Hahn coefficient: $\|D_\star\|_1 = 1$ and $\|Y\|_1 \leq 2 \sum_n n^{-2} < \infty$. Both coefficients are self-adjoint and trace class, and they do not commute. $A_\star$ is positive definite: if $v(x) = \alpha$, then $\langle A_\star x, x \rangle$ has leading term $\langle D_\star x_\alpha, x_\alpha \rangle_{H^2} > 0$. For each n , theorem 41.4 supplies one eigenvalue λ_n with standard part 2^{-n} ; since that residue is simple, λ_n is the unique ordinary analytic perturbation series in $s = t^\eta$, so $\lambda_n \in \mathbb{C}[[s]] \subseteq K$. The existence of these formal series does not give one common ordinary analytic radius as $n \rightarrow \infty$.

Theorem 43.1 (No global bounded-coefficient Hahn unitary). *For $A_\star$ of (43.1), no $U \in \mathcal{A}_\Gamma(H)$ with $U^*U = UU^* = I$ makes $U^*A_\star U$ diagonal relative to (e_n) . This holds even when the support of U may use arbitrary exponents of Γ , not only integer multiples of η .*

Proof. Suppose such U exists and put $\Lambda = U^*A_\star U$. If $\alpha = v(U)$, the coefficient of U^*U at 2α is $U_\alpha^*U_\alpha \neq 0$, so $2\alpha = 0$ and $\alpha = 0$; the two unitary identities make U_0 an ordinary unitary. Since $U_0^*D_\star U_0$ is diagonal and $D_\star$ has distinct eigenvalues, every column of U_0 is a phase multiple of one basis vector, and unitarity makes the assignment a permutation. Right multiplication by the inverse ordinary monomial unitary reduces to $U_0 = I$, $\Lambda_0 = D_\star$, with both U and Λ of nonnegative support.

Compare coefficients in $A_\star U = U\Lambda$. For every $0 < \gamma < \eta$, $\Lambda_\gamma = 0$ and U_γ is diagonal, by well-ordered induction over the union of the positive supports: at such γ ,

$$D_\star U_\gamma - U_\gamma D_\star = \Lambda_\gamma + \sum_{\alpha+\beta=\gamma, \alpha, \beta > 0} U_\alpha \Lambda_\beta,$$

every β on the right is smaller than γ , so the sum vanishes; the left side has zero diagonal while Λ_γ is diagonal, so $\Lambda_\gamma = 0$, and distinct diagonal entries make U_γ diagonal. This covers supports with infinitely many exponents below η . At exponent η the same equation acquires the term Y , and the lower-order products still vanish, so $[D_\star, U_\eta] + Y = \Lambda_\eta$. The $(n, n+1)$ entry gives

$$(U_\eta)_{n,n+1} = \frac{Y_{n,n+1}}{2^{-(n+1)} - 2^{-n}} = -\frac{2^{n+1}}{n^2}. \quad (43.2)$$

These entries are unbounded, and every entry of an ordinary bounded operator is bounded by its norm, so $U_\eta \notin \mathcal{B}(H)$, contradicting $U \in \mathcal{A}_\Gamma(H)$. $\square$

This is a global, arbitrary-support strengthening of the first-order obstruction proposition 24.1 of Part II, which rules out a bounded first coefficient X in a formal expansion $U(t) = \mathbf{1} + tX + \dots$ for $\text{diag}(1/n)$ and the unweighted adjacency operator. Here the perturbation is trace class, the exponents of U are arbitrary, and the conclusion is about any Hahn unitary in $\mathcal{A}_\Gamma(H)$.

Proposition 43.2 (The eigenvalue list is not a bounded coefficient). *For $A_\star$, all odd powers of $s = t^\eta$ in each λ_n vanish, and*

$$\lambda_1 = \frac{1}{2} + 4s^2 + O(s^4), \quad \lambda_n = 2^{-n} + 2^n \left(\frac{2}{n^4} - \frac{1}{(n-1)^4} \right) s^2 + O(s^4) \quad (n \geq 2). \quad (43.3)$$

The coefficient of s^2 is at least $2^{n-1}/n^4$ for $n \geq 16$. Consequently $\text{diag}(\lambda_n)$ is not an element of $\mathcal{A}_\Gamma(H)$, nor is any permutation of this diagonal list.

Proof. Write an eigenvector with n th coordinate normalized to one as $e_n + su^{(1)} + s^2u^{(2)} + \dots$. The first coefficient equation gives $(u^{(1)})_j = Y_{jn}/(d_n - d_j)$ for $j \neq n$, with $d_n = 2^{-n}$, and $[s]\lambda_n = Y_{nn} = 0$. The n th coordinate of the second equation gives $[s^2]\lambda_n = \sum_{j \neq n} Y_{nj}Y_{jn}/(d_n - d_j)$. Only adjacent indices occur: for $n = 1$ this is 4, and for $n \geq 2$ the two terms are $-2^n/(n-1)^4$ and $2^{n+1}/n^4$. With $Je_n = (-1)^n e_n$ one has $JD_\star J = D_\star$ and $JYJ = -Y$, so $A_\star(s)$ and $A_\star(-s)$ have the same uniquely labelled simple eigenbranch at residue d_n , and $\lambda_n(s) = \lambda_n(-s)$; this

proves the odd-coefficient assertion without any convergence argument in Γ . For $n \geq 16$, $(n/(n-1))^4 \leq (16/15)^4 < 3/2$, so the bracket in (43.3) is at least $1/(2n^4)$, and the resulting positive sequence is unbounded. A diagonal coefficient operator is bounded exactly when its entries are, and permutation does not change that. $\square$

43.2 Why Part II has a different answer

For the locally finite path matrix in (43.1), each fixed formal perturbation coefficient of an eigenvector has finite coordinate support: formal recursion propagates a bounded number of edges at a fixed order. The coefficients can therefore be assembled as row- and column-finite matrices, and the entries (43.2) are allowed there, since row- and column-finiteness does not bound their sizes. This is why theorem 43.1 does not contradict theorem 18.2: that theorem acts on $\mathbb{C}^{(\mathbb{N})}((t^\Gamma))$, not on $\ell^2(\mathbb{N})((t^\Gamma))$, in a coefficient algebra without ordinary boundedness, and theorem 43.1 proves that its conclusion cannot simply be transferred to $\mathcal{A}_\Gamma(H)$. Nor does the result deny all spectral representations in other categories: it rules out the specified global Hahn unitary and diagonal coefficient operator. A theory allowing unbounded coefficient operators, different vector spaces or a different summation notion would have to state its own domain and action requirements.

44 A trace-class determinant with an exact Fredholm alternative

44.1 Trace class is imposed coefficient by coefficient

Let $\mathcal{S}_1(H)$ be the ordinary complex trace-class ideal with its trace norm, and

$$\mathcal{T} = \mathcal{S}_1(H)((t^\Gamma)), \quad \mathcal{T}_+ = \{C \in \mathcal{T} : v(C) \geq 0\}.$$

$\mathcal{T}$ is a two-sided ideal of $\mathcal{A}_\Gamma(H)$ under finite coefficient convolution. It is a specified coefficient ideal; no identification with all nuclear or compactoid operators for a non-Archimedean norm is assumed. Define

$$\mathrm{tr}_\Gamma C = \sum_{\gamma} (\mathrm{tr} C_\gamma) t^\gamma. \quad (44.1)$$

Ordinary traces are taken before forming the Hahn series; the support can shrink by cancellation but cannot enlarge. Ordinary cyclicity at each of the finitely many coefficient products gives $\mathrm{tr}_\Gamma(XC) = \mathrm{tr}_\Gamma(CX)$ for $X \in \mathcal{A}_\Gamma(H)$ and $C \in \mathcal{T}$.

The following classical inputs are not claimed as new. The map $C \mapsto \det_H(I + C)$ on $\mathcal{S}_1(H)$ is entire; it is multiplicative on products of operators of the form $I + C$, agrees with finite-dimensional determinants, satisfies the finite-rank Sylvester identity, and is nonzero exactly when $I + C$ is invertible; near $C = 0$ it satisfies $\det_H(I + C) = \exp(\mathrm{tr} \log(I + C))$ [28, Section 3].

Definition 44.1 (Hahn Fredholm determinant). Every $C \in \mathcal{T}_+$ decomposes uniquely as $C = C_0 + C_+$ with $C_0 \in \mathcal{S}_1(H)$ and $v(C_+) > 0$. Define

$$\mathrm{Det}_\Gamma(I + C) = \sum_{k \geq 0} \frac{1}{k!} d^k \det_H(I + \cdot)(C_0)[C_+, \dots, C_+], \quad (44.2)$$

each multilinear expression being expanded into ordered coefficient words as in (39.1).

This is well defined by lemma 2.1, with support in $(\mathrm{supp} C_+)^*$. It is independent of any basis and needs no labelling of eigenvalues. The ordinary coefficient C_0 may have large trace norm and $I + C_0$ may be singular, so this is not merely a determinant on the near-identity group $I + \mathcal{T}_{>0}$.

Proposition 44.2 (Determinant identities). *For $C, C' \in \mathcal{T}_+$,*

$$\text{st Det}_\Gamma(I + C) = \det_H(I + C_0), \quad \text{Det}_\Gamma((I + C)(I + C')) = \text{Det}_\Gamma(I + C) \text{Det}_\Gamma(I + C'). \quad (44.3)$$

The determinant commutes with coefficient conjugation and adjunction and agrees with a finite matrix determinant on a finite-dimensional ordinary summand. If $v(C) > 0$,

$$\text{Det}_\Gamma(I + C) = \exp\left(\text{tr}_\Gamma \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} C^k\right), \quad (44.4)$$

all powers, logarithms and exponentials being strong Hahn sums.

Proof. The standard-part assertion is the constant Taylor coefficient. For multiplicativity apply lemma 39.1 to the ordinary identity $\det_H(I + C + C' + CC') = \det_H(I + C) \det_H(I + C')$ near the ordinary pair (C_0, C'_0) ; the nonconstant increments of $C + C' + CC'$ have positive support. The adjoint and finite-matrix assertions follow by the same comparison. For $v(C) > 0$, the ordinary logarithmic identity holds near zero in trace norm; compare Taylor coefficients at zero and lift. Positive-word finiteness justifies the operator logarithm and the scalar exponential without any claim of sequential convergence in the value group. $\square$

44.2 The Fredholm alternative

The nonzero-determinant criterion needs more than residue testing: a determinant with zero standard part can still be a nonzero Hahn number. The argument reduces the singular ordinary residue to one finite matrix.

Lemma 44.3 (Ordinary finite-rank correction). *For $C_0 \in \mathcal{S}_1(H)$ there is a finite-rank ordinary operator $F = \iota\kappa$, with $\iota : \mathbb{C}^m \rightarrow H$ and $\kappa : H \rightarrow \mathbb{C}^m$, such that $I + C_0 + F$ is invertible.*

Proof. $I + C_0$ is Fredholm of index zero. Decompose its domain as its finite-dimensional kernel plus the orthogonal complement, and its target as its closed range plus a finite-dimensional orthogonal complement of equal dimension. Choose an isomorphism from the kernel onto that complement and extend it by zero; adding it makes both block maps bijective and boundedly invertible. Any finite-rank operator factors through $\mathbb{C}^m$. $\square$

Theorem 44.4 (Hahn Fredholm alternative). *Let $C \in \mathcal{T}_+$. Then*

$$\text{Det}_\Gamma(I + C) \neq 0 \iff (I + C)^{-1} \in \mathcal{A}_\Gamma(H) \iff I + C : \mathcal{H}_\Gamma \rightarrow \mathcal{H}_\Gamma \text{ is bijective.} \quad (44.5)$$

Every such $I + C$ is algebraically Fredholm of index zero; if the determinant vanishes, $I + C$ has a nonzero finite-dimensional kernel and a cokernel of the same dimension. Explicitly, with $F = \iota\kappa$ as in lemma 44.3, $M_F = I + C + F$ and $L_F = I_m - \kappa M_F^{-1} \iota$,

$$\text{Det}_\Gamma(I + C) = \text{Det}_\Gamma(M_F) \det_m L_F, \quad \text{st Det}_\Gamma(M_F) \neq 0, \quad (44.6)$$

so one finite matrix decides the exact invertibility problem.

Proof. The constant coefficient of M_F is invertible, so M_F^{-1} exists in the nonnegative part of $\mathcal{A}_\Gamma(H)$ by lemma 2.3. Let $R_F = M_F^{-1} \iota$; then $I + C = M_F(I - R_F \kappa)$. Every coefficient of $R_F \kappa = M_F^{-1} F$ is trace class, with nonnegative support. Multiplicativity and the lifted finite-rank Sylvester identity give $\text{Det}_\Gamma(I + C) = \text{Det}_\Gamma(M_F) \text{Det}_\Gamma(I - R_F \kappa) = \text{Det}_\Gamma(M_F) \det_m(I_m - \kappa R_F)$, and $\text{Det}_\Gamma(M_F)$ has the nonzero standard part $\det_H$ of the constant coefficient. If L_F is invertible, the identity $(I - R_F \kappa)^{-1} = I + R_F L_F^{-1} \kappa$ constructs an inverse in $\mathcal{A}_\Gamma(H)$ (the entries of L_F^{-1} may have negative valuation, which $\mathcal{A}_\Gamma(H)$ permits); combined with M_F^{-1} this proves the first

implication, and an algebra inverse is a module inverse. If $\det_m L_F = 0$, choose $y \neq 0$ with $L_F y = 0$; then $y = \kappa R_F y$, the vector $x = R_F y$ is nonzero, and $(I - R_F \kappa)x = 0$, so $(I + C)x = 0$. Finally R_F and κ identify the kernel and cokernel of $N_F = I - R_F \kappa$ with those of L_F : on kernels they are inverse because $R_F \kappa = I - N_F$ and $\kappa R_F = I_m - L_F$, and on cokernels they are well defined and inverse because $N_F R_F = R_F L_F$ and $\kappa N_F = L_F \kappa$. Multiplication by the invertible M_F does not change the dimensions, so both are finite and equal. The empty correction when the constant coefficient is already invertible is covered by $\det_0 = 1$. $\square$

Remark 44.5 (The exact domain matters). The theorem applies to C of nonnegative valuation, not to every series in $\mathcal{T}$. It allows a singular ordinary residue and arbitrary positive Hahn corrections and produces inverses whose valuations may be negative. It does not extend (44.2) to negative-valuation trace-class series; section 46 explains why that cannot be done by the same infinite Taylor formula.

45 Trace and determinant without eigenvalue sums or products

The trace-class counterexample yields an obstruction stronger than the automatic failure of strong summability of infinitely many nonzero constants: even if ordinary infinite sums are allowed at each Hahn coefficient, spectral trace and product formulas fail. For $A_\star$ of (43.1) put $s = t^\eta$, $a_n = [s^2]\lambda_n$, and $g_n = 2^{n+1}/n^4$ for $n \geq 1$, $g_0 = 0$; proposition 43.2 says exactly that $a_n = g_n - g_{n-1}$.

Theorem 45.1 (Failure of the spectral trace sum). $\mathrm{tr}_\Gamma A_\star = 1$. *However, $\sum_{n \geq 1} \lambda_n$ does not exist even under the convention of taking an ordinary complex sum separately at every Hahn exponent: its second coefficient has the exact partial-sum law*

$$\sum_{n=1}^N a_n = \frac{2^{N+1}}{N^4} \rightarrow +\infty. \quad (45.1)$$

This rules out a universal eigenvalue-sum formula for the trace in the coefficientwise trace-class category $\mathcal{T}$.

Proof. $\mathrm{tr} D_\star = 1$ and the off-diagonal trace-class Y has trace zero. The second-coefficient partial sum telescopes to g_N . The individual a_n do not tend to zero, so no rearrangement makes their sum convergent; the spectral sum is undefined rather than equal to a different Hahn scalar. $\square$

For comparison, truncate to the first N basis vectors. The finite-matrix eigenbranches agree to second order with λ_n for $n < N$, but the last branch loses the edge to $N+1$, so its second coefficient is $a_N - g_N = -g_{N-1}$, and the finite-section second trace coefficient is $\sum_{n < N} a_n + (a_N - g_N) = 0$, as the finite trace identity requires. The missing boundary term g_N grows rather than tends to zero: stabilization of each fixed branch is insufficient to exchange the branch sum with formal coefficient extraction.

Put $\mathfrak{d}_0 = \det_H(I + D_\star) = \prod_{n \geq 1} (1 + 2^{-n}) > 0$ and $q_n = 1/(n^4(1 + 2^{-n})(1 + 2^{-(n+1)}))$; these products and sums are ordinary real ones.

Theorem 45.2 (Failure of a spectral product formula).

$$[s^2] \mathrm{Det}_\Gamma(I + A_\star) = -\mathfrak{d}_0 \sum_{n \geq 1} q_n < 0, \quad (45.2)$$

whereas the coefficient of s^2 in $\prod_{n=1}^N (1 + \lambda_n)$ tends to $+\infty$. This spectral product has no coefficientwise ordinary limit and cannot represent $\mathrm{Det}_\Gamma(I + A_\star)$.

Proof. Factor $I + D_\star + sY = (I + D_\star)(I + sK_0)$ with $K_0 = (I + D_\star)^{-1}Y$, an ordinary trace-class operator of trace zero. The determinant logarithm (44.4) gives $[s^2] \text{Det}_\Gamma(I + A_\star) = -\frac{1}{2}\mathfrak{d}_0 \text{tr}(K_0^2)$. The diagonal of K_0^2 consists of adjacent two-step contributions, and absolute convergence gives $\text{tr}(K_0^2) = 2 \sum_{n \geq 1} q_n$, finite and positive. Let $P_N = \prod_{n=1}^N (1 + 2^{-n})$. Since all first eigenvalue coefficients vanish, $[s^2] \prod_{n=1}^N (1 + \lambda_n) = P_N \sum_{n=1}^N a_n / (1 + 2^{-n})$, and summation by parts, or pairing the two ends of each edge, gives

$$\sum_{n=1}^N \frac{a_n}{1 + 2^{-n}} = \frac{g_N}{1 + 2^{-N}} - \sum_{n=1}^{N-1} q_n. \quad (45.3)$$

The first term tends to $+\infty$, the sum of the q_n converges, and $P_N \rightarrow \mathfrak{d}_0 > 0$. $\square$

The same missing edge that obstructs the trace sum supplies the divergent boundary term. Finite-section determinants remove that term and converge coefficientwise to (45.2); products of the labelled branches of the infinite operator do not. There is no conflict with ordinary Lidskii theory or the ordinary trace-class eigenvalue product: at each fixed ordinary real parameter, $D_\star + sY$ is an ordinary compact self-adjoint trace-class operator and the ordinary theory applies. What fails is interchanging a countable spectral sum with a formal perturbation expansion around infinitely many spectral gaps shrinking to zero, for want of a uniform analytic labelling.

46 Finite-coupling determinants and the accumulation boundary

46.1 One coherent entire family on ordinary coupling space

Assume now that $A = T + E$ satisfies the hypotheses of theorem 42.1 and also $A \in \mathcal{T}_+$, so T and every coefficient of E are trace class. Let $\mathcal{O}(\mathbb{C})$ be the ordinary entire functions. For an ordinary complex variable z , expand the ordinary determinant map at $-zT$ in the directions $-zE_\gamma$; the coefficient of any Hahn exponent is a finite sum of such derivatives and is entire in z , with common support in $(\text{supp } E)^*$. This defines a coherent family

$$\mathcal{F}_A \in \mathcal{O}(\mathbb{C})((t^\Gamma)), \quad \mathcal{F}_A(z) = \text{Det}_\Gamma(I - zA) \quad (z \in \mathcal{O}_K), \quad (46.1)$$

where at $z = z_0 + \delta$, with ordinary z_0 and $v(\delta) > 0$, the entire coefficients are evaluated by their Taylor series at z_0 using the positive support of δ ; lemma 39.1 identifies this with definition 44.1 applied to $-zA$.

Theorem 46.1 (Exact finite-coupling detection). *For every finite $z \in \mathcal{O}_K$, $\mathcal{F}_A(z) = 0$ if and only if $I - zA$ is not invertible in $\mathcal{A}_\Gamma(H)$. Its finite zeros are exactly $\{\lambda^{-1} : \lambda \in \sigma_{p,\Gamma}(A)\}$; in particular $\mathcal{F}_A(0) = 1$ and no infinitesimal coupling is a zero.*

Proof. For finite z , $-zA$ has nonnegative support with trace-class coefficients, so theorem 44.4 gives the equivalence. If $z = 0$ there is nothing to prove; if $v(z) > 0$, $I - zA$ has a Neumann inverse. For $v(z) = 0$, invertibility is equivalent to that of $A - z^{-1}I$, where z^{-1} has a nonzero standard part, and theorem 42.1 makes it singular exactly at the listed eigenvalues. Conversely every eigenvalue in (42.2) has a finite inverse with nonzero standard part. $\square$

The entire *ordinary coefficient functions* do not mean that the Taylor series can be evaluated at every infinite surcomplex argument.

Proposition 46.2 (Negative-valuation evaluation can fail). *Write the ordinary entire determinant of $D_\star$ as $\prod_{n \geq 1} (1 - 2^{-n}z) = \sum_{k \geq 0} c_k z^k$. Every c_k is nonzero, and for $\eta > 0$ the family $(c_k t^{-k\eta})_{k \geq 0}$ is not strongly Hahn summable; so the entire Taylor series does not define the determinant at $t^{-\eta}$ by Hahn substitution.*

Proof. Up to sign, c_k is the k th elementary symmetric sum of the positive numbers 2^{-n} , finite (at most $(\sum_n 2^{-n})^k/k!$) and strictly positive. The substituted terms have exponents $0, -\eta, -2\eta, \dots$, which are not well ordered. $\square$

This does not exclude a separately defined regularization or another function category at infinite arguments; it excludes the termwise extension by the entire Taylor series. Finite coupling z corresponds to spectral parameter $\lambda = z^{-1}$ away from the zero monad, and nonzero $\lambda \in \mathfrak{m}_K$ would require infinite coupling, exactly where that substitution need not exist.

46.2 No coherent analytic characteristic function through zero

Let $\mathbb{D}_r = \{z \in \mathbb{C} : |z| < r\}$ with ordinary $r > 0$. A *coherent Hahn analytic function* on its halo is a series $\mathcal{G}(z) = \sum_{\gamma \in S} g_\gamma(z) t^\gamma$ with $g_\gamma \in \mathcal{O}(\mathbb{D}_r)$ and one well-ordered support S , evaluated at $z = c + \delta$ by ordinary Taylor expansion of its coefficients at $c \in \mathbb{D}_r$ and positive-order Hahn substitution of δ . Its halo is $\mathbb{D}_r^\# = \{c + \delta : c \in \mathbb{D}_r, \delta \in \mathfrak{m}_K\}$.

Lemma 46.3 (A nonzero coherent germ cannot vanish on a whole monad). *If $\mathcal{G} \in \mathcal{O}(\mathbb{D}_r)((t^\Gamma))$ is nonzero, there is $\delta_0 \in \Gamma_{\geq 0}$ with $\mathcal{G}(x) \neq 0$ whenever $0 \neq x \in \mathfrak{m}_K$ and $v(x) > \delta_0$.*

Proof. Let m be the least vanishing order at 0 among the nonzero coefficient germs. Every coefficient is divisible by z^m , so $\mathcal{G} = z^m \mathcal{H}$ with $\mathcal{H}(0) \neq 0$. Let α be the least exponent of a nonzero coefficient function of $\mathcal{H}$ and $\beta = v(\mathcal{H}(0)) \geq \alpha$. All Taylor coefficients of $\mathcal{H}$ at zero have valuation at least α , so for $x \neq 0$ of positive valuation the strong substitution gives $v(\mathcal{H}(x) - \mathcal{H}(0)) \geq \alpha + v(x)$; the substitution is legitimate after extracting t^α , by lemma 2.1. If $v(x) > \beta - \alpha$, the bound exceeds β , so $\mathcal{H}(x)$ has the leading term of $\mathcal{H}(0)$; since $x^m \neq 0$, $\mathcal{G}(x) \neq 0$. Take $\delta_0 = \beta - \alpha$. $\square$

Theorem 46.4 (No analytic characteristic determinant at the accumulation point). *For A as in theorem 42.1, no coherent Hahn analytic function on $\mathbb{D}_r^\#$, for any ordinary $r > 0$, has zero set exactly $\sigma_\Gamma^{\text{adj}}(A) \cap \mathbb{D}_r^\#$.*

Proof. The desired zero set contains all of $\mathfrak{m}_K$ by corollary 40.3, and a nonzero coherent function cannot vanish there by lemma 46.3. The zero function fails too: a nonreal ordinary c with $0 < |c| < r$ lies outside the real ordinary spectrum of T , so $A - cI$ is invertible by the second case of theorem 42.1. $\square$

The theorem concerns a scalar analytic characteristic function of the *spectral parameter* near zero. It does not conflict with the finite-coupling family $\mathcal{F}_A$, which equals one at coupling zero, or with the finite polynomial $\det(\lambda I - A_c)$ on an isolated nonzero halo. It does not exclude set-theoretic indicators, meromorphic objects with a singularity at zero, or function classes whose coefficients do not share an ordinary neighbourhood; its content is the failure inside the named coherent analytic category.

47 Genuinely higher-rank examples

47.1 A scale invisible to every power of another scale

Take $\Gamma = \mathbb{Q}^2$ lexicographically, $\eta = (0, 1)$ and $\beta = (1, 0)$, so $n\eta < \beta$ for every ordinary integer n . The strong identity $(1 - t^\eta)^{-1} = \sum_{n \geq 0} t^{n\eta}$ holds, but its finite partial sums do not converge to it in the full valuation topology: the remainder after order N has valuation $(N + 1)\eta < \beta$

(compare section 23). All proofs above use finite coefficient words, not such convergence. For a noncommuting example at both scales take

$$A = D_\star + t^\eta Y + t^\beta |e_1\rangle\langle e_1|,$$

with $D_\star, Y$ from (43.1). The main spectral theorem and the determinant theorem apply; the obstruction at exponent η is unchanged, and the second eigenvalue coefficients at 2η are unchanged because $\beta > n\eta$ for every finite n . The failures of global diagonalization and of the spectral trace sum therefore persist in a genuinely higher-rank perturbation. Identifying (a, b) with $a\omega + b$ gives $t^\eta = \omega^{-1}$ and $t^\beta = \omega^{-\omega}$: the example has actual surcomplex coefficients involving both scales.

47.2 Finite clusters may need exponents outside the input monoid

Let the first two ordinary coordinates share the eigenvalue $c = 1$, let the remaining eigenvalues be $2^{-1}, 2^{-2}, \dots$, and perturb only the first block by

$$A_c = \begin{pmatrix} c + t^\eta & t^\beta \\ t^\beta & c - t^\eta \end{pmatrix}. \quad (47.1)$$

Its eigenvalues are exactly

$$\lambda_\pm = c \pm \sqrt{t^{2\eta} + t^{2\beta}} = c \pm t^\eta \sum_{k \geq 0} \binom{1/2}{k} t^{2k(\beta - \eta)}, \quad (47.2)$$

the binomial series being strongly summable because $\beta - \eta > 0$. The first two nonconstant exponents are η and $2\beta - \eta$; the second is not in the nonnegative integer monoid generated by η and β , although it lies in the original group. The cluster matrix has support controlled by the input positive monoid, its eigenvalues lie in the original divisible field, and their supports need not stay in that monoid; the last claim would be false and is not part of theorem 41.4. This is a repeated-residue phenomenon of the same kind as the finite example cited in remark 18.4. Under the surreal identification, $\lambda_\pm = 1 \pm \omega^{-1}(1 + \frac{1}{2}\omega^{-2(\omega-1)} - \frac{1}{8}\omega^{-4(\omega-1)} + \dots)$, with set-sized normal-form supports.

48 Part IV: proof dependencies, limits and further questions

48.1 Dependencies

The main spectrum theorem uses ordinary compact self-adjoint spectral theory, finite-word support, the inherited range factorization and finite Hermitian diagonalization. The obstruction uses only its simple-eigenvalue instance and a first-coefficient commutator equation. The spectral trace and product failures additionally use the explicit second-order recurrence and elementary finite summation identities. The determinant theorem uses the ordinary trace-class determinant as an analytic map, finite-word Taylor lifting, the ordinary index-zero Fredholm alternative and finite-dimensional linear algebra. The analytic accumulation obstruction uses only a common coefficient neighbourhood, the least ordinary vanishing order and a valuation comparison, and is independent of trace-class assumptions. The matrix identities, finite recurrence and support algebra are natural formalization targets; the ordinary compact and trace-ideal inputs would remain substantial analytic prerequisites in a Lean development. No Lean files are included and no theorem here is represented as machine checked.

48.2 Precisely limited claims

This part does not classify spectra of arbitrary elements of $\mathcal{A}_\Gamma(H)$; its complete formula requires an injective compact self-adjoint ordinary residue, a self-adjoint positive-order perturbation, a separable infinite-dimensional H and a divisible Γ . It does not supply a positive spectral measure, a general non-Archimedean Hilbert-space spectral theorem, or a nuclear-operator theory independent of the chosen coefficient model. Its determinant is defined on the nonnegative-valuation component of the coefficientwise trace-class ideal, not on every trace-class-valued Hahn series. The no-product theorem concerns products of the labelled Hahn eigenbranches and ordinary coefficientwise limits; it does not claim that all conceivable regularized products are impossible. The no-characteristic theorem is confined to the named coherent analytic category. The counterexample to global diagonalization is confined to bounded ordinary operator coefficients, not to the row- and column-finite algebra of Part II or to every possible unbounded-operator framework. No complete classification of bounded-coefficient Hahn operators, no equivalence with intrinsic nuclear or compactoid operator ideals, and no determinant for every negative-valuation trace-class series is claimed. The finite checks of section 49.4 are not evidence of priority and do not replace the infinite-dimensional arguments; the part has not been independently refereed.

48.3 Further questions

First, what uniform spectral-gap and coefficient-ideal hypotheses ensure that all cluster rotations assemble into a bounded-coefficient Hahn unitary? The first commutator equation requires a bounded solution of $[D, X] = -B_{\text{off}}$ for the diagonal residue D and the off-diagonal first perturbation coefficient B_{off} ; that necessary first-order condition is not asserted to be sufficient at every higher order. Second, which altered coefficient categories support a meaningful regularized spectral product compatible with the Fredholm determinant when the ordinary coefficientwise product fails? Third, how should the classification change when the compact residue has a nontrivial kernel, so that genuine infinitesimal eigenvalues can coexist with the range defect? These are directions beyond the proved statements, not missing steps in their proofs.

Shared back matter

49 The four exact check suites

The four parts arrived with separate verification programs, which are kept separate in `code/` and `data/` under the numbers 02 (Part I), 03 (Part II), 04 (Part III) and 05 (Part IV). No program checks any statement of another part.

Warning 49.1 (A coincidence, not a dependence). The Part I and Part II suites both report exactly **218 exact checks**, and both recorded runs used **Python 3.13.5** with **SymPy 1.14.0**. This is a coincidence of two independently written programs, and the shared tool versions are a coincidence of two runs on the same day. The two counts are of entirely different assertions: the Part I total is 218 assertions in six groups about finite convolutions, adjoints, truncated Neumann identities, the isolated-point resolvent, a finite rectangular analogue, finite sections and a backward-shift residual, while the Part II total is 218 checks about perturbation jets, an accumulating chain, sharp boundary coefficients, bidegree truncations, projection identities and a closed-form eigenpair. Neither number is evidence about the other suite, and neither program copied the other. The Part III and Part IV suites report 484 and 1,268 checks; their recorded runs also used Python 3.13.5 (Part IV with SymPy 1.14.0, Part III with the standard library only). The four counts are counts of program assertions, not of theorems, and are not comparable with one another.

49.1 The Part I suite

The program `code/02-hahn-hilbert-spectral-verify.py` uses exact rational and symbolic arithmetic. It checks finite-support convolution and adjoints, including a lexicographically ordered two-dimensional exponent group; truncated noncommutative Neumann identities; the isolated-point resolvent formula (8.4); the algebraic left-inverse and complement projection identities of theorem 7.1 in a finite rectangular analogue; the finite-section formulas of section 14; and the backward-shift residual of example 8.4. The recorded run passed 218 exact assertions in six groups; its output is `data/02-hahn-hilbert-spectral-verification.txt`.

The finite rectangular analogue is deliberately identified as such. An injective nonsurjective endomorphism cannot exist on a finite-dimensional space, so a finite square-matrix test cannot validate the infinite range defect. Nor can finite tests establish Baire category, arbitrary Hahn support well-ordering, an infinite cokernel dimension, or the spectral formula. Those assertions rest on the proofs, not on the program.

49.2 The Part II suite

The program `code/03-infinite-spectral-verify.py` passed **218 of 218** exact checks; the machine-readable record is `data/03-infinite-spectral-verification.json`. The suite verifies a complex Hermitian three-by-three example with two perturbation orders through degree six; the six-vertex accumulating-level chain through degree ten; the sharp finite-section coefficient for path distances one through four; a two-variable example through total degree four, including the coefficients in (23.2); projection idempotence and self-adjointness at every retained bidegree; the symbolic inverse-gap obstruction of proposition 24.1; and the recurrence, quadratic identity, normalization generating function, and Taylor coefficients of the closed-form eigenpair of proposition 22.1.

These are exact algebraic checks, not floating-point evidence for an infinite theorem. They are not a Lean formalization, do not certify novelty, and cannot replace the summability arguments.

The script raises an exception at the first failed check. The two-scale test uses finite bidegrees, not an allegedly finite initial segment of a lexicographic valuation group.

49.3 The Part III suite

The program `code/04-drazin-halos-verify.py` uses exact `fractions.Fraction` arithmetic and only the Python standard library. Its recorded run, `data/04-drazin-halos-verification.json`, passed **484** assertions in six categories: pure nilpotent inverses (30), mixed Drazin decompositions after a nonorthogonal rational change of basis with left and right residuals of truncated regular resolvents (175), nilpotent pole-order transformation under ramified polynomials (210), local ramified germs and their compositional inverses (18), nonmonomial shifted nilpotent inverses (24), and finite-order instances of the straightening inverse construction (27). An exact finite matrix identity is verified for the tested matrix only, and a truncated formal identity to its recorded order only. Neither the Volterra spectrum nor the increasing-Jordan-block counterexample is established by these tests; their evidence is the proof in section 35. The program prints its report and writes a file only when given `-output`.

49.4 The Part IV suite

The program `code/05-fredholm-verify.py` uses exact rational and symbolic arithmetic with SymPy and the deterministic seed 20260922. Its recorded run, `data/05-fredholm-verification.json`, passed **1,268** checks in eighteen categories. They cover exact eigenpair recursions through order six for 24 finite symmetric systems, with residuals, normalization and the second-order Rayleigh coefficient; truncated coefficient identities for $U^*U = I$ and $UP^\circ = PU$ in the direct rotation of lemma 41.2; the determinant–exponential-trace identity, finite-rank Sylvester identities and the singular-residue finite-rank correction; the forced generator entries (43.2), the second eigenvalue coefficients, vanishing odd coefficients and the growth bound of proposition 43.2; the telescoping laws (45.1) and (45.3) and their finite-section cancellations; and finite instances of $n\eta < \beta$ in lexicographic order. No finite run verifies arbitrary-rank support well-ordering, existence of all Hahn coefficients, continuum-dimensional cokernels, spectral exhaustion over the whole field, or the absence of all possible global unitaries. The program writes its report to `data/verification.json` relative to the working directory unless `-output` is given, so it should be run on a copy.

49.5 Building

The L^AT_EX source contains its bibliography and needs no external images, fonts, or bibliography database. A standard build is

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
python code/02-hahn-hilbert-spectral-verify.py
python code/03-infinite-spectral-verify.py
python code/04-drazin-halos-verify.py
python code/05-fredholm-verify.py --output rerun/05-verification.json
```

The scripts support Python 3.10 or later; those of Parts I, II and IV need SymPy, and that of Part III needs only the standard library. Some of them write their reports next to themselves or into the working directory, so run them on a copy of this directory. The delivered `build.sh` wrappers refer to the source packages' original unnumbered file names and do not run as placed. The source is not Lean code, and no proof-assistant certification is claimed for any part.

A Part I: repository and literature audit

A.1 Repository snapshot

The Part I coverage comparison is pinned to

608dd23c0539fce73723843949ee627641c27b30.

It inspected the repository tree, the project `README.md`, the full `docs/README.md` catalogue, the opening 250 lines of `docs/surcomplex/spectral-theory/article.tex`, and the opening 230 lines of the differential-equations article. This was a targeted scope audit, not a claim to have reread every archived source manuscript or verified every existing Lean file. The spectral opening explicitly states its finite-dimensional scope, while the catalogue maps the already-covered analytic, differential, Berkovich, foundational and computational strands. No repository files were modified by that audit.

A.2 Primary-source comparison and date boundary

The Part I literature search was carried out through September 22, 2026. The publisher records and accessible abstracts for [11, 12], the author-provided text associated with [10], and the arXiv records for [13, 14] were used to locate nearby topics. Ishizuka’s paper appears in volume 39(2), labeled 2024, with the publisher recording online publication on January 3, 2025. Both dates are recorded in the bibliography to avoid silently conflating the issue year with the online date.

The comparison supports the modest statement that the exact theorem package of Part I was not identified in the inspected sources. It does not establish that no earlier work contains any equivalent result. In particular, formal support inversion, failure of non-Archimedean orthogonality, and ordinary resolvent expansions have substantial pre-existing literatures.

B Part II: provenance and research scope

B.1 Repository snapshot

The Part II comparison is pinned to

a3124af79f66b8b9c196d76b4cbc5ac3938907c4

with commit timestamp 2026-09-22T15:58:14Z as returned by GitHub. It used the documentation catalogue and the spectral-theory `README` at that commit; the differential-equations and spectral-theory inventories were also examined to avoid repeating finite-dimensional work. The pinned spectral-theory scope explicitly restricts dimensions to ordinary finite integers. This is a targeted scope comparison, not a line-by-line audit of every theorem in the repository. Initially returned unpinned documentation was older; the provenance is based on the pinned snapshot above. No change was made to the GitHub repository.

The two audits used two different pinned revisions, and neither was repeated against the other’s pin when the two manuscripts were assembled here.

B.2 Primary sources consulted

Gonshor for foundational normal forms [5]; Neumann [6]; Higman [7]; Greenbaum, Li and Overton [15]; Kenmoe, Smerlak and Zadorin [16]; Giscard, Thwaite and Jaksch [17]; and Ara, Goodearl and O’Meara [18], particularly for the operator-category distinctions and failures of general infinite-dimensional spectral assertions.

The source list and comparison do not certify the absence of all prior versions of the new package. The elementary perturbation coefficients, the formal recursion mechanism, the Neumann support lemma, the Higman word lemma, the walk interpretation of products, and the broad phenomenon of positive infinite matrices with no eigenvectors are not claimed as discoveries.

B.3 Candidate contribution

The Part II candidate-original contribution is the global arbitrary-rank Hahn operator/module synthesis: unique normalized infinite diagonalization with support control, all coherent resolvents, a complete commuting projection calculus, marked-walk stability, and sharp finite-section error thresholds, under simple ordinary residues and positive global support. The explicit accumulating-level closed-form eigenpair of proposition 22.1 is proved directly and used as an independent check; no separate priority claim is made for that solvable model.

B.4 A proof not reprinted

The Part II source manuscript carried an appendix with the classical minimal-bad-sequence proof of Higman’s lemma, specialized to a well-ordered alphabet, so that the summability argument would be auditable. That proof is not reprinted here, because the identical argument is already written out in this collection, as `spec:lem:words` and `spec:lem:neumann` of the finite-dimensional spectral report [2]. Lemma 2.1 above states what both parts use, and strict positivity of the letters in an ordered group is an additional hypothesis used only after the word-order result, to pass from embeddings to comparisons of sums.

C Part III: repository and literature audit

C.1 Repository snapshot

The Part III source compared its results with the repository at

048b72cf7cbfc8ab246e4f73788c10460cb3f6e0,

reviewed on September 22, 2026 [4]. It consulted the root README, `docs/README.md`, `docs/manifest.tex`, the incoming-report inventory, and the README and source of this report as it then stood, with Parts I and II only. It identified Part I’s constant-normal spectral formula as the closest existing result and credited it as an overlap. It noted that the report’s related-work paragraph already mentioned Drazin spectra and cited Boasso, so that a claim that the repository contained no Drazin mention would have been false. A GitHub code search for “Drazin” returned no matches but marked its results incomplete, and was not used as evidence of absence; a search of the retrieved catalogue found no occurrence, and the retrieved spectral source contained the related-work mention. The novelty assessment rests on comparison of statements and hypotheses. No repository file was changed. The delivered audit is kept unchanged as `04-drazin-halos-RESEARCH_AUDIT.md`.

C.2 Primary sources

The bibliographic record of Drazin’s paper [19] (the audit does not claim a full reading of it); the relevant portions of Boasso’s two papers [20, 13], including definitions, the resolvent-pole characterization and spectral mapping; Koliha’s distinction between finite-index and generalized Drazin inverses [21]; Gonshor and van den Dries–Ehrlich for normal forms and field foundations [5, 22]; and Kaplan–Krapp–Serra for contemporary Hahn and surreal automorphism context [23],

not credited with the spectral statements. Targeted searches combining Hahn, Drazin, Laurent inverse, surreal spectrum and ramified spectral mapping did not locate the combined theorem package; this does not establish priority.

C.3 Statements of the source that are stale or had to be corrected

- *Divisibility in Part I.* The source says that Part I works “under a standing divisibility assumption” (its audit: “with divisibility as a standing convention”), and presents the absence of divisibility as one of the hypotheses Part III drops. At its pin 048b72c this was accurate: Part I then kept divisibility as a standing convention although its norm did not need it. Commit 6e34651 removed that assumption from Part I (section 5.1) before Part III was placed. Relative to the present Part I, Part III drops normality and the Hilbert and Banach structure, not divisibility (remark 33.2).
- *README line numbers.* The audit locates the normal formula at lines 67–70 of this report’s README at the pin. That was correct there; the README has since been rewritten, and the formula is theorem 8.2.
- *What the report contains.* The audit describes this report as two inequivalent theories, which was accurate at the pin. The report now also contains Parts III and IV, and the related-work paragraph it cites (section 1.8) now points to Part III.
- *Notation.* The source’s letters A , D , P , Q , B , V , h , r , K_Γ and $\mathfrak{m}_\Gamma$ are renamed as stated at the start of Part III; no statement changes.

D Part IV: repository and literature audit

D.1 Repository snapshot

The Part IV source is pinned to the same revision 048b72c as Part III and was inspected on September 22, 2026; its build and verification timestamps fall on September 23 in UTC. It retrieved, through a GitHub connector, the root README.md, docs/README.md, this report’s README and source, and the READMEs of the dynamics and analytic-geometry reports, and checked the introduction and relevant scope passages of this report. It identified Part I’s constant-normal formula and positive-order range-defect factorization as predecessors, not discoveries. The comparison was targeted, not a review of every file or proof, and no repository file was modified. The delivered record is kept unchanged as 05-fredholm-PROVENANCE.md.

D.2 Primary sources and search scope

Conway and Gonshor for surreal normal forms [24, 5]; Higman [7]; Poonen, Corollary 4, for Hahn-field algebraic closedness [25]; Kato for compact spectral and isolated-cluster perturbation inputs [26]; Simon for the direct-rotation formula [27]; Bornemann, Section 3, for ordinary trace-class determinant identities [28]; and Serre for prior non-Archimedean Fredholm theory in a different operator category [29]. The Bornemann, Simon and Poonen material was also inspected through page images. Targeted searches combined Hahn, surreal, compact operator, trace class, Fredholm determinant, Lidskii and noncommuting perturbation; exact-phrase searches for “Hahn series” with “Fredholm determinant” and for “Hahn” with “Lidskii” produced no directly matching treatment. That negative outcome is not a proof of priority, and unrelated search snippets were not used as evidence.

D.3 Statements of the source that are stale or had to be corrected

- *Trace class.* The source says this report “explicitly excludes a trace-class theory”. That was true at the pin and remains true of Parts I–III. The report now contains Part IV’s coefficientwise trace and Fredholm determinant, which is still not a general trace-class theory (section 1.1).
- *Constant spectra.* The source describes the bounded-coefficient part as computing the spectrum of constant normal operators. The report now also computes the adjointable-algebra spectrum of every constant bounded operator (Part III).
- *Divisibility.* The source’s divisibility hypothesis is its own and is kept. Unlike the Part III source, it makes no claim about Part I’s hypotheses on Γ .
- *Duplicated inputs.* The source re-proves the range-defect factorization, the support lemma and the Neumann inverse. They are printed once (theorem 7.1 and lemmas 2.1 and 2.3); the source’s different proof of the first is kept as remark 40.1, and its deduction of the word lemma from Higman’s theorem as a paragraph after lemma 2.1.
- *Notation.* The source’s $\mathcal{B}$, $\sigma_{\mathcal{B}}$, σ_{bij} and res are written $\mathcal{A}_{\Gamma}(H)$, $\sigma_{\Gamma}^{\text{adj}}$, $\sigma_{\Gamma}^{\text{alg}}$ and st ; its letters D , B , P_0 , W and M (rotation), U , V (finite-rank factors), M , L , R (Fredholm reduction), d_0 , d_A and G become $D_{\star}$, Y , P_c° , Ω and $I - (P - P^{\circ})^2$, ι , κ , M_F , L_F , R_F , $\mathfrak{d}_0$, $\mathcal{F}_A$ and $\mathcal{G}$; no statement changes.

E How this report was assembled

The report is built from **four** research manuscripts, all dated September 22, 2026. Two arrived together and were assembled first as Parts I and II; two arrived in a later batch and were added as Parts III and IV.

Part	Files	Pin	Contribution
I	02-hahn-hilbert-spectral-*	608dd23	Hahn–Hilbert spaces: automatic adjoints, defect rigidity, normal spectral formula, coercive defects, norm attainment
II	03-infinite-spectral-*	a3124af	Row- and column-finite Hahn matrices: diagonalization, synthesis, commutants, algebra-relative spectrum, walks
III	04-drazin-halos-*	048b72c	Drazin halos in any unital algebra, exact resolvent valuation, ramified spectral mapping, Volterra and block examples
IV	05-fredholm-*	048b72c	Compact self-adjoint operators under noncommuting perturbations; obstruction to global diagonalization; Hahn Fredholm determinant

The source manuscripts themselves are not shipped; their code, data and provenance records are, under the numbers above. Where the merge had to choose:

- Results duplicated between sources are printed once: the support lemma and Neumann inverse (all four sources), and the range-defect factorization (Parts I and IV), with the Part IV proof kept as a second route. The Part III normal-case corollary is kept as a second proof of the algebra half of theorem 8.2, not merged into it.
- Parts III and IV are kept as separate parts, not merged into Part I, because Part III’s spectrum is relative to an arbitrary algebra and Part IV assumes divisibility and a separable infinite-dimensional H , which Part I does not.

- Letters with an existing meaning in Parts I and II were renamed in the new parts (appendices C.3 and D.3); no existing label or statement of Parts I and II was changed, and cross-references to the new parts were added to them.
- The comparison of the algebra-relative spectra (proposition 3.1) and the cross-references between parts were made in assembly and are marked as such; they add no hypothesis to any source theorem and no result to any source.

F Dependency and proof-obligation ledgers

F.1 Part I

Result	Inputs	Independent restriction
Positive inner product and norm	Finite Hahn convolution; even leading exponent; binomial root	No divisibility needed
Automatic adjoint structure	Ordinary closed graph; Baire; Hahn support	Everywhere-defined adjoints
Defect splitting	Injective leading term; Neumann support lemma	No topology or commutation needed
Normal spectral formula	Defect splitting; ordinary normal spectral theorem	Constant normal operator
New eigenvalue exclusion	Orthogonal eigenspace projection	Normality is essential
Continuum cokernel	Leading-coefficient obstruction; power-series divergence test	Ordinary ℓ^2 coefficients
Closed coercive range	Continuous defect projection; kernel identity	Arbitrary value group
Least norm-bound criterion	Ordinary positive square root; leading norm ratio	Constant bounded operator

Table 3: Where each Part I hypothesis enters; “complete” does not replace “orthogonally complete.”

All general statements in theorem 1.1 have proofs in the text, relative to the named classical inputs. No computational observation is used as a premise of those proofs. The most consequential steps to audit independently are the Baire uniform-support argument, the complement projection in theorem 7.1, and the accumulation-point case in theorem 8.2; their details are given without an appeal to a numerical experiment.

F.2 Part II

The logical dependence of the Part II claims is as follows. The Hahn algebra and inner product depend on finite coefficient convolution, not on any spectral theorem. Ordered-word finiteness licenses the homogeneous construction. Simple residues make the divided commutator invertible on off-diagonal coefficients. Self-adjointness and positivity make the constructed eigenvalues real and permit near-one unitary normalization. The invertibility already proved for the normalization is what supplies two-sided unitarity. Synthesis uses vector-valued strong sums. Commutant classification uses only the pairwise nonzero eigenvalue differences. The full spectrum theorem adds a separate inverse-coherence argument. Locality uses the homogeneous formulas and finite weighted walks, and surcomplex transport uses the classical normal-form embedding.

The Part II results do not assert a solution to a named published open problem; uniqueness or priority of every underlying idea; existence of a bounded operator-norm diagonalizer; a Hilbert-space or probability interpretation; convergence of the Hahn series in the fine surreal topology or

in every intrinsic valuation topology; an operator-valued Hahn sum of all spectral projections; a spectral theorem for arbitrary repeated residues; coherence of arbitrary entrywise Hahn matrices; decidability for arbitrary surreal presentations; or a completed Lean formalization.

F.3 Part III

The positive-support lemma supports the straightening automorphism and the formal substitutions. The elementary Laurent recurrence supports the Drazin criterion. Straightening plus cyclic-support projection yields inverse descent, which yields the halo formula and the exact scale. The power image and the formal inverse germ, with the halo formula and the commuting-factor lemma, yield the ramified image, and the finite-union subgroup lemma turns that image into the necessary and sufficient criterion. Among the general results, only the Banach section and the pole-order profile (section 33 and proposition 34.8) use ordinary analytic spectral theory; the Volterra and block examples also use ordinary operator-norm estimates and convergence. Only the surreal specialization uses Conway normal forms. The steps most worth auditing independently are the surjectivity argument in lemma 31.1 and the converse direction of lemma 30.2.

The Part III results do not assert a classification of nonconstant Hahn operators, of all linear endomorphisms, or of bijectivity spectra beyond the worked examples; algebraicity for non-Banach algebras; a least field-valued norm, trace-class determinant or spectral measure; a solution of a named published conjecture; priority; or a Lean formalization.

F.4 Part IV

The main spectrum theorem uses ordinary compact self-adjoint spectral theory, finite-word support, Part I's range-defect theorem and finite Hermitian diagonalization over the divisible field. The obstruction uses only the simple-eigenvalue instance and one commutator equation; the trace and product failures add the explicit second-order recurrence and finite summation identities. The determinant theorem uses the ordinary trace-class determinant, Taylor lifting, the ordinary index-zero Fredholm alternative and finite-dimensional linear algebra; the accumulation obstruction uses only a common coefficient neighbourhood and a valuation comparison. The steps most worth auditing independently are the self-adjointness of the coefficientwise Riesz projection (proposition 41.1), the well-ordered induction in theorem 43.1, and the kernel-cokernel identification in theorem 44.4.

The Part IV results do not assert a spectral classification of arbitrary elements of $\mathcal{A}_\Gamma(H)$ or of compact residues with nontrivial kernel; anything without divisibility; a positive spectral measure or general non-Archimedean Hilbert-space spectral theorem; equivalence with intrinsic nuclear or compactoid ideals; a determinant for negative-valuation trace-class series; the impossibility of every regularized spectral product; anything outside the named coherent analytic category; a failure of global diagonalization outside bounded coefficients; a solution of a named published conjecture; priority; or a Lean formalization.

F.5 What this report as a whole does not claim

Nothing above unifies the four parts into a single theorem, and no hypothesis is transferred between them. Parts I–III allow nondivisible Γ by their respective proofs; Part IV assumes divisibility, and nothing here transfers its conclusions to nondivisible groups. The row- and column-finiteness of Part II is not available in Part I or Part IV. The spectra of table 1 are not proved to agree, and no claim is made that one determines another, beyond the identities of definitions in proposition 3.1 and the equalities proved for particular operators (theorems 8.2 and 42.1, section 35.2). Parts I–III are not trace-class theories; Part IV supplies a coefficientwise trace and a Fredholm determinant on the nonnegative-valuation component of $\mathcal{S}_1(H)((t^\Gamma))$,

and neither a general trace-class nor a nuclear-operator theory. Part II supplies a vectorwise Hahn-additive projection calculus, not a general topologically countably additive spectral-measure theorem; the counterexample in remark 19.4 states the precise limitation. No part supplies a positive spectral measure. The local cluster theory of Part IV does not assemble into a global bounded-coefficient spectral theorem (theorem 43.1).

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